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CSC 203

Discrete Structures



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Course Introduction

Welcome to the Discrete Structures course. This covers essential mathematical concepts that underpin computer science and technology. The course introduces key topics like propositional and predicate logic, proof techniques, combinatorics, and recurrence relations. By the end of this course, you will have developed strong problem-solving skills and be able to apply logic and counting techniques to real-world applications.

What You Will Learn in This Course

This course introduces the foundations of discrete mathematics and logical reasoning used in computer science. You will learn how to represent and analyse logical statements using propositional and predicate logic, explore sets and functions, and apply formal proof techniques. The course also covers counting methods in combinatorics, solving recurrence relations, and understanding the basics of discrete probability for problem-solving in computing.

Course Aims

This course builds a strong foundation in discrete structures for computer science, focusing on logic, set theory, proof techniques, and combinatorics. It equips you with problem-solving skills using counting methods, permutations, combinations, and recurrence relations, and helps you develop the ability to construct logical arguments and apply mathematical concepts to computing problems.

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Course Objectives

Throughout this course, you will achieve a set of specific and measurable objectives that contribute to your overall mathematical growth. By the course's conclusion, you will be able to:

1. Convert logical statements from informal language to propositional and predicate logic expressions.
2. Describe the strengths and limitations of propositional and predicate logic, outlining basic proof techniques.
3. Apply proof techniques correctly in constructing sound arguments and use the pigeonhole principle in formal proofs.
4. Compute permutations and combinations, interpret their meaning, and solve problems in context.
5. Map real-world problems to appropriate counting formalisms and solve recurrence relations.

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Study Session One

Propositional Logic

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Introduction

Real numbers are represented on the number line, where each point shows a unique real number in both directions, allowing both positive and negative numbers. Real numbers can be classified into different sets, such as natural numbers, integers, rational numbers, and irrational numbers. Trichotomy's law is also defined as stating that a real number can be either negative or positive.

Learning Outcomes for Study Session 1

When you have studied this session, you should be able to:

1.1 Analyse the number system, such as the classification of numbers, such as natural numbers, whole numbers, integers, and real numbers.

1.2 Examine Trichotomy's Law and the purpose of absolute numbers.

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1.1 Logical Statements

In this section, we shall introduce the basic properties of the set, which is generally called the $x - axis$. This set is denoted in mathematical analysis by the letter R and is called the **set of real numbers.** Members of R are called real numbers. We shall denote the set of natural numbers by $N = \{1, 2, 3, ...\}$. We shall also denote the set of all integers by $Z = \{..., -2, -1, 0, 1, 2, ...\}$ and the set of rational numbers (or fractions) by $Q = \left\{\frac{p}{q}: p, q \in Z, q \neq 0\right\}$.

1.2 Propositional Logic Expressions

In this section, we defined an order relation ($<$) with the following order axioms:

Axiom 1: (Trichotomy Law). For all $x, y \in R$, exactly one of the following three relations must hold:

- $x = y$
- $x < y$
- $y < x$.

It is conventional to agree; that $x < y$ means $y > x$. we also write $x \leq y$ for $x < y$ or $x = y$. similarly, $y \geq x$ means $y > x$ or $y = x$. numbers $x \in R$ such that $y < 0$ are called 'negative'. The trichotomy axioms assert that every real number other than zero is either positive or negative but never both, and zero is neither positive nor negative. Additionally, the trichotomy axioms assert that the order relation neglects no number.

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We now introduce two more order relations.

Axiom 2: For all $x, y \in R$,

$x < y$ if and only if $x + z < y + z$.

Axiom 3: For all $x, y \in R$, if $x > 0$ and $y > 0$, then,

$x + y > 0; \quad xy > 0$.

With all these order axioms, we prove the following proposition.

Proposition 1.2.1: Let $x, y \in R$,

(i) $x < y$ if and only if $x - y < 0$ and $y - x > 0$.

(ii) If $x < y$ and $y < z$.

Proof: $x < y$ if and only if $x + (-y) < y + (-y)$, by (axiom 2)

That is $x < y \; if \; and \; only \; if \; x - y < 0$.

Moreover, $x - y < 0$ if and only if $x - y + (y - x) < 0 + (y - x)$,

i.e: $x - y < 0$ If and only if $x - y + y - x < y - x$ If only if $0 < y - x$.

From the above inequalities, we obtain

$x < y$ if and only if $x - y < 0$ if and only if, $y - x > 0$, thus establishing (i).

To prove (ii), we use **Axiom 3** as follows:. Assume $x < y$ and $y < z$. this is the same as:

$y - x > 0$ and $z - y > 0$.

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By **axiom 3,** the sum of these two positive numbers is positive. Hence,

$(y - x) + (z - y) > 0,\ i.e.,\ z - x > 0,\ i.e: x < z.$

With **proposition 1.2.1** proved, we can now use obvious statements such as these:

For all $x, y, z \in R$,

a. $x < y$ and $y \leq z \Longrightarrow x < z$.

b. $x \leq y$ and $y \leq z \Longrightarrow x \leq z$.

We shall also need the notion of **the absolute value** of a real number. If we define the absolute value of a, denoted by $|a|$, by

$|a| = \{a\ if\ a > 0,\ 0\ if\ a = 0,\ \ -a\ if\ a < 0\}.$

For example, $|-3| = 3;\ |3| = 3$, and so on. Intuitively, the absolute value of a represents the distance from 0 to a. Some basic properties of the absolute value are given in the following theorem.

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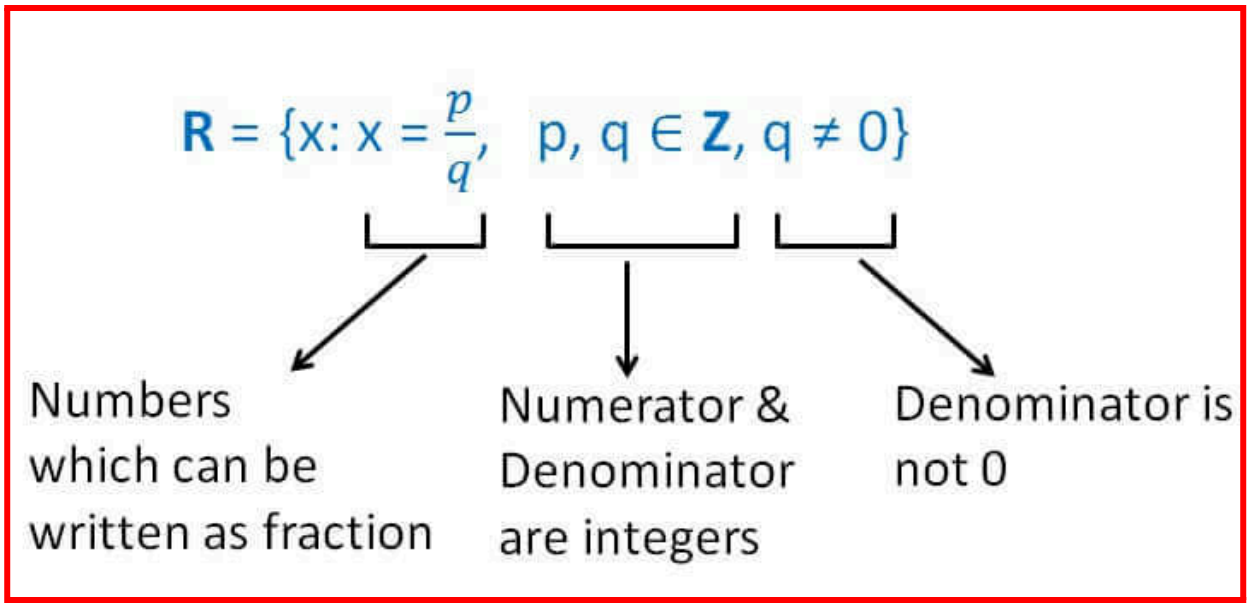


Figure 1.1. Propositional logic expressions

Theorem 1.2.1

(i) $|a| \geq 0$ for all $a \in R$; $|a| = 0$ if and only if $a = 0$.

(ii) $|a| \leq \epsilon$ if and only if $-\epsilon \leq a \leq \epsilon$.

(iii) $|ab| = |a||b|$ for all $a, b \in R$;

(iv) $|a + b| \leq |a| + |b|$ for all $a, b \in R$.

Proof:

(i) There are two cases: if $a \geq 0$, then $|a| = a \geq 0$. On the other hand, if $a < 0$, then $|a| = -a$. In both cases, $|a| \geq 0$.

(ii) Since $a = |a|$ or $a = -|a|$, it follows that $-|a| \leq a \leq |a|$.

Now, if $|a| < \epsilon$, then we have $-\epsilon \leq -|a| \leq a \leq |a| \leq \epsilon$.

And if $a < 0$, then $|a| = -a \leq \epsilon$. In both cases, $|a| \leq \epsilon$.

(iii) Consider four cases when a they b are both positive, both negative, or have opposite signs.

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From the inequalities $-|a| \leq a \leq |a|$ and $-|b| \leq b \leq |b|$, we obtain that

$-(|a| + |b|) \leq a + b \leq |a| + |b|$.

This implies that $|a + b| \leq |a - b|$ for all $a, b \in R$.

Remark: Condition (iv) of **Theorem 1.2.1** is often called the triangle inequality. It roughly says that the length of any side of a triangle is less than or equal to the sum of the lengths of the other two sides. A useful variant of the triangle inequality is

$||a| - |b|| \leq |a - b| \; for \; all \; a, b, \in R$.

In-Text Questions and Answers (ITQs and ITAs)

ITQ 1.1: According to the Trichotomy Law, what are the possible relationships between two real numbers x and y?

Answer: x = y, x < y, or y < x. The Trichotomy Law states that exactly one of these relations must hold for any real numbers x and y. This property ensures the total ordering of real numbers.

ITQ 1.2: What does the absolute value of a real number represent?

Answer: Its distance from zero on the number line. The absolute value of a number shows how far that number is from zero, regardless of its direction (positive or negative).

Conclusion

This study session on propositional logic has equipped you with a basic knowledge of formal logic. We have covered the foundation of logical

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statements, the power of constructing complex expressions using connectives, and the crucial role of logical implications and inference rules. Propositional logic provides a powerful framework for analysing arguments, reasoning about truth values, and simplifying complex statements.

Summary

1. Trichotomy's law shows that a real number can be either positive or negative.
2. Classification of real numbers.
3. Equation showing why the absolute value of a number is positive.

Self-Assessment Questions and Answers (SAQs and SAAs)

SAQ 1.1. Define the set of rational numbers Q.

Answer: $Q = \{p/q : p, q \in Z, q \neq 0\}$. Rational numbers are numbers that can be expressed as a fraction where both the numerator and denominator are integers, and the denominator is not zero.

SAQ 1.2. Why is 0 neither positive nor negative according to the Trichotomy Law?

Answer: This is because it does not satisfy $x < 0$ or $x > 0$. Zero serves as the dividing point on the number line and therefore holds a neutral value—neither positive nor negative.

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SAQ 1.3. Classify the number √2 in the real number system.

Answer: Irrational number. √2 cannot be expressed as a ratio of two integers, making it irrational but still a real number.

Multiple-Choice Questions and Answers (MCQs and MCAs)

1. **Which statement correctly expresses Trichotomy's Law?**

A. Every real number is either positive, negative, or zero.

B. Every real number can be both positive and negative.

C. Only integers can be positive or negative.

D. The sum of two real numbers is always positive.

Answer: A

Explanation: Trichotomy's Law states that for any two real numbers x and y, exactly one of the following is true: $x = y$, $x < y$, or $y < x$.

In particular, for any real number x, it must be positive, negative, or zero — never more than one at a time. This ensures that the real number line is totally ordered.

2. **Which of the following correctly describes the set of integers (Z)?**

A. All positive whole numbers only.

B. All positive and negative whole numbers, including zero.

C. All fractions and decimals.

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D. All real numbers except zero.

Answer: B

Explanation: The set of integers (Z) consists of all whole numbers and their negatives, including zero: ..., -3, -2, -1, 0, 1, 2, 3, ...

This set bridges the gap between natural numbers and rational numbers.

3. **What does the expression |a| represent for any real number a?**

A. The square of a.

B. The reciprocal of a.

C. The absolute value of a.

D. The additive inverse of a.

Answer: C

Explanation: The absolute value of a, denoted |a|, represents the distance of a from zero on the number line, irrespective of direction.

For instance, |3| = 3 and |-3| = 3.

It is always non-negative because distance cannot be negative.

4. **According to Proposition 1.2.1, which of the following is true?**

A. $x < y$ if and only if $x - y > 0$

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B. $x < y$ if and only if $x - y < 0$

C. $x < y$ if and only if $y - x < 0$

D. $x < y$ if and only if $x = y$

Answer: B

Explanation: Proposition 1.2.1 links inequalities to arithmetic subtraction. It states that $x < y$ holds if and only if $x - y < 0$.

In other words, if x is less than y, their difference must be negative, which aligns with the ordering of numbers on the number line.

5. **Which of the following inequalities represents the Triangle Inequality?**

A. $|a + b| \leq |a| + |b|$

B. $|a + b| \geq |a| + |b|$

C. $|a - b| = |a| + |b|$

D. $|a| = |b| + |a - b|$

Answer: A

Explanation: The Triangle Inequality is a fundamental rule in real analysis and geometry. It asserts that the absolute value of the sum of two numbers cannot exceed the sum of their absolute values.

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This mirrors the geometric idea that the length of one side of a triangle is always less than or equal to the sum of the other two sides.

Tutor-Marked Assessments (TMAs)

1. Prove that if $\epsilon > 0$ and $a, x, \in R$, then $|a - x| < \epsilon$ if and only if $x - \epsilon < a < x + \epsilon$.
2. Find all $x \in R$ that satisfy the following inequalities:

   (i) $|4x - 3| \leq 11$ (ii) $|x - 2| > |x + 1|$ (iii) $|2 - 5x| \leq 7$
3. Prove that for all $a, b, \in R$,

   $|a - b| \leq |a - b|$.
4. Prove that for all $a, b, c \in R$,

   $|a + b + c| \geq |a| - |b| - |c|$.
5. Solve the inequality

   $\left|x - \frac{1}{2}\right| > 1$.

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Study Session Two

Introduction to Predicate Logic

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Introduction

Welcome to this session on predicate logic!

In our journey through logic, we've explored propositional logic, which deals with simple statements and their logical connectives. Now, we're diving deeper into predicate logic, a more powerful form of logic that allows us to express complex statements about the relationships between objects and properties.

Predicate logic is incredibly useful in mathematics and computer science, where precise and formal reasoning is essential. It provides a framework for expressing statements about the properties of objects, relationships between them, and conditions that may or may not hold true.

In this lesson, we'll start by understanding the basics of predicate logic, including predicates, variables, and quantifiers. Predicates represent the properties or attributes of objects, while variables allow us to refer to unspecified objects. We'll then explore quantifiers, such as the universal quantifier ($\forall$) and the existential quantifier ($\exists$), which enable us to make statements about all objects in a domain or at least one object satisfying a condition.

Finally, we'll learn how to translate English sentences into predicate logic, allowing us to formalise natural language statements and reason about them using the principles of logic.

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Learning Outcomes for Study Session 2

When you have studied this session, you should be able to:

2.1 List the two major types of quantifiers in predicate logic.

2.2 Identify universal or existential quantifiers.

2.3 Make use of universal or existential quantifiers to form a logical statement.

2.4 Negate a quantified statement

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2.1 Universal Quantifier ∀

The universal quantification of $P(x)$ for a particular domain is the proposition that asserts that $P(x)$ is true for all values of x in this domain. Note that the domain specifies the possible values of the variable x.

The notation $\forall x\, P(x)$ denotes the universal quantification of P(x). Here, '∀' is called the **universal quantifier**. We read ∀x P(x) as 'for all x P(x)' or 'for every x P(x).'

An element for which P (x) is false is called a **counterexample of ∀x P(x).**

Example 1: Let $P(x)$ be the statement, '$x^2 + 2 \geq x$' What is the truth value of the quantification ∀x P(x), where the domain consists of all real numbers?

Solution: We observe that the statement P (x): $x^2 + 2 \geq x$ is true for all real numbers x; therefore, the quantification $\forall x\, P(x)$ is true.

Example 2: Let $Q(m)$ be the statement '$2m < 4$.' What is the truth value of the quantification $\forall m\, Q(m)$, where the domain consists of all integers?

Solution: We observe that the statement $Q(m)$: $2m < 4$ is not true for all integers m, For instance, $Q(0)$: $2(0) < 4$ is true but $Q(3)$: $2(3) < 4$ is not true. $Q(3)$ is a counterexample to the statement $\forall m\, P(m)$. Therefore, the quantification $\forall m\, P(m)$ is not true.

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Example 3: What does the statement $\forall x\, C(x)$ mean if $C(x)$ is 'Computer x is connected to the network' and the domain consists of all computers on the Miva Open University campus?

Solution: The statement $\forall x\, C(x)$ means that for every computer x on the Miva University campus, that computer x is connected to the network.

This statement can be expressed in English as 'Every computer on the Miva Open University campus is connected to the network.'

Example 4: What is the truth value of $\forall x(x^2 \geq x)$ if the domain consists of all real numbers? What is the truth value of this statement if the domain consists of all integers?

Solution: The universal quantification $\forall x(x^2 \geq x)$, where the domain consists of all real numbers, is false. For example, $(\frac{1}{2})^2$ is not greater than or equal to $\frac{1}{2}$. Note that $x^2 \geq x$ if and only if $x^2 - x = x(x-1) \geq 0$. Consequently, $x^2 \geq x$ if and only if $x \leq 0$ or $x \geq 1$. It follows that $\forall x(x^2 \geq x)$ is false if the domain consists of all real numbers (because the inequality is false for all real numbers x with $0 < x < 1$. However, if the domain consists of integers, $\forall x(x^2 \geq x$ is true, because there are no integers x with $0 < x < 1$.

2.2 Existential Quantifiers ∃

Existential quantification forms a proposition that is true if and only if P (x) is true for at least one value of x in the domain.

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The existential quantification of P (x) is the proposition 'There exists an element x in the domain such that P (x).' We use the notation $\exists x P(x)$ for the existential quantification of P(x). Here, '$\exists$' is called the existential quantifier.

Besides the phrase 'there exists', we can also express existential quantification in many other ways, such as by using the words 'for some', 'for at least one', or 'there is'.

The existential quantification $\exists x P(x)$ is read as

'There is an x such that P(x).' or

'There is at least one x such that P(x).' or

'For some x P(x).'

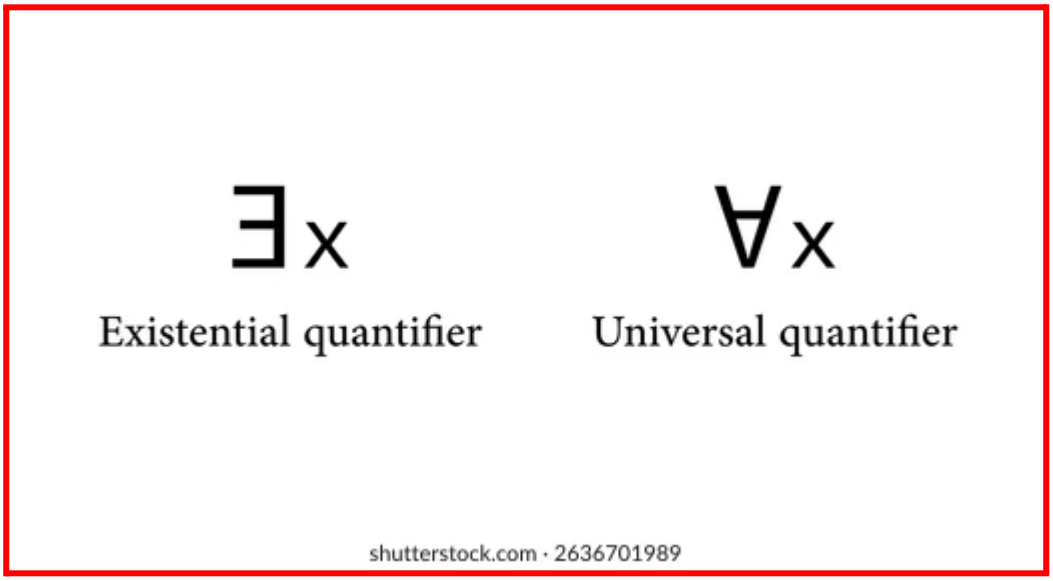


Figure 2.2. Existential vs universal quantifier

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In-Text Questions and Answers (ITQs and ITAs)

ITQ 2.1: What is the main difference between propositional logic and predicate logic?

Answer: Predicate logic deals with objects, properties, and relationships between them using variables and quantifiers; propositional logic only handles simple true/false statements.

ITQ 2.2: How is an existential quantifier written and interpreted?

Answer: Written as ∃x P(x) and read as 'There exists an x such that P(x).

2.2.1 Examples on Existential Quantification

Example 1: Let $P(x)$ denote the statement '$x < 0$.' What is the truth value of the quantification $\exists x\, P(x)$, where the domain consists of all real numbers?

Solution: We observe that '$x < 0$' is sometimes true.

For instance, when $x = -2$.

Therefore, the existential quantification of $P(x)$, which is $\exists x\, P(x)$, is true.

Example 2: Let Q(x) denote the statement 'x = x + 4'. What is the truth value of the quantification $\exists x\, Q(x)$, where the domain consists of all real numbers?

Solution: $Q(x)$ is false for every real number x.

Therefore, the existential quantification of $Q(x)$, which is $\exists x\, Q(x)$, is false.

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2.3 Negating Quantified Expressions

We will often want to consider the negation of a quantified expression. For instance, consider the negation of the statement:

'Every student in your class has taken a course in calculus.'

This statement is a universal quantification, namely,

$\forall x P(x)$,

where P (x) is the statement 'x has taken a course in calculus' and the domain consists of the students in your class. The negation of this statement is *'It is not the case that every student in your class has taken a course in calculus.'* This is equivalent to *'There is a student in your class who has not taken a course in calculus.'* And this is simply the existential quantification of the negation of the original propositional function, namely,

$\exists x \neg P(x)$.

This example illustrates the following logical equivalence:

$\neg \forall x P(x) \equiv \exists x \neg P(x)$.

To show that $\neg \forall x P(x)$ and $\exists x P(x)$ are logically equivalent no matter what the propositional function P (x) is and what the domain is, first note that $\neg \forall x P(x)$ is true if and only if $\forall x P(x)$ is false. Next, note that $\forall x P(x)$ is false if and only if there is an element x in the domain for which P (x) is false. This holds if and only if there is an element x in the domain for which $\neg P(x)$ is true. Finally, note that there is an element x in the domain for which $\neg P(x)$ is true if and only if $\exists x \neg P$

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(x) is true. Putting these steps together, we can conclude that $\neg \forall x P(x)$ is true if and only if $\exists x \neg P(x)$ is true. It follows that $\neg \forall x P(x)$ and $\exists x \neg P(x)$ are logically equivalent.

Suppose we wish to negate an existential quantification. For instance, consider the proposition, 'There is a student in this class who has taken a course in calculus.' This is the existential quantification $\exists x Q(x)$, where $Q(x)$ is the statement 'x has taken a course in calculus.; The negation of this statement is the proposition, 'It is not the case that there is a student in this class who has taken a course in calculus.' This is equivalent to 'Every student in this class has not taken calculus,' which is just the universal quantification of the negation of the original propositional function, or, phrased in the language of quantifiers,

$$\forall x \neg Q(x).$$

This example illustrates the equivalence

$$\neg \exists x Q(x) \equiv \forall x \neg Q(x).$$

To show that $\neg \exists x Q(x)$ and $\forall x \neg Q(x)$ are logically equivalent no matter what $Q(x)$ is and what the domain is, first note that $\neg \exists x Q(x)$ is true if and only if $\exists x Q(x)$ is false. This is true if and only if no x exists in the domain for which $Q(x)$ is true. Next, note that no x exists in the domain for which $Q(x)$ is true if and only if $Q(x)$ is false for every x in the domain. Finally, note that $Q(x)$ is false for every x in the domain if and only if $\neg Q(x)$ is true for all x in the domain, which holds if and only if $\forall x \neg Q(x)$ is true. Putting these steps together, we see that $\neg \exists x Q(x)$ is true if and only if $\forall x \neg Q(x)$ is true. We conclude that $\neg \exists x Q(x)$ and $\forall x \neg Q(x)$ are logically equivalent.

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Example 1: What are the negations of the statements 'There is an honest politician' and 'All Americans eat cheeseburgers'?

Solution: Let H (x) denote 'x is honest.' Then the statement 'There is an honest politician' is represented by ∃xH (x), where the domain consists of all politicians. The negation of this statement is ¬∃xH (x), which is equivalent to ∀x¬H (x). This negation can be expressed as 'Every politician is dishonest.' (**Note:** In English, the statement 'All politicians are not honest' is ambiguous In common usage, this statement often means 'Not all politicians are honest.' Consequently, we do not use this statement to express this negation.)

Conclusion

In conclusion, universal quantifiers (∀) assert that a statement holds true for all elements in a given domain, while existential quantifiers (∃) indicate that there exists at least one element for which the statement is true. Negating quantified expressions involves interchanging universal and existential quantifiers and negating the proposition within them. Understanding these concepts is crucial for reasoning about statements involving generalisations across domains and their negations.

Summary

1. Universal quantification (∀x P(x)) asserts that a proposition P(x) is true for all values of x in a specified domain, denoted as 'for all x P(x)' or 'for every x P(x).'

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2. A counterexample of ∀x P(x) is an element for which P(x) is false within the given domain.
3. Existential quantification (∃x P(x)) signifies that a proposition P(x) is true for at least one value of x in the domain, expressed as 'there exists an x such that P(x)' or 'for some x P(x).'
4. The negation of a universal quantification ¬∀xP(x) is logically equivalent to the existential quantification of the negation ∃x ¬P(x).
5. Similarly, the negation of an existential quantification ¬∃xQ(x) is equivalent to the universal quantification of the negation ∀x ¬Q(x).
6. Universal quantification (∀x C(x)) of a statement C(x) like 'Computer x is connected to the network' implies that every computer in the specified domain (e.g., all computers on a campus) is connected to the network.
7. Existential quantification (∃x Q(x)) denotes that there exists at least one x in the domain for which statement Q(x) holds true, for example, 'There exists an honest politician.'

Self-Assessment Questions and Answers (SAQs and SAAs)

SAQ 2.1: Let $P(x)$ denote the statement 'x=0.' What is the truth value of the quantification $\exists x\, P(x)$, where the domain consists of all real numbers?

Answer: We observe that 'x=0' is true for a single value of 0. Therefore, the existential quantification of $P(x)$, which is $\exists x\, P(x)$, is true.

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SAQ 2.2: Let $Q(x)$ denote the statement 'x −1 = x.' What is the truth value of the quantification $\exists x\, Q(x)$, where the domain consists of all real numbers?

Answer: We observe that 'x−1=x' is never true. Therefore, the existential quantification of Q(x), which is $\exists x\, Q(x)$, is false.

SAQ 2.3: What is the difference between ∀x P(x) and ∃x P(x)?

Answer: ∀x P(x) means P(x) is true for every x in the domain. ∃x P(x) means P(x) is true for at least one x in the domain.

Multiple-Choice Questions and Answers (MCQs and MCAs)

1. **What is the logical equivalence of ¬ ∃x P(x)?**

A. ∃x ¬P(x)

B. ∀x P(x)

C. ∀x ¬P(x)

D. ¬ ∀x ¬P(x)

Answer: C

Explanation: Negating an existential quantifier changes it into a universal quantifier with negation inside:

¬ ∃x P(x) ≡ ∀x ¬P(x).

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Meaning: "There does not exist an x for which P(x) is true" ⇒ "P(x) is false for all x."

2. **If ∀x P(x) is false, what must be true?**

A. P(x) is true for all x.

B. ∃x ¬P(x) is true.

C. ¬∃x ¬P(x) is false.

D. P(x) is false for all x.

Answer: B

Explanation: If ∀x P(x) is false, it means not all x satisfy P(x); therefore, there exists at least one x for which P(x) is false.

That is, ∃x ¬P(x) is true.

3. **Which of the following correctly represents 'Some computers on campus are connected to the network'?**

A. ∀x C(x)

B. ∃x C(x)

C. ∃x ¬C(x)

D. ∀x ¬C(x)

Answer: B

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Explanation: 'Some computers' corresponds to existential quantification.

Thus, $\exists x\, C(x)$ — 'There exists at least one computer that is connected to the network.'

4. **Which of the following correctly expresses the statement 'Every student passed the test'?**

A. $\forall x\, P(x)$

B. $\exists x\, P(x)$

C. $\forall x\, \neg P(x)$

D. $\neg \exists x\, P(x)$

Answer: A

Explanation: 'Every student passed' means that for all students x, P(x) ('x passed') is true.

Thus, $\forall x\, P(x)$ accurately captures this statement.

5. **What does the universal quantifier ($\forall$) indicate?**

A. The statement is true for some values of x.

B. The statement is false for all x.

C. The statement is true for all x in the domain.

D. The statement is sometimes true.

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Answer: C

Explanation: The universal quantifier (∀) declares that a predicate P(x) holds true for every element in the domain.

Example: ∀x ($x^2 \geq 0$) — every real number squared is non-negative.

Tutor-Marked Assessments (TMAs)

1. Show that ¬∀x P(x) ≡ ∃x ¬P(x) using logical reasoning.
2. Define and give examples of the two types of quantifiers used in predicate logic.
3. Negate the statement 'Some computers are infected with malware' and express the result in predicate logic.
4. Give one real-world example each of a universal and existential statement in computer science.

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Study Session Three

Set Operations and Relations

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Introduction

Welcome to this session on set operations and relations!

In our journey through sets and operations, we've explored set operations and relations, which are fundamental concepts in mathematics that play a crucial role in various areas of study, including algebra, logic, and computer science. Sets are collections of distinct objects, while relations establish connections or associations between elements of different sets. Understanding set operations and relations allows us to analyse and manipulate data, solve problems, and build sophisticated mathematical models.

In this exploration, we'll delve into the basic principles of set operations, such as union, intersection, and complement, which enable us to combine, compare, and classify sets. Additionally, we'll explore relations, which describe the interactions or connections between elements of sets, including properties like reflexivity, symmetry, and transitivity.

By mastering these concepts, we gain powerful tools for organising data, reasoning about relationships, and solving complex mathematical problems across various disciplines.

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Learning Outcomes for Study Session 3

When you have studied this session, you should be able to:

3.1 Perform the basic fundamental set operations, including union, intersection, and complement, and learn how to apply them to manipulate and analyse sets effectively.

3.2 Describe different types of relations, such as reflexive, symmetric, transitive, and equivalence relations, and grasp their significance in representing connections between elements of sets

3.3 Apply set operations and relations in practical problem-solving scenarios

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3.1 Set Operations

In this session, we will explore the fundamental operations involving sets. A set is a well-defined collection of distinct objects, considered as a single entity. These objects can be anything: numbers, letters, symbols, or even other sets. Sets are denoted by uppercase letters such as A, B, and C. An element is an individual object within a set. The elements of the set are written in lowercase letters. We denote the membership of an element x in a set A as $x \in A$. Conversely, if an element is not in a set, we use the notation $x \notin A$.

3.1.1 Types of Sets

3.1.1.1 Definition of Sets

A set is a well-defined collection of distinct objects, considered as a single entity.

- Examples of sets are sets of natural numbers, sets of real numbers, etc.

3.1.1.2 Non-Empty and Empty Sets

- A set with elements or members is called a non-empty set. A non-empty set with one member is called a singleton set.
- A set without elements is known as empty set, denoted by ϕ or $\{\}$

3.1.1.3 Finite and Infinite Sets

- A set is finite if all its elements can be listed . The set of students at MIVA Open University is an example of a finite set.

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- A set is infinite if all its elements cannot be listed . The infinite sets always begin or end with three dots to show their continuity; an example of an infinite set is the set of natural numbers. 1, 2, 3, …
- A universal set ξ is a set that contains all sets in consideration.

However, we shall discuss the concept of cardinality of sets and power sets.

- The cardinality of sets is the number of all elements present in a given set. n(A) stands for the cardinality of sets a. It is important to know that the cardinality of empty sets is zero

Example 1: Give that set $M = \{a, b, c, 1, 2. 3\}$. Determine its cardinality.

Solution: The cardinality of set M, i.e., $n(M) = 6$. If you count the elements of set M, you will notice that it contains six elements.

- The power set of set M is all possible subsets of the set M. The power set of set M is represented by $P(M) = 2^n$. where n is the cardinality of the set under consideration. It is important to know that every set is a subsets of itself and empty sets are also subsets of all sets

Example 2: Give a set $A = \{k, l, m\}$. Determine the P(A).

$P(A) = \{\phi, \{k\}, \{l\}, \{m\}, \{k, l\}, \{k, m\}, \{l, m\} \{k, l, m\}\}$

Set A has eight possible subsets, Using $P(A) = 2^n$, n=3

Hence $P(A) = 2^3 = 8$.

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3.1.2 Set Operations

Union of Sets (∪): The union of two sets A and B is the set containing all elements that are in either A or B, or both.

$A = \{-3, -1, 0, 1, 3, 4\}$

$B = \{-2, 0, 1, 3, 5, 6, 7\}$

$A \cup B = \{-3, -1, -2, 0, 1, 3, 4, 5, 6, 7\}$.

Intersection of Sets (∩): The intersection of sets A and B is the set containing all elements that are common to both sets A and B.

$A \cap B = \{0, 1, 3\}$

Difference of Two Sets: The difference of sets A and B is the set containing all elements that are in set A but not in set B.

$A = \{-3, -1, 0, 1, 3, 4\}$

$B = \{-2, 0, 1, 3, 5, 6, 7\}$

$A - B = \{-3, -1, 4\}$

Also,

$B - A = \{-2, 5, 6, 7\}$

Complement : The complement of a set *A* with respect to the universal set μ is a set that contains all the elements in μ but not in *A*. The complement of set A is denoted by A^c. If, $A \subseteq \mu$ and $x \epsilon \mu$, this implies

$x \epsilon A^1$ but $x \notin A$.

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Cartesian Product(×) : The Cartesian product of sets A and B is the set of all ordered pairs (a,b) where a is in A and b is in B.

Example 3: What is the Cartesian product of A = {1, 2} and B = {a, b, c}?

Solution: The Cartesian product A × B is A × B = {(1, a), (1, b), (1, c), (2, a), (2, b), (2, c)}.

In-Text Questions and Answers (ITQs and ITAs)

ITQ 3.1: What makes a relation reflexive?

Answer: A relation R on set A is reflexive if every element is related to itself. That is, for all a in A, (a, a) ∈ R. Example: R = {(1,1), (2,2), (3,3)} on A = {1,2,3} is reflexive.

ITQ 3.2: How do you find the power set of a given set A = {p, q}?

Answer: The power set is the set of all possible subsets of A. Since A has 2 elements, the power set has $2^2 = 4$ subsets: P(A) = {⌀, {p}, {q}, {p, q}}.

3.1.3 Properties of Sets Operations

- **Commutative**: $A \cup B = B \cup A$, $A \cap B = B \cap A$
- **Associative**: $(A \cup B) \cup C = A \cup (B \cup C)$, $(A \cap B) \cap C = A \cap (B \cap C)$
- **Distributive**: $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$, $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$

Equality of Sets : Two sets, A and B, are equal if they have the same number of elements. Therefore, $B \subseteq A$ and $A \subseteq B$. The idea of equality can be used to establish the validity of the three properties of sets that were discussed earlier.

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3.2 Relations

Relations and their types are one of the important concepts in set theory. Sets, relations, and functions are interlinked topics. Sets denote the collection of ordered elements, whereas relations and functions define the operations performed on sets.

A relation between sets A and B is any subset of their Cartesian product, A × B. These relations can be represented as sets of ordered pairs or through diagrams like directed graphs. Specifically, a binary relation from A to B is a set R of ordered pairs where the first element of each pair comes from A and the second element comes from B. We denote the relationship between elements a and b as "a R b" if (a, b) is in R, and 'a R b' if (a, b) is not in R. Additionally, when (a, b) belongs to R, we say that 'a is related to b by R.'

Example 4: Let A = {z, 1, 2} and B = {k, b}; then {(z, k), (z, b), (1, k), (2, b)} is a relation from A to B.

This means, for instance, that z R k but that 1 R b.

3.2.1 Properties of Relations

Let's discuss some properties of relations along with examples:

Reflexive Property:

- **Definition**: A relation R on a set A is reflexive if every element in A is related to itself.

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- **Example 5:** Consider the relation R on the set of integers defined by $R=\{(x,x)|\ x$ is an integer$\}$. Here, every integer is related to itself, so R is reflexive.

Symmetric Property:

- **Definition**: A relation R on a set A is symmetric if whenever (a,b) is in R, then (b,a) is also in R.
- **Example 6:** Let A be the set of all people, and define a relation R as "is a sibling of". If person x is a sibling of person y, then person y is also a sibling of person x. So, the relation "is a sibling of" is symmetric.

Transitive Property:

- **Definition**: A relation R on a set A is transitive if whenever (a,b) and (b,c) are in R, then (a,c) is also in R.
- **Example 7:** Consider the relation R on the set of real numbers defined by $R=\{(x,y)\ |\ x < y\}$. If $x < y$ and $y < z$, then it follows that $x < z$. So, the relation R is transitive.

Antisymmetric Property:

- **Definition**: A relation R on a set A is antisymmetric if whenever (a,b) and (b,a) are in R and $a \neq$ b, then (a,b) is not in R.
- **Example 8:** Let A be the set of all students, and define a relation R as "is older than". If student x is older than student y and student y is older than student x, then x and y must be the same person. So, the relation "is older than" is antisymmetric.

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Equivalence Relation Property:

- **Definition**: A relation R on a set A is an equivalence relation if it is reflexive, symmetric, and transitive.
- **Example 9:** Consider the relation R on the set of integers defined by R=$\{(x,y) | \ x-y$ is divisible by 5$\}$. This relation is reflexive, symmetric, and transitive, as (x,y) is always true (reflexive); if $x-y$ is divisible by 5, then y$-x$ is also divisible by 5 (symmetric); and if $x-y$ and y$-z$ are both divisible by 5, then $x-z$ is also divisible by 5 (transitive). Therefore, R is an equivalence relation.
- Understanding these properties helps in analysing and categorising relations, providing insights into their behaviour and characteristics.
- **Example 10:** Let A = {1, 2, 3} and B = {1, 2, 3, 4}. The relations R_1 = {(1, 1), (2, 2), (3, 3)} and R_2 = {(1, 1), (1, 2), (1, 3), (1, 4)} can be combined to obtain $R_1 \cup R_2$ = {(1, 1), (1, 2), (1, 3), (1, 4), (2, 2), (3, 3)}, $R_1 \cap R_2$ = {(1, 1)}, $R_1 - R_2$ = {(2, 2), (3, 3)}, $R_2 - R_1$ = {(1, 2), (1, 3), (1, 4)}

3.3.2 Set Operations and Relations in Practical Problem-Solving Scenarios

Data Analysis and Management: In database systems, set operations like union, intersection, and difference are used to manipulate and query datasets efficiently. Relations, modelled as tables, establish connections between entities, facilitating data retrieval and analysis.

1. **Network Analysis:** In social networks or communication systems, set operations are employed to analyse connections between individuals or

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devices. Relations, represented as edges in graphs, help identify patterns of interaction or communication flow, aiding in network optimisation and troubleshooting

2. **Engineering and Optimisation:** Set operations and relations are utilised in optimisation problems, where sets represent feasible solutions and relations capture constraints. Techniques like linear programming leverage these concepts to optimise resource allocation, production schedules, and logistics.
3. **Mathematical Modelling:** Sets and relations are fundamental in the mathematical modelling of real-world phenomena. For instance, in epidemiology, sets represent populations, and relations describe interactions between individuals, helping predict disease spread and inform intervention strategies.
4. **Decision Support Systems:** Set operations and relations are integral components of decision support systems, enabling complex decision-making processes. Relational models dependencies and preferences, while set operations facilitate scenario analysis

Example 11: You have a set of students enrolled in a mathematics course and another set of students enrolled in a physics course. Some students are enrolled in both courses. How can you represent this situation using sets and relations?

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Solution: Let's denote Set M as the set of students enrolled in the mathematics course, Set P as the set of students enrolled in the physics course, and Set U as the set of all students.

The relation R between M and P represents the students who are enrolled in both courses. So, $R \subseteq M \times P$.

Example 12: You are organising a sports event with various teams participating. Each team has a captain. How can you represent this using sets and relations?

Solution: Let's define Set T as the set of teams participating in the sports event and Set C as the set of team captains.

We can establish a relation R between T and C where each tuple (t,c) in R indicates that team t has captain c. So, $R \subseteq T \times C$. Sensitivity testing aids in optimal decision selection.

Example 13: In a social network, you want to manage friendships between users. How can you represent this using sets and relations?

Solution: Let's define a set U representing the users in the social network.

We can define a relation F between the set U and itself, where each tuple (u,v) in F indicates that user u is friends with user v. So, $F \subseteq U \times U$.

Conclusion

In conclusion, sets and relations are fundamental mathematical concepts with widespread applications in various fields, including computer science, engineering, economics, and more. Sets provide a way to organise and group

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elements based on common properties, while relations describe the connections or interactions between elements of different sets.

Summary

Key points about sets and relations include:

1. **Sets**: Sets are collections of distinct elements with no inherent order. They allow us to organise and manipulate data in a structured manner. Operations on sets, such as union, intersection, and complement, are essential tools for analysing and manipulating data.
2. **Relations:** Relations define connections or associations between elements of different sets. They can represent various types of relationships, such as friendships between individuals, assignments of tasks to employees, or dependencies between components in a system.
3. **Representation:** Sets and relations can be represented using mathematical notation, such as set builder notation or tuple notation. This allows for precise specification and manipulation of relationships in mathematical terms.
4. **Applications:** Sets and relations find applications in diverse fields, including database management, network analysis, graph theory, and formal languages. They provide powerful tools for modelling real-world scenarios, analysing complex systems, and solving practical problems.
5. **Properties:** Sets and relations have certain properties, such as reflexivity, symmetry, transitivity, and equivalence, which help in understanding and analysing their behaviour. These properties often have implications in

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various applications, such as in establishing algorithms or designing efficient data structures.

6. **Complexity:** While sets and relations offer elegant and versatile frameworks for modelling relationships, dealing with complex relationships and large datasets can pose challenges. Advanced techniques, such as relational algebra, graph theory algorithms and set theory principles, are often employed to address these challenges.

Overall, sets and relations provide a solid foundation for understanding and manipulating relationships within and between datasets, making them indispensable tools in both theoretical and practical aspects of mathematics and other disciplines.

Self-Assessment Questions and Answers (SAQs and SAAs)

SAQ 3.1: Explain the difference between a subset and a proper subset and give an example of each.

SAA 3.1: The difference between a subset and a proper subset lies in whether the subset contains all the elements of the original set.

A subset of a set B (denoted by $A \subseteq B$) simply means that every element in set A is also an element in set B.

A proper subset of a set B (denoted by $A \subset B$) is a more restrictive type of subset. There must be at least one element in B that is not present in A.

Example: Let Set $A = \{1, 2, 3\}$; Subset: $B = \{1, 2\}$—This is a subset because every element in B (1 and 2) is also present in A. Proper Subset: $C = \{2, 3\}$—This is a

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proper subset because all elements in C (2 and 3) are also in A, but C is not equal to A (it's missing the number 1).

SAQ 3.2: Given sets $A=\{1,2,3\}$ and $B=\{3,4,5\}$, find the union and intersection of these sets.

SAA 3.2: For sets A = {1, 2, 3} and B = {3, 4, 5}, Union (U): A U B = {1, 2, 3, 4, 5}; Intersection (∩): A ∩ B = {3}.

SAQ 3.3: Given the relation $R=\{(1,2),(2,3),(3,1)\}$, determine if R is reflexive, symmetric, and transitive.

SAA 3.3: A relation R is reflexive if for every element a in the set, (a, a) belongs to R. In this case, our set includes elements 1, 2, and 3.

- *(1, 1) ∉ R (1 is not related to itself in set R)*
- *(2, 2) ∉ R (2 is not related to itself in set R)*
- *(3, 3) ∉ R (3 is not related to itself in set R)*

Since none of the elements have a pair with themselves in the relation, R is not reflexive.

A relation R is symmetric if for every (a, b) in R, (b, a) is also in R. Let's check each element in R:

- *(1, 2) is in R, but (2, 1) is not in R.*
- *(2, 3) is in R, but (3, 2) is in R (already present).*
- *(3, 1) is in R, but (1, 3) is not in R.*

We see that (1, 2) doesn't have a corresponding (2, 1) in R, and (1, 3) is missing its counterpart. Therefore, R is not symmetric.

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A relation R is transitive if for every (a, b) and (b, c) in R, (a, c) is also in R. Let's examine the pairs:

(1, 2) is in R, and (2, 3) is in R. However, (1, 3) is not in R. Here, we see that even though there's a path from 1 to 3 (1 → 2 → 3), there's no direct relation between 1 and 3 in set R. Therefore, R is not transitive.

Multiple-Choice Questions and Answers (MCQs and MCAs)

1. **A relation that is reflexive, symmetric, and transitive is called:**

A. Partial order

B. Equivalence relation

C. Antisymmetric relation

D. Binary relation

Answer: B

Explanation: An equivalence relation satisfies all three: reflexive, symmetric, and transitive properties — forming equivalence classes.

2. **Which of these relations is antisymmetric?**

A. 'is equal to' on numbers

B. 'is older than' among people

C. 'is sibling of' among people

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D. 'has same height as' among students

Answer: B

Explanation: Antisymmetry means if a is related to b and b to a, then a = b. 'Older than' satisfies this since both cannot be true simultaneously for different people.

3. **A relation R on set A is symmetric if:**

A. Every element is related to itself

B. (a, b) ∈ R implies (b, a) ∈ R

C. (a, b) ∈ R implies (a, a) ∈ R

D. (a, b) ∈ R implies (b, b) ∈ R

Answer: B

Explanation: 'Symmetry' means 'mutual relation' — if a is related to b, then b must be related to a.

4. **Which of the following is not a property of set operations?**

A. Commutative

B. Associative

C. Reflexive

D. Distributive

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Answer: C

Explanation: Reflexivity is a property of relations, not set operations. Sets use commutative, associative, and distributive properties.

5. **If A = {1, 2} and B = {a, b, c}, what is A × B?**

A. {(1, a), (2, b)}

B. {(a, 1), (b, 2)}

C. {(1, a), (1, b), (1, c), (2, a), (2, b), (2, c)}

D. {(a, b), (b, c)}

Answer: C

Explanation: Cartesian product forms all ordered pairs (a, b) such that a ∈ A and b ∈ B. Since A has 2 elements and B has 3, total pairs = 2×3 = 6.

Tutor-Marked Assessments (TMAs)

1. Explain the difference, intersection, and complement of sets with relevant examples.
2. Let A = {2, 4, 6} and B = {4, 6, 8}. Find (i) A ∪ B (ii) A∩B (iii) A−B and explain what each represents.
3. Explain the properties of relations (reflexive, symmetric, transitive, antisymmetric) with real-world examples.
4. Discuss the applications of set operations and relations in real-world systems such as database management or network analysis

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5. Define a relation and explain with an example how it differs from a function.

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Study Session Four

Field and Ordered Field

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Introduction

Our goal with this session is to develop a comprehensive understanding of the fundamental concepts that characterise the set of real numbers. In mathematics, the real numbers form a crucial and ubiquitous field, known as an ordered field. As we delve into this study session, we will revisit and review the core elements of this field and ordered field, exploring their properties and significance in various mathematical contexts.

By the end of this session, you will have a solid grasp of the foundational principles that underlie the set of real numbers and how they form an ordered field. This knowledge will serve as a stepping stone for more advanced mathematical studies and equip you with valuable tools for practical applications in various disciplines.

Learning Outcomes for Study Session 4

When you have studied this session, you should be able to:

4.1 Define and identify an ordered field.

4.2 Analyse and apply properties of real numbers.

4.3 Compare and contrast real numbers with other number sets.

4.4 Construct and justify mathematical arguments using real numbers.

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4.1 Fields

In mathematics, a field is a fundamental algebraic structure that encompasses two operations, addition and multiplication, and satisfies specific axioms. The concept of a field lays the groundwork for understanding various mathematical structures, including the set of real numbers, rational numbers, and complex numbers.

4.1.1 Definition of a Field

A set F is considered a field if it satisfies the following axioms for addition and multiplication.

4.1.1.1 Addition Axioms

1. **Closure Under Addition:** For any two elements a and b in F, their sum (a + b) is also in F.
2. **Associativity of Addition:** For any elements a, b, and c in F, the equation (a + b) + c = a + (b + c) holds.
3. **Existence of an Additive Identity:** There exists an element 0 in F, such that for any element a in F, a + 0 = a.
4. **Existence of Additive Inverses:** For every element a in F, there exists an element -a in F, such that a + (-a) = 0.
5. **Commutativity of Addition**: For any elements a and b in F, the equation a + b = b + a holds.

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4.1.1.2 Multiplication Axioms

1. **Closure under Multiplication:** For any two elements **a** and **b** in **F**, their product (**a** * **b**) is also in **F**.
2. **Associativity of Multiplication:** For any elements **a**, **b**, and **c** in **F**, the equation (**a** * **b**) * **c** = **a** * (**b** * **c**) holds.
3. **Existence of a Multiplicative Identity:** There exists an element **1** in **F** such that for any element **a** in **F**, **a** * **1** = **a**.
4. **Existence of Multiplicative Inverses:** For every non-zero element **a** in **F**, there exists an element a^{-1} in **F** such that **a** * a^{-1} = **1**.
5. **Commutativity of Multiplication:** For any elements **a** and **b** in **F**, the equation **a** * **b** = **b** * **a holds.**

4.1.1.3 Distributive Axiom

For any elements **a**, **b**, and **c** in **F**, the distributive property

(**a** * (**b** + **c**) = **a** * **b** + **a** * **c**) holds.

Examples of Fields

1. The set of real numbers (denoted by $\mathbb{R}$) is a field. It includes all rational and irrational numbers and satisfies all the field axioms.
2. The set of rational numbers (denoted by $\mathbb{Q}$) is a field. It includes all numbers expressible as fractions and satisfies all the field axioms.
3. The set of complex numbers (denoted by $\mathbb{C}$) is a field. It includes all numbers of the form a + bi, where a and b are real numbers, and i is the imaginary unit ($i^2 = -1$). It satisfies all the field axioms.

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Non-Examples of Fields

1. The set of natural numbers (denoted by $\mathbb{N}$) is not a field because it does not have additive inverses (e.g., there is no additive inverse for the number 1).
2. The set of integers (denoted by $\mathbb{Z}$) is not a field because it does not have multiplicative inverses (e.g., there is no multiplicative inverse for the number 2).

4.2 Ordered Fields

An ordered field **F** consists of a set **F** on which are defined binary operations $+$ of addition and $\times$ of multiplication, together with an ordering relation $<$, where these binary operations and ordering relation satisfy the following axioms:

4.2.1 The Trichotomy Law

If u and v are elements of, F then one and only one of the three statements is true:

a. $u < v$,

b. $u = v$

c. $u > v$

4.2.2 The Transitivity of the Ordering

If u, v and w are elements of F and

if $u < v$ and $v < w$ then $u < w$.

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4.2.3 Preservation of Order Under Addition in Field F

If u, v and w are elements of F and

if $u < v$ then $u + w < v + w$.

In-Text Questions and Answers (ITQs and ITAs)

ITQ 4.1: Why is the set of integers ($\mathbb{Z}$) not a field?

Answer: This is because integers do not have multiplicative inverses within the set (for example, 2 has no multiplicative inverse in $\mathbb{Z}$).

ITQ 4.2: What two operations must be defined for a set to qualify as a field?

Answer: A set must have addition and multiplication operations that satisfy the field axioms — closure, associativity, commutativity, existence of identity and inverses, and distributivity.

4.2.4 Positivity Preservation in Field F under Multiplication

If u and v are elements of F which satisfy $0 < u$ and $0 < v$ then $0 < u \times v$.

The operations of subtraction and division are defined on an ordered field F in terms of the operations of addition and multiplication on that field in the obvious fashion:

$u - v = u + (-v)$

for all elements u and v of F, moreover,

$\frac{u}{v} = uv^{-1}$

provided that $v \neq 0$.

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Example 1

The rational numbers, with standard ordering and the standard operations of addition, subtraction, multiplication, and division, constitute an ordered field.

Example 2

Let $\mathbb{Q}(\sqrt{2})$ denote the set of all numbers that can be represented in the form $b + c\sqrt{2}$, where b and c are rational numbers. The sum and difference of any two numbers belonging to $\mathbb{Q}(\sqrt{2})$ themselves belong to $\mathbb{Q}(\sqrt{2})$. Also, the product of any two numbers $\mathbb{Q}(\sqrt{2})$ itself belongs to $\mathbb{Q}(\sqrt{2})$ because, for any rational numbers b, c, e, and f,

$(b + c\sqrt{2})(e + f\sqrt{2}) = (be + 2cf) + (bf + ce)\sqrt{2}$,

and both be + 2cf and bf + ce are rational numbers. The reciprocal of any non-zero element of $\mathbb{Q}(\sqrt{2})$ itself belongs to $\mathbb{Q}(\sqrt{2})$, because

$$\frac{1}{b + c\sqrt{2}} = \frac{b - c\sqrt{2}}{b^2 - 2c^2}$$

for all rational numbers b and c. It is then a straightforward exercise to verify that $\mathbb{Q}(\sqrt{2})$ is an ordered field.

The absolute value |x| of an element number x of an ordered field F is defined by

$$|x| = \begin{cases} x & \text{if } x \geq 0; \\ -x & \text{if } x < 0. \end{cases}$$

Note that |x| ≥ 0 for all x and that |x| = 0 if and only if x = 0. Also

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$|x + y| \leq |x| + |y|$ and $|xy| = |x||y|$

for all elements x and y of the ordered field F.

Conclusion

In conclusion, this study session on fields and ordered fields has equipped you with a basic understanding of the fundamental axioms that govern their operations. Additionally, the addition and multiplication axioms were explained in detail alongside the distributive property. Finally, the ordering principles within an ordered field were defined. Remember, the trichotomy law guarantees a clear distinction between elements, while transitivity allows us to chain inequalities for efficient comparison.

Summary

We covered the essential axioms for addition and multiplication, along with the powerful distributive property that ties them together.

1. **Addition and Multiplication Axioms:** These axioms establish the fundamental rules for performing arithmetic operations within a field. They ensure closure, commutativity, associativity, the existence of inverses (except for additive/multiplicative identity), and the ability to distribute multiplication over addition.
2. **The Trichotomy Law:** This elegant principle states that for any two elements a and b in an ordered field, exactly one of the following relationships must hold true: $a < b$, $a > b$, or $a = b$. This property ensures a clear ordering and comparison between elements.

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3. **Transitivity of Ordering:** Building upon the trichotomy law, transitivity guarantees that if $a < b$ and $b < c$, then $a < c$. This allows us to "chain" inequalities, establishing a clear order among elements.
4. **Preservation of Order under Addition:** This crucial property ensures that adding a fixed element to both sides of an inequality preserves the order. In simpler terms, if $a < b$, then adding the same element c to both sides maintains the inequality: $(a + c) < (b + c)$.
5. **Positivity Preservation under Multiplication:** This property guarantees that multiplying a positive element by another element in the field preserves the sign of the result. In simpler terms, if $a > 0$ and b is any element, then the product $(a)(b)$ will also be positive (a positive times anything is positive).

Self-Assessment Questions and Answers (SAQs and SAAs)

SAQ 4.1. Prove that if $a < b$, then $a+c < b+c$ in an ordered field.

Answer: By the Preservation of Order under Addition axiom, adding the same element c

c to both sides of an inequality maintains the order. So if $a < b$, then $a+c < b+c$. This is crucial for consistency in comparisons within ordered fields.

SAQ 4.2. Explain the difference between a field and an ordered field.

Answer: A field focuses only on algebraic operations (addition and multiplication), while an ordered field adds an ordering relation ($<$) that satisfies

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trichotomy, transitivity, and order preservation. The ordering allows comparison between elements, as in real numbers.

SAQ 4.3. What does the distributive property link together in a field?

Answer: The distributive property connects addition and multiplication, stating $a(b+c)=ab + ac$. This ensures multiplication distributes over addition, maintaining consistency between the two operations.

Multiple-Choice Questions and Answers (MCQs and MCAs)

1. **Which of the following sets forms a field?**

A. $\mathbb{Z}$ (Integers)

B. $\mathbb{N}$ (Natural numbers)

C. $\mathbb{R}$ (Real numbers)

D. Even numbers

Answer: C

Explanation: $\mathbb{R}$ satisfies all field axioms, including the existence of additive and multiplicative inverses (except for 0). $\mathbb{Z}$ and $\mathbb{N}$ fail because they lack inverses.

2. **Which of the following is not required for a set to be a field?**

A. Associativity

B. Commutativity

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C. Closure

D. Ordering relation

Answer: D

Explanation: An ordering relation ($<$) is specific to ordered fields, not all fields. A standard field only needs the algebraic axioms.

3. **Why is $\mathbb{Z}$ not a field?**

A. It is not closed under addition

B. It lacks additive identity

C. It lacks multiplicative inverses

D. It fails the distributive law

Answer: C

Explanation: Integers have no multiplicative inverses (e.g., $\frac{1}{2} \notin \mathbb{Z}$). This violates the field axiom.

4. **If $a > 0$ and $b > 0$, what must be true in an ordered field?**

A. $a+b < 0$

B. $ab > 0$

C. $ab < 0$

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D. $a - b > 0$

Answer: B

Explanation: Positivity preservation states that multiplying two positive elements yields a positive product.

5. **In an ordered field, if $a < b$ and $b < c$, then what follows?**

A. $a = c$

B. $a < c$

C. $b < a$

D. $c < a$

Answer: B

Explanation: The Transitivity of Ordering axiom guarantees that inequalities can be chained logically.

Tutor-Marked Assessments (TMAs)

1. Discuss the difference between algebraic properties (field axioms) and ordering properties (ordered field axioms), using examples.
2. Prove that the set of rational numbers $\mathbb{Q}$ is a field under addition and multiplication. Clearly identify each axiom that holds.
3. Explain the importance of the distributive property in connecting addition and multiplication in fields. Give a practical numerical example.

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4. Prove that in an ordered field, if $a < b$ and $0 < c$, then $ac < bc$.

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Study Session Five

Proof Techniques & Mathematical Induction

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Introduction

In the realm of mathematics, proving the truth of statements is fundamental. A proof is a structured argument that establishes the validity of a given statement. Throughout mathematical discourse, various techniques are employed to rigorously demonstrate the veracity of propositions. In this lesson, we will delve into these proof techniques, understanding their nuances and applications.

We begin by elucidating the concept of proof and then explore different proof techniques, including direct proof, proof by contraposition, proof by contradiction, and proof by cases. Each technique offers a distinct approach to proving mathematical statements, catering to different scenarios and complexities. Through examples and explanations, we will dissect these techniques, providing insight into their mechanics and effectiveness.

Furthermore, we will explore the notion of existence and uniqueness proofs, which play a pivotal role in mathematical reasoning. Existence proofs establish the presence of solutions or instances that satisfy given conditions, while uniqueness proofs demonstrate that such solutions are singular. These techniques broaden our understanding of mathematical truths and empower us to tackle a diverse array of problems with confidence.

By the end of this session, you will possess a comprehensive understanding of various proof techniques and their applications. Armed with this knowledge, you will be well-equipped to navigate the intricacies of mathematical reasoning, confidently justifying the truth of theorems and propositions.

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Learning Outcomes for Study Session 5

When you have studied this session, you should be able to:

5.1 Itemise proof techniques

5.2 Prove by direct or contraposition

5.3 Prove by contradiction or by cases

5.4 State the existence or uniqueness of a proof

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5.1 Basic Structure of Proof Techniques

A proof is a valid argument that establishes the truth of a mathematical statement. A proof can use the hypotheses of the theorem, if any, axioms assumed to be true, and previously proven theorems. Using these ingredients and rules of inference, the final step of the proof establishes the truth of the statement being proved.

Proof techniques are methods for constructing proofs. They include:

- Direct proof
- Proof by contraposition
- Proof by contradiction
- Proof by cases
- Proof by induction

5.2 Direct Proof Technique

A direct proof shows that a conditional statement p → q is true by showing that if p is true, then q must also be true. We assume that p is true and use axioms, definitions, and previously proven theorems together with rules of inference to show that q must also be true.

5.2.1 Example on Direct Proof Technique

Example 1: Give a direct proof of the theorem, "If n is an odd integer, then n^2 is odd."

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Solution: We can write the theorem "If n is an odd integer, then n^2 is odd" in logical notation as $p \rightarrow q$. Thus, the hypothesis p: n is an odd integer, Our goal or target is- q: n^2 is odd n odd implies n = 2k + 1, for some integer k ($k \in \mathbb{Z}$) square both sides: $n = 2k + 1$ becomes $n^2 = (2k + 1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2) + 1$

Therefore, $n^2 = 2q + 1$ where $q = (2k^2 + 2) \in \mathbb{Z}$. Hence, n^2 is odd.

Example 2: Show that the square of an even number is an even number using a direct proof

Solution: We can write the problem as "If n is an even integer, then n^2 is even" in logical notation as $p \rightarrow q$. Thus, the hypothesis p: n is an even integer. Our goal or target is - **q:** n^2 is also even.

n even implies n = 2k for some integer k ($k \in \mathbb{Z}$), square both sides

$n = 2k$ becomes $n^2 = (2k)^2 = 4k^2 = 2(2k^2)$

Therefore, $n^2 = 2q$ where $q = 2k^2 \in \mathbb{Z}$. Hence n^2 is even

5.3 Proof by Contraposition

Proofs by contraposition make use of the fact that the conditional statement p → q is equivalent to its contrapositive, ¬q → ¬p. This means that the conditional statement p → q can be proved by showing that its contrapositive, ¬q → ¬p, is true.

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In a proof by contraposition of p → q, we take ¬q as a premise, and using axioms, definitions, and previously proven theorems, together with rules of inference, we show that ¬p must follow.

Example 3: Prove that if n is an integer and 3n + 2 is odd, then n is odd.

Proof: The first step is to assume that the conclusion of the conditional statement "If 3n + 2 is odd, then n is odd" is false; namely, assume that n is even. Then, by the definition of an even integer, n = 2k for some integer k. Substituting 2k for n, we obtain $3n + 2 = 3(2k) + 2 = 6k + 2 = 2(3k + 1)$. This tells us that 3n + 2 is even. This is the negation of the premise of the theorem

Example 4: Prove that if n is an integer and n is a perfect square, then n+2 is not.

Proof: The first step is to assume that the conclusion of the conditional statement "If n is a perfect square, then n+2 is not" is false; namely, assume that n+2 is a perfect square. Then, by the definition of a perfect square integer, n + 2 = k^2 for some integer k. This implies that n = k^2−2

This tells us that n is not a perfect square. This is the negation of the premise of the conditional statement.

The proof is complete

5.4 Proof by Contradiction

We have another technique of indirect proof known as proof by contradiction.

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If r is a statement, then r ∧ ¬r is a contradiction. This technique of proof capitalises on the fact that a statement and its negation are contradictions. So we find a contradiction that helps us prove the given statement.

Proof by contradiction can be used to prove conditional statements. In such proofs, we first assume the negation of the conclusion. We then use the premises of the theorem and the negation of the conclusion to arrive at a contradiction.

The reason that such proofs are valid rests on the logical equivalence of p → q and (p ∧ ¬q) → F. To see that these statements are equivalent, simply note that each is false in exactly one case, namely when p is true and q is false.

Note that we can rewrite a proof by contraposition of a conditional statement as a proof by contradiction. In a proof of p → q by contraposition, we assume that ¬q is true. We then show that ¬p must also be true. To rewrite a proof by contraposition of p → q as a proof by contradiction, we suppose that both p and ¬q are true. Then, we use the steps from the proof of ¬q → ¬p to show that ¬p is true. This leads to the contradiction p ∧ ¬p, completing the proof.

Example 5: Prove that √2 is irrational by giving a proof by contradiction.

Solution: Let p be the proposition "√2 is irrational." To start a proof by contradiction, we suppose that ¬p is true. Note that ¬p is the statement "It is not the case that √2 is irrational," which says that √2 is rational. We will show that assuming that ¬p is true leads to a contradiction.

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If √2 is rational, there exist integers a and b with √2 = a/b, where b is not equal to 0 and a and b have no common factors (so that the fraction a/b is in lowest terms.) (Here, we are using the fact that every rational number can be written in lowest terms.) Because √2 = a/b, when both sides of this equation are squared, it follows that 2 = $\frac{a^2}{b^2}$

Hence, $2b^2 = a^2$.

By the definition of an even integer, it follows that a^2 is even. We next use the fact that if a^2 is even, a must also be even. Furthermore, because a is even, by the definition of an even integer, a = 2c for some integer c. Thus,

$2b^2 = 4c^2$. Dividing both sides of this equation by 2 gives $b^2 = 2c^2$.

By the definition of even, this means that b^2 is even. Again using the fact that if the square of an integer is even, then the integer itself must be even, we conclude that b must be even as well.

We have now shown that the assumption of ¬p leads to the equation √2 = a/b, where a and b have no common factors but both a and b are even; that is, 2 divides both a and b. Note that the statement that √2 = a/b, where a and b have no common factors, means, in particular, that 2 does not divide both a and b. Because our assumption of ¬p leads to the contradiction that 2 divides both a and b and 2 does not divide both a and b, ¬p must be false. That is, the statement p, "√2 is irrational," is true. We have proved that √2 is irrational.

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Example 6: Show that at least four of any 22 days must fall on the same day of the week.

Solution: Let p be the proposition "At least four of 22 chosen days fall on the same day of the week." Suppose that ¬p is true. This means that at most three of the 22 days fall on the same day of the week. Because there are seven days of the week, this implies that at most 21 days could have been chosen, as for each of the days of the week, at most three of the chosen days could fall on that day. This contradicts the premise that we have 22 days under consideration.

5.5 Proof By cases

Apart from these techniques of proof we discussed, there is yet another approach known as proof by cases

Sometimes we cannot prove a theorem using a single argument that holds for all possible cases. We can prove the theorem by considering different cases separately.

Example 7: Use a proof by cases to show that $|xy|=|x||y|$, where x and y are real numbers. (Recall that $|a|$, the absolute value of a, equals a when $a \geq 0$ and equals $-a$ when $a \leq 0$.)

Solution: In our proof of this theorem, we remove absolute values using the fact that $|a| = a$ when $a \geq 0$ and $|a| = -a$ when $a < 0$. Because both $|x|$ and $|y|$ occur in our formula, we will need four cases:

- X and Y are both nonnegative.
- X is nonnegative and Y is negative.

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- X is negative and y is nonnegative, and
- X is negative and Y is negative. We denote by p1, p2, p3, and p4 the proposition stating the assumption for each of these four cases, respectively.

(Note that we can remove the absolute value signs by making the appropriate choice of signs within each case.)

Case (i): We see that p1 → q because xy ≥ 0 when x ≥ 0 and y ≥ 0, so that |xy| = xy = |x||y|.

Case (ii): To see that p2 → q, note that if x ≥ 0 and y < 0, then xy ≤ 0, so that |xy| = −xy = x(−y) = |x||y|. (Here, because y < 0, we have |y| = −y.)

Case (iii): To see that p3 → q, we follow the same reasoning as the previous case with the roles of x and y reversed.

Case (iv): To see that p4 → q, note that when x < 0 and y < 0, it follows that xy > 0.

Hence, |xy| = xy = (−x)(−y) = |x||y|.

Because |xy|=|x||y| holds in each of the four cases and these cases exhaust all possibilities, we can conclude that |xy|=|x||y| whenever x and y are real numbers

Example 8: Prove that if n is an integer, then n^2 ≥ n.

Solution: We can prove that n^2 ≥ n for every integer by considering three cases: when n = 0, when n ≥ 1, and when n ≤ −1. We split the proof into three cases

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because it is straightforward to prove the result by considering zero, positive integers, and negative integers separately.

Case (i): When n = 0, because $0^2 = 0$, we see that $0^2 \geq 0$. It follows that $n^2 \geq n$ is true in this case.

Case (ii): When n ≥ 1, when we multiply both sides of the inequality n ≥ 1 by the positive integer n, we obtain n · n ≥ n · 1. This implies that $n^2 \geq n$ for n ≥ 1.

Case (iii): In this case, n ≤ −1. However, $n^2 \geq 0$. It follows that $n^2 \geq n$. Because the inequality $n^2 \geq n$ holds in all three cases, we can conclude that if n is an integer, then $n^2 \geq n$.

5.6 Proof by Induction

Let $P(n)$ be a given statement involving the natural number n such that

- The statement is true for $n = 1$, i.e., $P(1)$ is true (or true for any fixed natural number) and
- If the statement is true for $n = k$ (where k is a particular but arbitrary natural number), then the statement is also true for $n_ = k + 1$, i.e., the truth of $P(k)$ implies the truth of $P(k + 1)$. Then $P(n)$ is true for all natural numbers n

Example 9: $1 + 3 + 5 + \ldots + (2n - 1) = n^2$

Solution: Let the given statement $P(n)$ be defined as $P(n)$: $1 + 3 + 5 + \ldots + (2n - 1) = n^2$, for $n \in N$. Note that this $P(1)$ is true since:

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$P(1): 1 = 1^2$

Assume that $P(k)$ is true for some $k \in N$, i.e.,

$P(k): 1 + 3 + 5 + \ldots + (2k - 1) = k^2$

Now, to prove that $P(k + 1)$ is true, we have:

$1 + 3 + 5 + \ldots + (2k - 1) + (2k + 1) = k^2 + (2k + 1) = k^2 + 2k + 1 = (k + 1)^2$

$= k^2 + (2k + 1)$

Thus, $P(k + 1)$ is true. whenever $P(k)$ is true.

Hence, the principle of mathematical induction $P(n)$ is true for all $n \in N$.

Example 10:

$\sum_{n=1}^{n-1} t(t + 1) = \frac{n(n-1)(n+1)}{3}$, for all natural numbers $n \geq 2$.

Solution: Let the given statement $P(n)$, be given as:

$P(n): \sum_{t=1}^{n-1} r(t + 1) = \frac{n(n-1)(n+1)}{3}$, $for\ all\ natural\ numbers\ n \geq 2$

We observe that:

$P(2): \sum_{t=1}^{2-1} t(t + 1) = \sum_{t=1}^{1} t(t + 1) = 1.2 = \frac{1.2.3}{3} = \frac{2.(2-1)(2+1)}{3}$

Thus, $P(n)$ in true for $n = 2$

Assume that $P(n)$ is true for $n = k \in N$

i.e.,

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$$P(k): \sum_{t=1}^{k-1} t(t+1) = \frac{k(k-1)(k+1)}{3}$$

To prove that $P(k+1)$ is true, we have

$$\sum_{t=1}^{(k+1-1)} t(t+1) = \sum_{t=1}^{k} t(t+1)$$

$$= \sum_{t=1}^{k-1} t(t+1) + k(k+1)$$

$$= \frac{k(k-1)(k+1)}{3} + k(k+1)$$

$$= k(k+1)\left[\frac{k-1+3}{3}\right]$$

$$= \frac{k(k+1)(k+2)}{3} = \frac{(k+1)((k+1)-1))((k+1)+1)}{3}$$

Thus, $P(k+1)$ is true, whenever $P(k)$ is true.

Hence, by the Principle of Mathematical induction $P(n)$ is true for all natural numbers $n \geq 2$

Example 11:

$\left(1-\frac{1}{2^2}\right)\cdot\left(1-\frac{1}{3^2}\right)\ldots\left(1-\frac{1}{n^2}\right) = \frac{n+1}{2n}$, for all natural numbers $n \geq 2$.

Solution: Let the given statement be $P(n)$, i.e.,

$$P(n): \left(1-\frac{1}{2^2}\right)\cdot\left(1-\frac{1}{3^2}\right)\ldots\left(1-\frac{1}{n^2}\right) = \frac{n+1}{2n},\ for\ all\ natural\ numbers,\ n \geq 2$$

We observe that $P(2)$ is true, since:

$$\left(1-\frac{1}{2^2}\right) = 1-\frac{1}{4} = \frac{4-1}{4} = \frac{3}{4} = \frac{2+1}{2\times 2}$$

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assume that $P(n)$ is true for some $k \in N$, i.e.,

$$P(k):\ 1 - \frac{1}{2^2}\cdot 1 - \frac{1}{3^2}\ldots 1 - \frac{1}{k^2} = \frac{k+1}{2k}$$

now, to prove that $P(k+1)$ is true, we have

$$1 - \frac{1}{2^2}\cdot 1 - \frac{1}{3^2}\ \ldots 1 - \frac{1}{k^2}\cdot 1 - \frac{1}{(k+1)^2} = \frac{k+1}{2k}\left(1 - \frac{1}{(k+1)^2}\right) = \frac{k^2+2k}{2k(k+1)} = \frac{(k+1)+1}{2(k+1)}$$

Thus, $P(k+1)$ is true whenever $P(k)$ is true.

Hence, the principle of mathematical induction $P(n)$ is true for all natural numbers, $n \geq 2$.

Example 12: $2^{2n} - 1$ is divisible by 3 .

Solution: Let the statement $P(n)$ given as:

$P(n)$: $2^{2n} - 1$ is divisible by 3 for every natural number n.

We observe that $P(1)$ is true, since:

$2^2 - 1 = 4 - 1 = 3.1\ is\ divisible\ by\ 3.$

Assume that $P(n)$ is true for some natural number k, i.e., $P(k)$: $2^{2k} - 1$ is divisible by 3, i.e., $2^{2k} - 1 = 3q$, where $q \in N$

Now, to prove that $P(k+1)$ is true, we have:

$$P(k+1): 2^{2(k+1)} - 1 = 2^{2k+2} - 1 = 2^{2k}\cdot 2^2 - 1 = 2^{2k}\cdot 4 - 1 = 3\cdot 2^{2k} + \left(2^{2k} - 1\right)$$

$$= 3.2^{2k} + 3q = 3\left(2^{2k} + q\right) = 3m,\ where\ m \in N$$

Thus $P(k+1)$ is true, whenever $P(k)$ is true.

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Hence, by the Principle of Mathematical Induction $P(n)$ is true for all natural numbers n.

Example 13: $2n + 1 < 2^n$, for all natural numbers $n \geq 3$.

Solution: Let $P(n)$ be the given statement, i.e., $P(n)$: $(2n + 1) < 2^n$ for all natural numbers, $n \geq 3$. We observe that $P(3)$ is true, since:

$2.3 + 1 = 7 < 8 = 2^3$

Assume that $P(n)$ is true for some natural number k, i.e., $2k + 1 < 2^k$

To prove $P(k + 1)$ is true, we have to show that $2(k + 1) + 1 < 2^{k+1}$. Now, we have:

$2(k + 1) + 1 = 2k + 3 = 2k + 1 + 2 < 2^k + 2 < 2^k . 2 = 2^{k+1}$.

Thus $P(k + 1)$ is true, whenever $P(k)$ is true.

Hence, the principle of mathematical induction $P(n)$ is true for all natural numbers, $n \geq 3$.

Example 14: The distributive law from algebra says that for all real numbers c, a_1 and a_2, we have $c(a_1 + a_2) = ca_1 + ca_2$.

Use this law and mathematical induction to prove that, for all natural numbers, if $c, a_1, a_2, \ldots, a_n$ are any real numbers, then:

$c(a_1 + a_2 + \ldots + a_n) = ca_1 + ca_2 + \ldots + ca_n$

Solution: Let $P(n)$ be the given statement, i.e.,

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$P(n)$: $c(a_1 + a_2 + \ldots + a_n) = ca_1 + ca_2 + \ldots ca_n$ for all natural numbers $n \geq 2$, for $c, a_1, a_2, \ldots a_n \in R$.

We observe that that $P(2)$ is true since:

$c(a_1 + a_2) = ca_1 + ca_2$

(by distributive law)

Assume that $P(n)$ is true for some natural number k, where $k > 2$, i.e.:

$P(k)$: $c(a_1 + a_2 + \ldots + a_k) = ca_1 + ca_2 + \ldots + ca_k$

Now to prove $P(k + 1)$ is true, we have:

$$P(k+1) : c(a_1 + a_2 + \ldots + a_k + a_{k+1})$$
$$= c((a_1 + a_2 + \ldots + a_k) + a_{k+1}) \quad = c(a_1 + a_2 + \ldots + a_k) + ca_{k+1}$$
$$= ca_1 + ca_2 + \ldots + ca_k + ca_{k+1}$$

(by distributive law)

Thus $P(k + 1)$ is true, whenever $P(k)$ is true.

Hence, by the principle of mathematical induction, it $P(n)$ is true for all natural numbers $n \geq 2$.

Example 15:

Prove by induction that for all natural numbers n:

$sin\alpha + sin(\alpha + \beta) + sin(\alpha + 2\beta) + \ldots + sin(\alpha + (n-1)\beta)$

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$$= \frac{sin\left(\alpha+\frac{n-1}{2}\beta\right)sin\left(\frac{n\beta}{2}\right)}{sin\left(\frac{\beta}{2}\right)}$$

Solution: Consider

$$:P(n): sin\alpha + sin(\alpha + \beta) + sin(\alpha + 2\beta) + \ldots + sin(\alpha + (n-1)\beta)$$

$$= \frac{sin\left(\alpha+\frac{n-1}{2}\beta\right)sin\left(\frac{n\beta}{2}\right)}{sin\left(\frac{\beta}{2}\right)},\ for\ all\ natural\ number\ n.$$

We observe that $P(1)$ is true, since:

$$P(1): sin\alpha = \frac{sin(\alpha+0)sin\frac{\beta}{2}}{sin\frac{\beta}{2}}$$

assume that $P(n)$ is true for some natural numbers k, i.e.,.

$$P(k): sin\alpha + sin(\alpha + \beta) + sin(\alpha + 2\beta) + \ldots + sin(\alpha + (k-1)\beta)$$

$$= \frac{sin\left(\alpha+\frac{k-1}{2}\beta\right)sin\left(\frac{k\beta}{2}\right)}{sin\left(\frac{\beta}{2}\right)}$$

Now, to prove that $P(k+1)$ is true, we have:

$$P(k+1): sin\alpha + sin(\alpha + \beta) + sin(\alpha + 2\beta) + \ldots$$

$$+ sin(\alpha + (k-1)\beta) + sin(\alpha + k\beta) = \frac{sin\left(\alpha+\frac{k-1}{2}\beta\right)sin\left(\frac{k\beta}{2}\right)}{sin\left(\frac{\beta}{2}\right)} + sin(\alpha + k\beta)$$

$$= \frac{sin\left(\alpha+\frac{k-1}{2}\beta\right)sin\frac{k\beta}{2}+sin(\alpha+k\beta)sin\frac{\beta}{2}}{sin\frac{\beta}{2}} = \frac{cos\left(\alpha-\frac{\beta}{2}\right)-cos\left(\alpha+k\beta-\frac{\beta}{2}\right)+cos\left(\alpha+k\beta-\frac{\beta}{2}\right)-cos\left(\alpha+k\beta+\frac{\beta}{2}\right)}{2sin\frac{\beta}{2}}$$

$$= \frac{cos\left(\alpha-\frac{\beta}{2}\right)-cos\left(\alpha+k\beta+\frac{\beta}{2}\right)}{2sin\frac{\beta}{2}} = \frac{sin\left(\alpha+\frac{k\beta}{2}\right)sin\left(\frac{k\beta+\beta}{2}\right)}{sin\frac{\beta}{2}} = \frac{sin\left(\alpha+\frac{k\beta}{2}\right)sin(k+1)\left(\frac{\beta}{2}\right)}{sin\frac{\beta}{2}}$$

Thus $P(k+1)$ is true whenever $P(k)$ is true.

Hence, by the Principle of Mathematical Induction $P(n)$ is true for all natural numbers n.

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Example 16: Prove by the Principle of Mathematical Induction that $1\times1! + 2\times2! + 3\times3! + \ldots + n \times n! = (n + 1)! - 1$ for all natural numbers n.

Solution: Let $P(n)$ be the given statement, that is, $P(n)$: $1\times1! + 2\times2! + 3\times3! + \ldots + n \times n! = (n + 1)! - 1$ for all natural numbers n

Note that $P(1)$ is true, since:

$P(1)$: $1\times1! = 1 = 2 - 1 = 2! - 1$.

assume that $P(n)$ is true for some natural number k, i.e., $P(k)$: $1\times1! + 2\times2! + 3\times3! + \ldots + k \times k! = (k + 1)! - 1$

To prove $P(k + 1)$ is true, we have:

$P(k + 1)$: $1\times1! + 2\times2! + 3\times3! + \ldots + k \times k! + (k + 1)\times(k + 1)!$

$= (k + 1)! - 1 + (k + 1)! \times(k + 1)$

$= (k + 1 + 1)(k + 1)! - 1 \quad = (k + 2)(k + 1)! - 1$

$= ((k + 2)! - 1$

Thus $P(k + 1)$ is true, whenever $P(k)$ is true. Therefore, by the Principle of Mathematical Induction $P(n)$ is true for all natural numbers n.

Example 17: Show by the Principle of Mathematical Induction that the sum S_n of the n term of the series $1^2 + 2\times2^2 + 3^2 + 2\times4^2 + 5^2 + 2\times6^2$ ... is given by:

$$S_n = \{\frac{n(n+1)^2}{2},\ if\ n\ is\ even\ \ \frac{n^2(n+1)}{2},\ if\ n\ is\ odd$$

Solution:

Here,

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$$P(n): S_n = \{\frac{n(n+1)^2}{2},\ when\ n\ is\ even\ \ \frac{n^2(n+1)}{2},\ when\ n\ is\ odd$$

Also, note that any term T_n of the series is given by:

$$T_n = \{n^2\ if\ n\ is\ odd\ \ 2n^2\ if\ n\ is\ even$$

We observe that $P(1)$ is true since:

$$P(1): S_1 = 1^2 = 1 = \frac{1\cdot 2}{2} = \frac{1^2\cdot(1+1)}{2}$$

Assume that $P(k)$ is true for some natural number k, i.e.,.

Case 1: When it k is odd, then $k + 1$ is even. We have:

$$P(k+1): S_{k+1} = 1^2 + 2\times 2^2 + \ldots + k^2 + 2\times(k+1)^2 = \frac{k^2(k+1)}{2} + 2\times(k+1)^2$$

$$= \frac{(k+1)}{2}\left[k^2 + 4(k+1)\right]\left(as\ k\ is\ odd\ , 1^2 + 2\times 2^2 + \ldots + k^2 = k^2\frac{(k+1)}{2}\right)$$

$$= \frac{k+1}{2}\left[k^2 + 4k + 4\right] = \frac{k+1}{2}(k+2)^2 = (k+1)\frac{[(k+1)+1]^2}{2}$$

So $P(k+1)$ is true, whenever $P(k)$ is true in the case when k is odd.

Case 2: When k is even, then $k + 1$ is odd. Now,

$$P(k+1): 1^2 + 2\times 2^2 + \ldots + 2.k^2 + (k+1)^2$$

$$= \frac{k(k+1)^2}{2} + (k+1)^2\ (as\ k\ is\ even,\ 1^2 + 2\times 2^2 + \ldots + 2k^2 = k\frac{(k+1)^2}{2})$$

$$= \frac{(k+1)^2(k+2)}{2} = \frac{(k+1)^2((k+1)+1)}{2}$$

Therefore, $P(k+1)$ is true whenever $P(k)$ is true for the case when k is even. Thus $P(k+1)$ is true whenever $P(k)$ is true for any natural numbers k. Hence, $P(n)$ true for all natural numbers.

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In-Text Questions and Answers (ITQs and ITAs)

ITQ 5.1: What is the primary goal of a mathematical proof?

Answer: The goal is to establish the truth of a statement through logical reasoning using axioms, definitions, and previously proven theorems. A proof transforms assumptions (premises) into verified conclusions through valid inference.

ITQ 5.2: How does a proof by contraposition differ from a direct proof?

Answer: A direct proof assumes the hypothesis (p) is true and shows that the conclusion (q) follows, while a proof by contraposition proves the contrapositive (¬q → ¬p) instead of p → q, as they are logically equivalent.

Conclusion

In this study session on proofs, you have learnt various methods of crafting rigorous mathematical arguments. We learnt the fundamental structure of a pr an example of a statement $P(n)$ that is true for all n. Justify your answer.

Summary

1. Proof techniques are essential skills for mathematicians to construct valid arguments.
2. Basic proof techniques include direct proof, contraposition, contradiction, and cases.
3. Direct proof involves showing that if a certain condition is true, then a consequent result must also be true.

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4. Contraposition allows proving a conditional statement by showing the truth of its contrapositive, which is logically equivalent.
5. Proof by contradiction involves assuming the negation of the statement to be proven and deriving a contradiction.
6. Proof by cases is employed when a single argument doesn't suffice, requiring separate consideration of different scenarios.
7. Induction is a powerful method where the truth of a statement for one value implies its truth for the next, starting from a base case.
8. Each technique has its strengths and is chosen based on the nature of the theorem or problem being addressed.

Self-Assessment Answers and Answers (SAQs and SAAs)

SAQ 5.1. P(n): $n^2 - n > 10$

Answer:

- P(1) is not true: 1^2 - 1 = 0 (not greater than 10)
- P(2) is not true: 2^2 - 2 = 2 (not greater than 10)
- P(3) is not true: 3^2 - 3 = 6 (not greater than 10)

However, for n ≥ 4:

- P(4) is true: 4^2 - 4 = 12 (greater than 10)
- P(5) is true: 5^2 - 5 = 20 (greater than 10)

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P(n) is true for any n ≥ 4: As n increases, n^2 grows much faster than n. Therefore, n^2 - n will always be greater than 10 for n ≥ 4.

SAQ 5.2. P(n): n ≤ n + 1

Answer: This statement is true regardless of the value of n because, by definition, any positive integer is less than or equal to itself plus one.

Here's a breakdown for different values of n:

n = 1: 1 ≤ 1 + 1 (True)

n = 2: 2 ≤ 2 + 1 (True)

n = 3: 3 ≤ 3 + 1 (True)

This pattern continues for all positive integers n. No matter how large n becomes, adding 1 will always result in a number greater than n. Therefore, P(n) holds true for all positive integers n.

SAQ 5.3. What are the two main types of existence proofs? Briefly describe each.

Answer:

- **Constructive Existence Proof:** Explicitly provides an example of an object that satisfies the required properties.
- **Non-Constructive Existence Proof:** Demonstrates that an object must exist without providing a specific example, often using logical reasoning or the pigeonhole principle.

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Multiple Choice Questions and Answers (MCQs and MCAs)

1. Which of the following best describes a direct proof?

A. It assumes the conclusion is true and deduces the hypothesis

B. It assumes the hypothesis is true and deduces the conclusion

C. It assumes both the hypothesis and conclusion are false

D. It divides a proof into multiple cases

Answer: B

Explanation: A direct proof begins by assuming the hypothesis (p) is true, then uses logical reasoning to show that the conclusion (q) must also be true.

2. The contrapositive of "If p then q" is:

A. If q then p

B. If ¬p then ¬q

C. If ¬q then ¬p

D. If q then ¬p

Answer: C

Explanation: The contrapositive of p → q is ¬q → ¬p. Both statements are logically equivalent and thus interchangeable in proofs.

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3. What logical step is essential in a proof by contradiction?

A. Assuming the hypothesis is true

B. Assuming the conclusion is true

C. Assuming the negation of the statement and finding a contradiction

D. Dividing the proof into several cases

Answer: C

Explanation: Proof by contradiction assumes the negation of the statement to be proved and derives a logical inconsistency, proving the original statement true.

4. Which of these is NOT a valid proof technique?

A. Proof by induction

B. Proof by cases

C. Proof by negation

D. Direct proof

Answer: C

Explanation: "Proof by negation" is not a standard proof method; contradiction, contraposition, and induction are recognized techniques.

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5. What does proof by cases ensure in mathematical reasoning?

A. That all possible scenarios are covered

B. That only one case is valid

C. That contradictions are eliminated

D. That assumption is false

Answer: A

Explanation: Proof by cases ensures that every possible condition is considered, and the statement holds true in each case — thus, it is universally true.oof, focusing on techniques like direct proof and proof by contradiction, each offering a solid approach to conquering different problems. We have also seen how proof by induction tackles statements about natural numbers, while proof by cases dissects problems with various conditions.

Tutor-Marked Assessments (TMAs)

Use the principle of mathematical induction in the following exercises.

1. A sequence a_1, a_2, a_3 … is defined by letting $a_1 = 3$ and $a_k = 7a_{k-1}$ for all natural numbers $k \geq 2$. Show that $a_n = 3.7^{n-1}$ for all natural numbers.
2. A sequence b_0, b_1, b_2 … is defined by letting $b_0 = 5$ and $b_k = 4 + b_{k-1}$ for all natural numbers k. Show that $b_n = 5 + 4n$ for all natural numbers n using mathematical induction.

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3. A sequence d_1, d_2, d_3 … is defined by letting $d_1 = 2$ and $d_k = \frac{d_{k-1}}{k}$ for all natural numbers, $k \geq 2$. Show that $d_n = \frac{2}{n!}$ for all $n \in N$.

4. Use a proof by contraposition to prove: "If 3n + 2 is odd, then n is odd."

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Study Session Six

Inclusion-Exclusion Principle

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Introduction

Counting problems across various domains often finds resolution through the principle of inclusion–exclusion. For instance, it serves as a key tool in determining the number of primes less than a positive integer. Additionally, many problems involving the enumeration of onto functions from one finite set to another can be effectively addressed using this principle. One notable example is the hatcheck problem, renowned for its solution derived through inclusion–exclusion. This problem poses the query regarding the probability of no individual receiving their rightful hat back from a hat check attendant distributing hats randomly.

Learning Outcomes for Study Session 6

When you have studied this session, you should be able to:

6.1 Explain the fundamental concept of the Inclusion-Exclusion principle.

6.2 Apply the principle to solve counting problems involving the union of sets.

6.3 Apply the Inclusion-Exclusion principle.

6.4 Describe set relationships with Venn diagrams.

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6.1 Inclusion Principle

The number of ways to select an element from A or from B, that is, the number of ways to select an element from their union, is the sum of the number of ways to select an element from A and the number of ways to select an element from B, minus the number of ways to select an element that is in both A and B. Because there are |A ∪ B| ways to select an element in either A or in B, and |A∩ B| ways to select an element common to both sets, we have |A ∪ B|=|A|+|B|−|A ∩ B|

6.2 The Principle of Inclusion–Exclusion

Let A_1, A_2 ,...,A_n be finite sets.

Then

$$\left|A_1 \cup A_2 \cup \ldots \cup A_n\right| = \sum_{1\le i\le n}\left|A_i\right| - \sum_{1\le i\le j\le n}\left|A_i \cap A_j\right| + \sum_{1\le i<j<k\le n}\left|A_i \cap A_j \cap A_k\right| - \cdots$$

$$(-1)^{n+1}\left|A_1 \cap A_2 \cap \ldots \cap A_n\right|.$$

6.3 The Proof

This can prove the formula by indicating that an element in the union is counted exactly once by the right-hand side of the equation. Suppose that a is a member of exactly r of the sets A_1, A_2,...,A_n where 1 ≤ r ≤ n. This element is counted C(r, 1) times by $\left|A_i\right|$. It is counted C(r, 2) times by $\left|A_i \cap A_j\right|$. In general, it is counted C(r, m) times by the summation involving m of the sets Ai. Thus, this element is counted exactly C(r, 1) − C(r, 2) + C(r, 3) −···+ $(-1)^{r+1}$C(r, r) times by the

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expression on the right-hand side of this equation. Our goal is to evaluate this quantity. By Corollary 2 of Section 6.4, we have

$$C(r, 0) - C(r, 1) + C(r, 2) - \cdots + (-1)^r C(r, r) = 0.$$

Hence,

$$1 = C(r, 0) = C(r, 1) - C(r, 2) + \cdots + (-1)^{r+1} C(r, r).$$

Therefore, each element in the union is counted exactly once by the expression on the right-hand side of the equation. This proves the principle of inclusion-exclusion.

The inclusion-exclusion principle gives a formula for the number of elements in the union of n sets for every positive integer n. There are terms in this formula for the number of elements in the intersection of every nonempty subset of the collection of the n sets. Hence, there are 2^{n-1} terms in this formula.

6.4 An Alternative Form of Inclusion–Exclusion

There is an alternative form of the principle of inclusion-exclusion that is useful in counting problems. In particular, this form can be used to solve problems that ask for the number of elements in a set that have none of n properties P_1, P_2,.... P_n.

Let A_i be the subset containing the elements that have property P_i. The number of elements with all the properties P_{i1} , P_{i2},...P_{ik} will be denoted by $N(P_{i1}$, P_{i2},... $P_{ik})$

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Writing these quantities in terms of sets, we have

$$\left|A_{i1} \cap A_{i2} \ldots \cap A_{ik}\right| = N(P_{i1}, P_{i2}, \ldots P_{ik})$$

If the number of elements with none of the properties $P_1, P_2, \ldots P_n$ is denoted by $N(P_1, P_2, \ldots P_n)$ and the number of elements in the set is denoted by N, it follows that $N(P_1, P_2, \ldots P_n) = N - \left|A_1 \cup A_2 \cup \ldots \cup A_n\right|$. From the inclusion-exclusion principle, we see that

$$NN(P^1_1, P^1_2, \ldots P^1_n), = N - \sum_{1 \le i \le j \le n} N(P_i) + \sum_{1 \le i \le j \le n} N(P_i P_j) - \sum_{1 \le i \le j \le n} N(P_i P_j P_j) + \cdots + (-1)^n N(P_1, P_2, \ldots P_n)\ (-1).$$

6.5 Two Finite Sets

Example 1: In a discrete mathematics class, every student is majoring in computer science, mathematics or both. The number of students having computer science as a major (possibly along with mathematics) is 30; the number of students having mathematics as a major (possibly along with computer science) is 15; and the number of students majoring in both computer science and mathematics is 10. How many students are in the class?

Solution: $|A \cup B| = |A| + |B| - |A \cap B| = 30 + 15 - 10 = 3$

Example 2: Suppose that there are 1807 freshmen at your school. Of these, 453 are taking a course in computer science, 567 are taking a course in mathematics, and 299 are taking courses in both computer science and

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mathematics. How many are not taking a course either in computer science or in mathematics?

Solution: To find the number of freshmen who are not taking a course in either mathematics or computer science, subtract the number that are taking a course in either of these subjects from the total number of freshmen. Let A be the set of all freshmen taking a course in computer science, and let B be the set of all freshmen taking a course in mathematics. It follows that $|A| = 453$, $|B| = 567$, and $|A \cap B| = 299$. The number of freshmen taking a course in either computer science or mathematics

$$|A \cup B| = |A| + |B| - |A \cap B| = 453 + 567 - 299 = 721$$

Consequently, there are $1807 - 721 = 1086$ freshmen who are not taking a course in computer science or mathematics.

6.6 Pigeonhole Principle and its Applications

In this section, we shall look into the pigeonhole principle and its applications.

Suppose that a flock of 30 pigeons flies into a set of 29 pigeonholes to roost. Because there are 30 pigeons but only 29 pigeonholes, at least one of these 29 pigeonholes must have at least two pigeons in it. To see why this is true, note that if each pigeonhole had at most one pigeon in it, at most 29 pigeons, one per hole, could be accommodated.

THE PIGEONHOLE PRINCIPLE If n is a positive integer and n + 1 or more objects are placed into n boxes, then there is at least one box containing two or more of the objects.

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Proof: Suppose that for each element y in the codomain of f we have a box that contains all elements x of the domain of f such that f (x) = y. Because the domain contains n + 1 or more elements and the codomain contains only n elements, the pigeonhole principle tells us that one of these boxes contains two or more elements x of the domain. This means that f cannot be one-to-one.

Suppose N objects are placed into k boxes; then there is at least one box containing at least N/k objects.

A common type of problem that asks for the minimum number of objects such that at least r of these objects must be in one of k boxes when these objects are distributed among the boxes is given by:

$$N=k(r-1)+1$$

Example 3: Among 100 people, determine the least number of people who were born in the same month.

Solution: This is a generalised Pigeonhole principle problem; at least r of these people must have been born in the same k months. When 100 people are taken into consideration, this is given by:

$$N=k(r-1)+1$$

$$N= 100\ ,\ r=12,\ r=?$$

$$100=k(12-1)+1$$

$$100=11k+1$$

k=9 . It means at least 9 people were born in the same month

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Example 4: What is the minimum number of computers that must be provided at MIVA Open University if at least 200 international students will be proctors for online examinations in 12 courses?

Solution: By applying the formula:

$$N = = k(r - 1) + 1$$

The minimum number of computers needed to ensure that at least 200 students would be proctoring the examination is the smallest integer N .

k=12, r=200

N= 12(200-1) + 1

N=12(199)+1

N = 2, 389 computers.

In-Text Questions and Answers (ITQs and ITAs)

ITQ 6.1: What does the Inclusion–Exclusion Principle correct when counting elements in overlapping sets?

Answer: It corrects overcounting by adding and subtracting intersections of sets so that each element is counted exactly once.

ITQ 6.2: What is the Pigeonhole Principle?

Answer: If n+1 or more objects are placed into n boxes, at least one box must contain two or more objects.

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Conclusion

This study session of counting with overlapping sets has a special focus on the Inclusion-Exclusion Principle. The Inclusion-Exclusion Principle refines this by strategically adding and subtracting set cardinalities based on the number of sets involved. We learnt its application to two finite sets and its more general alternative form, emphasising how this principle corrects for overcounting and allows us to accurately determine the size of unions in complex scenarios.

Summary

1. This study session showed how one set can be a subset of another.
2. Basic operations are done on sets, i.e., the union, intersection and difference of sets.
3. Venn diagram and how it shows possible memberships between two sets.

Self-Assessment Questions and Answers (SAQs and SAAs)

SAQ 6.1: Apply the Pigeonhole Principle: What is the minimum number of people required to guarantee that at least 5 were born in the same month?

Answer: Using $N=k(r-1)+1=12(5-1)+1=49$. Thus, at least 49 people.

SAQ 6.2: Distinguish between the basic and the generalized Pigeonhole Principle.

Answer: The basic version ensures that when more objects than boxes exist, one box holds at least two. The generalised version determines the minimum

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number of objects needed to ensure r or more in at least one box, using $N=k(r-1)+1$.

SAQ 6.3: Explain in your own words why the Inclusion–Exclusion Principle is necessary when counting elements in sets.

Answer: The principle is needed because simple addition of set cardinalities leads to double-counting of elements that appear in more than one set. The Inclusion–Exclusion Principle alternates between adding and subtracting intersections to count every element exactly once.

Multiple Choice Questions and Answers (MCQs and MCAs)

1. **If there are 12 courses and at least 200 students per course, how many computers must be available according to the generalised pigeonhole principle?**

A. 2,389

B. 2,400

C. 2,500

D. 2,388

Answer: A

Explanation: $N=12(200-1)+1=2,389$.

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2. What is the Inclusion–Exclusion Principle mainly used for in combinatorics?

A. Summation

B. Enumeration

C. Factorisation

D. Approximation

Answer: B

Explanation: It is a counting technique (enumeration) that ensures correct totals when sets overlap.

3. According to the Pigeonhole Principle, if 30 pigeons roost in 29 holes, what happens?

A. One pigeon is left out

B. At least one hole contains two pigeons

C. No pigeon finds a hole

D. All holes have one pigeon each

Answer: B

Explanation: Since 30 > 29, at least one hole must contain more than one pigeon.

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4. What is the total number of terms in the Inclusion–Exclusion formula for 4 sets?

A. 8

B. 15

C. 16

D. 10

Answer: B

Explanation: $2^4-1=15$ terms correspond to all non-empty intersections.

5. The expression $N=k(r-1)+1$ represents:

A. The Inclusion–Exclusion Principle

B. The Generalized Pigeonhole Principle

C. The Binomial Theorem

D. The Principle of Duality

Answer B

It calculates the minimum number of objects needed to ensure at least r objects fall into one of k boxes.

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Tutor-Marked Assessments (TMAs)

1. How many elements are in $M \cup N$ if there are 15 elements in A1, 20 elements in N and a) $M \cap N = \varnothing$? b) $|M \cap N| = 1$? c) $|M \cap N| = 6$? d) $M \subseteq N$?
2. There are 347 students at a college who have taken a course in calculus, 213 who have taken a course in discrete mathematics, and 189 who have taken courses in both calculus and discrete mathematics. How many students have taken a course in either calculus or discrete mathematics?
3. In a bushel containing 200 apples, 30 of them are afflicted with worms and 20 exhibit bruises. The apples eligible for sale are those that are devoid of both worms and bruises. Given that there are 15 bruised apples that also harbour worms, we need to determine the count of apples that meet the sale criteria.
4. Consider a drawer containing a dozen brown socks and a dozen black socks, all of them unmatched. A man is reaching into the drawer randomly in the dark.
    - How many socks must he draw to ensure that he has at least two socks of the same colour?
    - How many socks must he draw to ensure that he has at least two black socks?
5. Use the Inclusion–Exclusion Principle to find the number of students who study at least one of the following: Physics (40), Chemistry (35), Biology (30), given 10 study all three, 15 study both Physics and Chemistry, 8 study Chemistry and Biology, and 6 study Physics and Biology.

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Study Session Seven

Mathematical Induction

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Introduction

Mathematical induction is one of the techniques that can be used to prove a variety of mathematical statements that are formulated in terms of n, where n is a positive integer. But what makes induction so powerful and distinct from other proof techniques?

Imagine a line of dominoes standing closely together. If you knock over the first domino, and you know that each domino, when it falls, will knock over the next one, then you can be certain that all dominoes will fall. Mathematical induction operates on the same fundamental principle. It allows us to prove that a statement holds for an infinite set of natural numbers by establishing just two things: that it holds for the first number (the base case), and that whenever it holds for an arbitrary number, it must also hold for the next number (the inductive step).

This technique is particularly valuable because many mathematical statements—especially those involving sums, divisibility, inequalities, and properties of sequences—are naturally structured around the natural numbers. Directly proving such statements for every positive integer would be impossible, as it would require an infinite number of separate proofs. Induction provides an elegant and rigorous way to establish an infinite chain of truths from a finite set of logical steps.

In this session, we will begin by formally stating the principle of mathematical induction and carefully examining its logical foundation. We will then work

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through a wide range of examples, from simple summation formulas to more complex proofs involving divisibility, inequalities, and even trigonometric identities. You will learn how to structure an induction proof correctly, how to handle statements that do not start at n = 1, and how to identify situations where induction is the most appropriate proof technique. Finally, we will explore the limitations of induction—situations where it may not apply or where additional considerations are needed. By the end of this session, you will have developed the skills to confidently construct induction proofs and recognise their proper application in mathematical reasoning.

Learning Outcomes for Study Session 7

When you have studied this session, you should be able to:

7.1 State the principle of mathematical induction.

7.2 Apply mathematical induction to prove statements.

7.3 Identify the limitations of mathematical induction.

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7.1 The principle of mathematical induction

Let $P(n)$ be a given statement involving the natural number n such that

- The statement is true for $n = 1$, i.e., $P(1)$ is true (or true for any fixed natural number) and
- If the statement is true for $n = k$ (where k is a particular but arbitrary natural number), then the statement is also true for $n_ = k + 1$, i.e., the truth of $P(k)$ implies the truth of $P(k + 1)$. Then $P(n)$ is true for all natural numbers n.

Example 1:

$$1 + 3 + 5 + \ldots + (2n - 1) = n^2$$

Solution: Let the given statement $P(n)$ be defined as $P(n)$: $1 + 3 + 5 + \ldots + (2n - 1) = n^2$, for $n \in N$. Note that $P(1)$ is true, since: $P(1)$: $1 = 1^2$

Assume that $P(k)$ is true for some $k \in N$, i.e.,

$$P(k): 1 + 3 + 5 + \ldots + (2k - 1) = k^2$$

Now, to prove that $P(k + 1)$ is true, we have:

$$1 + 3 + 5 + \ldots + (2k - 1) + (2k + 1) = k^2 + (2k + 1) = k^2 + 2k + 1 = (k + 1)^2$$

$$= k^2 + (2k + 1)$$

Thus, $P(k + 1)$ is true. whenever $P(k)$ is true.

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Hence, by the Principle of Mathematical Induction, $P(n)$ is true for all $n \in N$.

Example 2:

$\sum_{n=1}^{n-1} t(t+1) = \frac{n(n-1)(n+1)}{3}$, for all natural numbers $n \geq 2$.

Solution: Let the given statement $P(n)$, be given as:

$$P(n): \sum_{t=1}^{n-1} r(t+1) = \frac{n(n-1)(n+1)}{3},\ for\ all\ natural\ numbers\ n \geq 2$$

We observe that:

$$P(2): \sum_{t=1}^{2-1} t(t+1) = \sum_{t=1}^{1} t(t+1) = 1.2 = \frac{1.2.3}{3} = \frac{2.(2-1)(2+1)}{3}$$

Thus, $P(n)$ in true for $n = 2$

Assume that $P(n)$ is true for $n = k \in N$

$$\text{I.e. } P(k): \sum_{t=1}^{k-1} t(t+1) = \frac{k(k-1)(k+1)}{3}$$

To prove that $P(k+1)$ is true, we have

$$\sum_{t=1}^{(k+1-1)} t(t+1) = \sum_{t=1}^{k} t(t+1)$$

$$= \sum_{t=1}^{k-1} t(t+1) + k(k+1)$$

$$= \frac{k(k-1)(k+1)}{3} + k(k+1)$$

$$= k(k+1)\left[\frac{k-1+3}{3}\right]$$

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$$= \frac{k(k+1)(k+2)}{3} = \frac{(k+1)((k+1)-1))((k+1)+1)}{3}$$

Thus, $P(k+1)$ is true, whenever $P(k)$ is true.

Hence, by the Principle of Mathematical Induction, $P(n)$ is true for all natural numbers $n \geq 2$

Example 3:

$\left(1-\frac{1}{2^2}\right)\cdot\left(1-\frac{1}{3^2}\right)\ldots\left(1-\frac{1}{n^2}\right)=\frac{n+1}{2n}$, for all natural numbers, $n \geq 2$.

Solution: Let the given statement be $P(n)$, i.e.,

$$P(n): \left(1-\frac{1}{2^2}\right)\cdot\left(1-\frac{1}{3^2}\right)\ldots\left(1-\frac{1}{n^2}\right)=\frac{n+1}{2n},\ for\ all\ natural\ numbers,\ n \geq 2$$

We observe that $P(2)$ is true, since:

$$\left(1-\frac{1}{2^2}\right)= 1-\frac{1}{4}=\frac{4-1}{4}=\frac{3}{4}=\frac{2+1}{2\times 2}$$

assume that $P(n)$ is true for some $k \in N$, i.e.,

$$P(k):\ 1-\frac{1}{2^2}\cdot 1-\frac{1}{3^2}\ldots 1-\frac{1}{k^2}=\frac{k+1}{2k}$$

now, to prove that $P(k+1)$ is true, we have

$$1-\frac{1}{2^2}\cdot 1-\frac{1}{3^2}\ldots 1-\frac{1}{k^2}\cdot 1-\frac{1}{(k+1)^2} = \frac{k+1}{2k}\left(1-\frac{1}{(k+1)^2}\right)=\frac{k^2+2k}{2k(k+1)}=\frac{(k+1)+1}{2(k+1)}$$

Thus, $P(k+1)$ is true whenever $P(k)$ is true.

Hence, by the Principle of Mathematical Induction $P(n)$ is true for all natural numbers, $n \geq 2$.

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Example 4:

$2^{2n} - 1$ is divisible by 3 .

Solution: Let the statement $P(n)$ given as:

$P(n)$: $2^{2n} - 1$ is divisible by 3 for every natural number n.

We observe that $P(1)$ is true, since:

$2^2 - 1 = 4 - 1 = 3.1$ *is divisible by* 3.

Assume that $P(n)$ is true for some natural number k, i.e., $P(k)$: $2^{2k} - 1$ is divisible by 3, i.e., $2^{2k} - 1 = 3q$, where $q \in N$

Now, to prove that $P(k + 1)$ is true, we have:

$$P(k+1): 2^{2(k+1)} - 1 = 2^{2k+2} - 1 = 2^{2k} \cdot 2^2 - 1 = 2^{2k} \cdot 4 - 1 = 3 \cdot 2^{2k} + \left(2^{2k} - 1\right)$$

$$= 3.2^{2k} + 3q = 3\left(2^{2k} + q\right) = 3m, \text{ where } m \in N$$

Thus $P(k + 1)$ is true, whenever $P(k)$ is true.

Hence, by the Principle of Mathematical Induction $P(n)$ is true for all natural numbers n.

Example 5:

$2n + 1 < 2^n$, for all natural numbers $n \geq 3$.

Solution: Let $P(n)$ be the given statement, i.e., $P(n)$: $(2n + 1) < 2^n$ for all natural numbers, $n \geq 3$. We observe that $P(3)$ is true, since:

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$$2.3 + 1 = 7 < 8 = 2^3$$

Assume that $P(n)$ is true for some natural number k, i.e., $2k + 1 < 2^k$

To prove $P(k + 1)$ is true, we have to show that $2(k + 1) + 1 < 2^{k+1}$. Now, we have:

$$2(k + 1) + 1 = 2k + 3 = 2k + 1 + 2 < 2^k + 2 < 2^k.2 = 2^{k+1}.$$

Thus $P(k + 1)$ is true, whenever $P(k)$ is true.

Hence, the principle of mathematical induction $P(n)$ is true for all natural numbers, $n \geq 3$.

Example 6:

The distributive law from algebra says that for all real numbers c, a_1 and a_2, we have $c(a_1 + a_2) = ca_1 + ca_2$.

Use this law and mathematical induction to prove that, for all natural numbers, $n \geq 2$, if $c, a_1, a_2, \ldots, a_n$ are any real numbers, then:

$$c(a_1 + a_2 + \ldots + a_n) = ca_1 + ca_2 + \ldots + ca_n$$

Solution: Let $P(n)$ be the given statement, i.e.,

$P(n)$: $c(a_1 + a_2 + \ldots + a_n) = ca_1 + ca_2 + \ldots ca_n$ for all natural numbers $n \geq 2$, for $c, a_1, a_2, \ldots a_n \in R$.

We observe that that $P(2)$ is true since:

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$c(a_1 + a_2) = ca_1 + ca_2$ (by distributive law)

Assume that $P(n)$ is true for some natural number k, where $k > 2$, i.e.,

$$P(k): c(a_1 + a_2 + \ldots + a_k) = ca_1 + ca_2 + \ldots + ca_k$$

Now to prove $P(k + 1)$ is true, we have:

$$P(k+1): c(a_1 + a_2 + \ldots + a_k + a_{k+1})$$
$$= c((a_1 + a_2 + \ldots + a_k) + a_{k+1}) \quad = c(a_1 + a_2 + \ldots + a_k) + ca_{k+1}$$
$$= ca_1 + ca_2 + \ldots + ca_k + ca_{k+1}$$

(by distributive law)

Thus $P(k + 1)$ is true, whenever $P(k)$ is true.

Hence, by the principle of mathematical induction, $P(n)$ is true for all natural numbers $n \geq 2$.

Example 7:

Prove by induction that for all natural numbers n:

$$sin\alpha + sin(\alpha + \beta) + sin(\alpha + 2\beta) + \ldots + sin(\alpha + (n-1)\beta)$$
$$= \frac{sin\left(\alpha + \frac{n-1}{2}\beta\right) sin\left(\frac{n\beta}{2}\right)}{sin\left(\frac{\beta}{2}\right)}$$

Solution:

Consider $:P(n): sin\alpha + sin(\alpha + \beta) + sin(\alpha + 2\beta) + \ldots + sin(\alpha + (n-1)\beta)$

$= \frac{sin\left(\alpha + \frac{n-1}{2}\beta\right) sin\left(\frac{n\beta}{2}\right)}{sin\left(\frac{\beta}{2}\right)}$, $for\ all\ natural\ number\ n.$

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We observe that $P(1)$ is true, since: $P(1): sin\alpha = \frac{sin(\alpha+0)sin\frac{\beta}{2}}{sin\frac{\beta}{2}}$; assume that $P(n)$ is true for some natural numbers k, i.e.,

$P(k): sin\alpha + sin(\alpha + \beta) + sin(\alpha + 2\beta) + \ldots + sin(\alpha + (k-1)\beta)$

$= \frac{sin\left(\alpha+\frac{k-1}{2}\beta\right)sin\left(\frac{k\beta}{2}\right)}{sin\left(\frac{\beta}{2}\right)}$

Now, to prove that $P(k+1)$ is true, we have:

$P(k+1): sin\alpha + sin(\alpha + \beta) + sin(\alpha + 2\beta) + \ldots$

$+ sin(\alpha + (k-1)\beta) + sin(\alpha + k\beta) = \frac{sin\left(\alpha+\frac{k-1}{2}\beta\right)sin\left(\frac{k\beta}{2}\right)}{sin\left(\frac{\beta}{2}\right)} + sin(\alpha + k\beta)$

$= \frac{sin\left(\alpha+\frac{k-1}{2}\beta\right)sin\frac{k\beta}{2}+sin(\alpha+k\beta)sin\frac{\beta}{2}}{sin\frac{\beta}{2}} = \frac{cos\left(\alpha-\frac{\beta}{2}\right)-cos\left(\alpha+k\beta-\frac{\beta}{2}\right)+cos\left(\alpha+k\beta-\frac{\beta}{2}\right)-cos\left(\alpha+k\beta+\frac{\beta}{2}\right)}{2sin\frac{\beta}{2}}$

$= \frac{cos\left(\alpha-\frac{\beta}{2}\right)-cos\left(\alpha+k\beta+\frac{\beta}{2}\right)}{2sin\frac{\beta}{2}} = \frac{sin\left(\alpha+\frac{k\beta}{2}\right)sin\left(\frac{k\beta+\beta}{2}\right)}{sin\frac{\beta}{2}} = \frac{sin\left(\alpha+\frac{k\beta}{2}\right)sin(k+1)\left(\frac{\beta}{2}\right)}{sin\frac{\beta}{2}}$

Thus $P(k+1)$ is true whenever $P(k)$ is true.

Hence, by the Principle of Mathematical Induction $P(n)$ is true for all natural numbers n.

Example 8:

Prove by the Principle of Mathematical Induction that $1\times1! + 2\times2! + 3\times3! + \ldots + n \times n! = (n+1)! - 1$ for all natural numbers n.

Solution: Let $P(n)$ be the given statement, that is, $P(n): 1\times1! + 2\times2! + 3\times3! + \ldots + n \times n! = (n+1)! - 1$ for all natural numbers n

Note that $P(1)$ is true, since:

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$P(1)$: $1\times1! = 1 = 2 - 1 = 2! - 1.$

assume that $P(n)$ is true for some natural number k, i.e.,

$P(k)$: $1\times1! + 2\times2! + 3\times3! + \ldots + k \times k! = (k + 1)! - 1$

To prove $P(k + 1)$ is true, we have:

$$P(k + 1)\text{: } 1\times1! + 2\times2! + 3\times3! + \ldots + k \times k! + (k + 1)\times(k + 1)!$$
$$= (k + 1)! - 1 + (k + 1)! \times(k + 1)$$
$$= (k + 1 + 1)(k + 1)! - 1 \quad = (k + 2)(k + 1)! - 1$$
$$= ((k + 2)! - 1$$

Thus, $P(k + 1)$ is true, whenever $P(k)$ is true. Therefore, by the Principle of Mathematical Induction $P(n)$ is true for all natural numbers n.

Example 9:

Show by the Principle of Mathematical Induction that the sum S_n of the n term of the series $1^2 + 2\times2^2 + 3^2 + 2\times4^2 + 5^2 + 2\times6^2$ … is given by:

$$S_n = \{\frac{n(n+1)^2}{2},\ if\ n\ is\ even\ \ \frac{n^2(n+1)}{2},\ if\ n\ is\ odd$$

Solution:

Here,

$$P(n)\text{: } S_n = \{\frac{n(n+1)^2}{2},\ when\ n\ is\ even\ \ \frac{n^2(n+1)}{2},\ when\ n\ is\ odd$$

Also, note that any term T_n of the series is given by:

$$T_n = \{n^2\ if\ n\ is\ odd\ \ 2n^2\ if\ n\ is\ even$$

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We observe that $P(1)$ is true since:

$$P(1): S_1 = 1^2 = 1 = \frac{1 \cdot 2}{2} = \frac{1^2 \cdot (1+1)}{2}$$

Assume that $P(k)$ is true for some natural number k, i.e.,.

Case 1: When k is odd, then $k + 1$ is even. We have:

$$P(k+1): S_{k+1} = 1^2 + 2\times 2^2 + \ldots + k^2 + 2\times(k+1)^2 = \frac{k^2(k+1)}{2} + 2\times(k+1)^2$$

$$= \frac{(k+1)}{2}\left[k^2 + 4(k+1)\right]\left(as\ k\ is\ odd, 1^2 + 2\times 2^2 + \ldots + k^2 = k^2\frac{(k+1)}{2}\right)$$

$$= \frac{k+1}{2}\left[k^2 + 4k + 4\right] = \frac{k+1}{2}(k+2)^2 = (k+1)\frac{[(k+1)+1]^2}{2}$$

So $P(k+1)$ is true, whenever $P(k)$ is true in the case when k is odd.

Case 2: When k is even, then $k + 1$ is odd. Now,

$$P(k+1): 1^2 + 2\times 2^2 + \ldots + 2.k^2 + (k+1)^2$$

$$= \frac{k(k+1)^2}{2} + (k+1)^2 \left(as\ k\ is\ even,\ 1^2 + 2\times 2^2 + \ldots + 2k^2 = k\frac{(k+1)^2}{2}\right)$$

$$= \frac{(k+1)^2(k+2)}{2} = \frac{(k+1)^2((k+1)+1)}{2}$$

Therefore, $P(k+1)$ is true whenever $P(k)$ is true for the case when k is even. Thus $P(k+1)$ is true whenever $P(k)$ is true for any natural numbers k. Hence, $P(n)$ true for all natural numbers.

In-Text Exercises

1. Give an example of a statement $P(n)$ which is true for all $n \geq 4$ but $P(1)$, $P(2)$ and $P(3)$ are not true. Justify your answer.

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2. Give an example of a statement $P(n)$, which is true for all n. Justify your answer. Prove each of the statements in Exercises 3–16 by the Principle of Mathematical Induction:

3. $4^n - 1$ is divisible by 3 for each natural number n.

4. $2^{3n} - 1$ is divisible by 7 for all natural numbers n.

5. $n^3 - 7n + 3$ is divisible by 3 for all natural numbers n.

6. $3^{2n} - 1$ is divisible by 8 for all natural numbers n. 7. For any natural number, it $n, 7^n - 2^n$ is divisible by 5.

7. For any natural number $n, x^n - y^n$ is divisible by $x - y$, where x and y are any integers with $x \neq y$.

8. $n^3 - n$ is divisible by 6 for each natural number $n \geq 2$.

9. $n(n^2 + 5)$ is divisible by 6 for each natural number n.

10. $n^2 < 2^n$ for all natural numbers $n \geq 5$.

11. $2n < (n + 2)!$ for all natural numbers n.

12. $\sqrt{n} < \frac{1}{\sqrt{1}} + \frac{1}{\sqrt{2}} + \ldots + \frac{1}{\sqrt{n}}$, for all natural numbers $n \geq 2$.

13. $2 + 4 + 6 + \ldots + 2n = n^2 + n$ for all natural numbers n.

14. $1 + 2 + 2^2 + \ldots + 2^n = 2^{n+1} - 1$ for all natural numbers n.

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15. $1 + 5 + 9 + \ldots + (4n - 3) = n(2n - 1)$ for all natural numbers n.

Use the principle of mathematical induction in the following exercises.

16. A sequence $a_1, a_2, a_3 \ldots$ is defined by letting $a_1 = 3$ and $a_k = 7a_{k-1}$ for all natural numbers $k \geq 2$. Show that $a_n = 3.7^{n-1}$ for all natural numbers.

17. A sequence $b_0, b_1, b_2 \ldots$ is defined by letting $b_0 = 5$ and $b_k = 4 + b_{k-1}$ for all natural numbers k. Show that $b_n = 5 + 4n$ for all natural numbers n using mathematical induction.

18. A sequence $d_1, d_2, d_3 \ldots$ is defined by letting $d_1 = 2$ and $d_k = \frac{d_{k-1}}{k}$ for all natural numbers, $k \geq 2$. Show that $d_n = \frac{2}{n!}$ for all $n \in N$.

19. Prove that for all $n \in N$

$cos\alpha + cos(\alpha + \beta) + cos(\alpha + 2\beta) + \ldots + cos(\alpha + (n-1)\beta)$

$= \frac{cos\left(\alpha + \left(\frac{n-1}{2}\right)\beta\right) sin\left(\frac{n\beta}{2}\right)}{sin\frac{\beta}{2}}$

21 Prove that $cos\theta cos2\theta cos2^2\theta \ldots cos2^{n-1}\theta = \frac{sin2^n\theta}{2^n sin\theta}$, for all $n \in N$.

22 Prove that $sin\theta + sin2\theta + sin3\theta + \ldots + sinn\theta = \frac{sin\frac{n\theta}{2} sin\frac{(n+1)}{2}\theta}{sin\frac{\theta}{2}}$, for all $n \in N$.

In-Text Questions and Answers (ITQs and ITAs)

ITQ 7.1: What is the first step (the base case) in a proof by mathematical induction, and why is it essential?

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Answer: The base case is to verify that the statement P(n) is true for the initial value, usually n=1 (or another specified starting point like n=2, n=3, etc.). It is essential because it provides the foundation upon which the inductive step builds. Without a true base case, the entire "domino effect" of induction has nothing to start from.

ITQ 7.2: In the inductive step, what is the statement we assume to be true, and what is the statement we must prove?

Answer: We assume the Induction Hypothesis: that P(k) is true for some arbitrary natural number k. We must then prove that P(k+1) is true, based on this assumption.

Conclusion

In this study session, we have demystified the principle of mathematical induction, a method for proving a base case and an inductive step. By demonstrating that the statement holds true for the initial value (often n = 1) and that it implies the statement for the next value (n + 1), we can establish its validity for all natural numbers.

Summary

1. Mathematical induction is a proof technique used to show the validity of statements for all natural numbers.
2. It involves three steps: the base step, the induction hypothesis, and the induction step.

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3. The base step establishes the truth of the statement for the initial value.
4. The induction hypothesis assumes that the statement is true for an arbitrary value, usually denoted by "k."
5. It can be used to prove various statements, such as those for sequences, divisibility, inequalities, and properties of numbers.

Self-Assessment Questions and Answers (SAQs and SAAs)

SAQ 7.1: State the principle of mathematical induction in your own words.

Answer: It is a method for proving that a statement is true for every natural number. First, you show it's true for the first number (e.g., n=1). Then, you prove that if it is true for any arbitrary number 'k', then it must also be true for the very next number 'k+1'. This creates a chain reaction that proves the statement for all numbers.

SAQ 7.2: A student correctly proves the base case P(1) for a statement. They then assume P(k) is true and successfully prove that P(2k) is true. Is this a valid proof by induction? Why or why not?

Answer: No, this is not valid. The principle of mathematical induction requires proving that the truth of P(k) implies the truth of P(k+1), the next integer. Proving P(2k) skips many numbers (e.g., from P(1) you prove P(2), but then you need P(3), not P(4)). This does not establish the necessary chain reaction.

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SAQ 7.3: Can mathematical induction be used to prove statements that are true only for even numbers? Explain.

Answer: Yes. The base case would start at the first even number (e.g., n=2). The inductive step would then prove that if the statement is true for an arbitrary even number k, it is true for k+2 (the next even number). The principle remains the same, but the step size changes.

Multiple-Choice Questions and Answers (MCQs and MCAs)

1. The Principle of Mathematical Induction is used to prove statements for:

a) All real numbers

b) All integers

c) A finite set of numbers

d) All natural numbers

Answer: d

Explanation: Induction is designed to prove that a statement holds for every member of the countably infinite set of natural numbers.

2. The first step in a proof by induction is to:

a) Assume P(k) is true for some k

b) Prove P(1) is true (or the first relevant case)

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c) Prove P(k+1) is true

d) State the conclusion

Answer: b

Explanation: The base case is the essential starting point that anchors the entire proof.

3. Which of the following is the correct goal when proving the inductive step?

a) Prove P(1) is true.

b) Prove P(k) is true.

c) Prove that P(k) ⇒ P(k+1) is true.

d) Prove P(k+1) ⇒ P(k) is true.

Answer: c

Explanation: The inductive step aims to show that the truth of the statement for one number logically forces it to be true for the next.

4. If you prove P(1) and you prove that P(k) ⇒ P(k+2), what have you proven?

a) P(n) is true for all natural numbers n.

b) P(n) is true for all odd natural numbers.

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c) P(n) is true for all even natural numbers.

d) Nothing of value.

Answer: B

Explanation: The base case P(1) is true, and the step shows it's true for 1, 3, 5, 7, etc. This is a valid form of induction for odd numbers.

5. A student proves the base case P(1) and then, in the inductive step, correctly proves that if P(k) is true then P(k+1) is true. This means:

a) P(n) is probably true for all n.

b) P(n) is true for n=1 and n=k+1 only.

c) P(n) is true for all natural numbers n.

d) The induction hypothesis was correct.

Answer. c.

Explanation: This is the complete and correct application of the principle, leading to the conclusion that the statement is universally true for the set of natural numbers.

Tutor-Marked Assessments (TMAs)

1. Use the principle of mathematical induction to prove that $11^n - 4^n$ is divisible by 7 for all natural numbers n.

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2. Prove by induction that $5^{2n}-1$ is divisible by 24 for all natural numbers n. (Hint: 25=24+1).
3. Describe the logical "domino effect" analogy for mathematical induction.
4. Looking at Example 9, why was it necessary to consider two separate cases (when k is even and when k is odd) in the inductive step?
5. If you prove P(1) and you prove that $P(k) \Rightarrow P(k+2)$, what have you proven?

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Study Session Eight

The Binomial Theorem

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Introduction

This session focuses on the binomial theorem, binomial expansion, binomial coefficients, and the Pascal triangle. In this session, we'll delve into the fundamental concepts surrounding algebraic expressions with two terms, understanding their expansion, coefficients, and a fascinating historical tool for calculating binomial coefficients, the Pascal Triangle.

From understanding the binomial theorem's statement to mastering the techniques of binomial expansion, we'll equip ourselves with the tools necessary to tackle complex problems with confidence. Moreover, we'll explore the elegant simplicity of Pascal's Triangle, a mathematical marvel that aids in determining binomial coefficients effortlessly.

By the end of this section, you'll grasp the intricacies of these mathematical principles and be equipped with the skills to apply them confidently in problem-solving scenarios.

Learning Outcomes for Study Session 8

When you have studied this session, you should be able to:

8.1 State the binomial theorem.

8.2 Identifying a binomial expression.

8.3 Expand a given binomial expression with a power of a natural number.

8.4 Apply the binomial theorem to expand expressions of the form $a+b^n$, where n is a positive integer power.

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8.5 Identify the middle term in the binomial expansion for even and odd values of n.

8.6 Determine the general term in the binomial expansion and its use in finding specific terms or coefficients.

8.7 Utilise the binomial series to approximate expressions when x is numerically less than one.

8.8 Apply the binomial series to derive approximations for given expressions under specific conditions.

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8.1 Binomial Theorem

A binomial expression is usually of the form $(a + b)^n$ where a, b are algebraic terms and n is a real number. If n is a natural number, one may decide to expand the bracket by multiplying two brackets, then use the result to multiply another bracket until the power of n is exhausted. This approach could be time-consuming in cases of large values of n and it may never work when n is not a natural number. In this section, we present the approaches to expanding the binomial expression for any value of n in the set of real numbers.

Binomial Theorem: Let a and b be variables, and let n be a nonnegative integer.

Then, $(a + b)^n = \sum_{j=0}^{n} \frac{n}{j} a^{(n-j)} b^j$

$= \frac{n}{0} a^{(n-0)} b^0 + \frac{n}{1} a^{(n-1)} b^1 + \frac{n}{2} a^{(n-2)} b^2 + \cdots + \frac{n}{n} a^{(n-n)} b^n$

Where $\frac{n}{m}$, read as n combination m, is known as the binomial coefficient, and it is defined as

$$\frac{n}{m} = \frac{n!}{(n-m)!\, m!},\ n!\ (n\ factorial) = n \times (n-1) \times (n-2) \times \cdots \times 2 \times 1$$

$$(a + b)^n = a^n + \frac{n}{1!} a^{n-1} b + \frac{n(n-1)}{2!} a^{n-2} b^{2+\ldots+\frac{n(n-1)(n-2)\ldots(n-r+1)}{r!}} a^{n-r} b^r + \ldots + b^n$$

Note: Since

$$\left(\frac{n}{r}\right) = \frac{n!}{r!(n-r)!}$$

So,

$$\left(\frac{n}{0}\right) = \frac{n!}{0!(n-0)!} = \frac{n!}{1 \times n!} = 1$$

$$\left(\frac{n}{1}\right) = \frac{n!}{1!(n-1)!} = \frac{n(n-1)!}{1!(n-1)!} = \frac{n(n-1)}{2!}$$

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$$\binom{n}{2} = \frac{n!}{2!(n-2)!} = \frac{n(n-2)!}{2!(n-2)!} = \frac{n(n-1)}{2!}$$

$$\binom{n}{3} = \frac{n!}{3!(n-3)!} = \frac{n(n-1)(n-2)(n-3)!}{3!(n-3)!} = \frac{n(n-1)(n-2)}{3!}$$

$$\binom{n}{r} = \frac{n(n-1)(n-2)!\ldots(n-r+1)(n-r)!}{r!(n-r)!}$$

$$= \frac{n(n-1)(n-2)!\ldots(n-r+1)}{r!}$$

$$\binom{n}{n} = \frac{n!}{n!(n-n)!} = \frac{n!}{n!\times 1} = 1.$$

The following points can be observed in the expansion of $(a + b)^n$

- There are $(n + 1)$ terms in the expansion
- The 1st term is a^n and $(n + 1)$th term or the last term is b^n
- The exponent of a decreases from n to zero
- The exponent of b increases from 0 to n.
- The sum of the exponents of a and b in any term is equal to the index n.
- The co-efficient of the term equidistant from the beginning and end of the expansion is equal to

  $\binom{n}{r} = \binom{n}{n-r}$.

We now apply this theorem to solve the following problems:

Example 1: What is the expansion of $(x + y)^5$?

Solution: We use the formula

$$(a + b)^n = \sum_{j=0}^{n} \frac{n}{j} a^{(n-j)} b^j$$

$$= \frac{n}{0} a^{(n-0)} b^0 + \frac{n}{1} a^{(n-1)} b^1 + \frac{n}{2} a^{(n-2)} b^2 + \cdots + \frac{n}{n} a^{(n-n)} b^n$$

Here $a=x$, $b=y$ and $n=5$. Thus,

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$$(x+y)^5 = \sum_{j=0}^{5} \frac{5}{0} a^{(5-j)} b^j$$

$$= \frac{5}{0}x^{(5-0)}y^0 + \frac{5}{1}x^{(5-1)}y^1 + \frac{5}{2}x^{(5-2)}y^2 + \frac{5}{3}x^{(5-3)}y^3 + \frac{5}{4}x^{(5-4)}y^4 + \frac{5}{5}x^{(5-5)}y^5$$

$$= \frac{5}{0}x^5 + \frac{5}{1}x^4y^1 + \frac{5}{2}x^3y^2 + \frac{5}{3}x^2y^3 + \frac{5}{4}x^1y^4 + \frac{5}{5}x^0y^5$$

$$= \frac{5}{0}x^5 + \frac{5}{1}x^4y + \frac{5}{2}x^3y^2 + \frac{5}{3}x^2y^3 + \frac{5}{4}xy^4 + \frac{5}{5}y^5$$

Applying the definition of factorial of numbers to simplify the binomial coefficients, we have

$\frac{5}{0} = \frac{5!}{(5-0)!\,0!} = \frac{5!}{5!\,0!}\frac{1}{1\cdot 0!} = \frac{1}{1\cdot 1} = 1$ (Observe that $0! = 1$)

$\frac{5}{1} = \frac{5!}{(5-1)!\,1!} = \frac{5!}{4!\,1!}\frac{5\times 4!}{4!\cdot 0!} = \frac{5}{1\cdot 1} = 5$

$\frac{5}{2} = \frac{5!}{(5-2)!\,2!} = \frac{5\times 4\times 3!}{3!\,2!}\frac{5\times 4}{1\cdot 2!} = \frac{5\times 4}{1\cdot 2} = 10$

$\frac{5}{3} = \frac{5!}{(5-3)!\,3!} = \frac{5\times 4\times 3!}{2!\,3!}\frac{5\times 4\times 1}{2\times 1\cdot 1} = \frac{20}{2} = 10$

$\frac{5}{4} = \frac{5!}{(5-4)!\,4!} = \frac{5\times 4!}{1!\,4!} = \frac{5}{1\cdot 1} = \frac{5}{1} = 5$

$\frac{5}{5} = \frac{5!}{(5-5)!\,5!} = \frac{5!}{0!\,5!}\frac{1}{1\cdot 0!} = \frac{1}{1\cdot 1} = 1$

Therefore,

$$\frac{5}{0}x^5 + \frac{5}{1}x^4y + \frac{5}{2}x^3y^2 + \frac{5}{3}x^2y^3 + \frac{5}{4}xy^4 + \frac{5}{5}y^5$$

$$= 1x^5 + 5x^4y + 10x^3y^2 + 10x^2y^3 + 5xy^4 + 1y^5$$

$$= x^5 + 5x^4y + 10x^3y^2 + 10x^2y^3 + 5xy^4 + y^5$$

Hence

$$(x+y)^5 = x^5 + 5x^4y + 10x^3y^2 + 10x^2y^3 + 5xy^4 + y^5$$

Example 2:

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What is the expansion of $(2x - 3)^4$? Hence, or otherwise, evaluate $(-2.998)^4$.

Solution: We use the formula $(a + b)^n = \sum_{j=0}^{n} \frac{n}{j} a^{(n-j)} b^j$

$= \frac{n}{0} a^{(n-0)} b^0 + \frac{n}{1} a^{(n-1)} b^1 + \frac{n}{2} a^{(n-2)} b^2 + \cdots + \frac{n}{n} a^{(n-n)} b^n$

Here a=2x, b=-3 and n=4. Thus, $(2x - 3)^4 = \sum_{j=0}^{4} \frac{4}{0} (2x)^{(4-j)} (-3)^j$

$= \frac{4}{0}(2x)^{(4-0)}(-3)^0 + \frac{4}{1}(2x)^{(4-1)}(-3)^1 + \frac{4}{2}(2x)^{(4-2)}(-3)^2$

$+ \frac{4}{3}(2x)^{(4-3)}(-3)^3 + \frac{4}{4}(2x)^{(4-4)}(-3)^4$

$= \frac{4}{0}(2x)^4 + \frac{4}{1}(2x)^3(-3) + \frac{4}{2}(2x)^2(-3)^2 + \frac{4}{3}(2x)^1(-3)^3 + \frac{4}{4}(2x)^0(-3)^4$

$= \frac{4}{0}2^4x^4 + \frac{4}{1}2^3x^3(-3) + \frac{4}{2}2^2x^2(-3)^2 + \frac{4}{3}2x(-3)^3 + \frac{4}{4}(1)(-3)^4$

$= \frac{4}{0}16x^4 + \frac{4}{1}8x^3(-3) + \frac{4}{2}4x^2(9) + \frac{4}{3}2x(-27) + \frac{4}{4}(1)(81)$

$= (1)16x^4 + (4)8x^3(-3) + (6)4x^2(9) + (4)2x(-27) + (1)(1)(81)$

Hence, $(2x - 3)^4 = 16x^4 - 96x^3 + 216x^2 - 216x + 81$

We can now use this expansion to simplify $(-2.998)^4$.

Comparing $(2x - 3)^4$ and $(-2.998)^4$ implying 2x-3=-2.998$\Rightarrow$x=0.001.

$(2x - 3)^4 = 16x^4 - 96x^3 + 216x^2 - 216x + 81$

$[(-2.998)^4 = [(2(0.001) - 3)]^4$

$= 16(-2.998)^4 - 96(-2.998)^3 + 216(-2.998)^2 - 216(-2.998) + 81$

=80.7842

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Hence, $(-\ 2.998)^4\ = 80.7842$

8.2 The General Term

The term $\left(\frac{n}{r}\right)a^{n-r}b^r$ in the expansion of the binomial theorem is called the **general term** or $(r\ +\ 1)$th term. It is denoted by T_{r+1}. Hence,

$$T_{r+1} = \left(\frac{n}{r}\right)a^{n-r}b^r$$

Note that the general term is used to find out the specified term or the required coefficient of the term in the binomial expansion.

Example 3: Find the coefficient of the 4th term in the expansion of $(2\ -\frac{1}{4x^2})^6$

Solution:

The 4th term is $\frac{6}{3}2^3\left(-\frac{1}{4x^2}\right)^3 = \frac{6}{3}8\left(-\frac{1}{4^3x^{2\times3}}\right) = 20\ \cdot\ 8\ \cdot\left(-\frac{1}{64x^6}\right) = \ -\frac{5}{2x^6}$ Hence, the coefficient of the 4th term is $-\frac{5}{2}$

Example 4:

Evaluate the $5th$ term in the expansion of $(4x\ -\ 3y)^8$

Solution: The $5th$ term is

$$\frac{8}{5}(4x)^3(-\ 3y)^5 = \frac{8}{5}4^3x^3\left(-\ 3^5y^5\right) = 56\ \cdot\ 64x^3\ \cdot\left(-\ 243y^5\right) = \ -\ 870912x^3y^5$$

Example 5: Find the eighth term in the expansion of $\left(2x^2\ -\frac{1}{x^2}\right)^{12}$

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Solution: The general term is $T_{r+1} = \left(\frac{n}{r}\right)a^{n-r}b^{r}$

Here, $T_8 = ?, a = 2x^2, b = -\frac{1}{x^2}, n = 12, r = 7.$

Therefore,

$$T_{7+1} = \left(\frac{12}{7}\right)\left(2x^2\right)^{12-7}\left(-\frac{1}{x^2}\right)^7$$

$$T_8 = \frac{12.11.10.9.8.7.6}{7.6.5.4.3.2.1}\left(2x^2\right)^5\frac{(-1)^7}{x^{14}}$$

$$= 793\times32x^{10}\frac{(-1)}{x^{14}}$$

$$= -\frac{25344}{x^4}$$

$$\therefore T_8 = -\frac{25344}{x^4}$$

8.3 Middle Term in the Expansion $(a + b)^n$

In the expansion, $(a + b)^n$, there are $(n + 1)$ terms.

Case I: If n is even, then $(n + 1)$ will be odd, so $\left(\frac{n}{2} + 1\right)$th term will be the only one middle term in the expansion. For example, if $n = 8$ (even), the number of terms will be 9 (odd); therefore, $\left(\frac{8}{2} + 1\right) = 5^{\text{th}}$ will be the middle term.

Case II: If n is odd, then $(n + 1)$ will be even; in this case, there will not be a single middle term, but $\left(\frac{n+1}{2}\right)$th and $\left(\frac{n+1}{2} + 1\right)$th term will be the two middle terms in the expansion. For example, for $n = 9$ (odd), the number of terms is 10 i.e., $\left(\frac{9+1}{2}\right)$th and $\left(\frac{9+1}{2} + 1\right)$th i.e., $5th$ and $6th$ terms are taken as middle terms and these middle terms are found by using the for the general term.

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Example 6: Find the middle term of $\left(1 - \frac{x^2}{2}\right)^{14}$

Solution: we have $n = 14$, then number of terms is 15.

$\therefore\left(\frac{14}{2} + 1\right)$ i.e., $8th$ will be the middle term.

$a = 1,\ b = -\frac{x^2}{2}, n = 14,\ r = 7,\ T_8 = ?$

$T_{r+1} = \left(\frac{n}{r}\right)a^{n-r}b^r$

$T_{r+1} = \left(\frac{14}{7}\right)(1)^{14-7}\left(-\frac{x^2}{2}\right)^7 = \frac{14!}{7!7!}(-1)^7\frac{x^{14}}{2^7}$

$= \frac{14.13.12.11.10.9.8.7!}{7.6.5.4.3.2.1\ 7!}\frac{(-1)}{128}x^{14}$

$= -(2)(13)(11)(2)(3)\frac{1}{128}x^{14}$

$\therefore T_8 = -\frac{429}{16}x^{14}$

Example 7: Find the coefficient of x^{19} in $\left(2x^3 - 3x\right)^9$.

Solution: Here, $a = 2x^5, b = -3x, n = 9$. First, we find r.

Since, $T_r + 1 = \left(\frac{n}{r}\right)a^{n-r}b^r$

$= \left(\frac{9}{r}\right)\left(2x^3\right)^{9-r}(-3x)^r$

$= \left(\frac{9}{r}\right)2^{9-r}(-3)^r x^{27-3r}x^r$

$= \left(\frac{9}{r}\right)2^{9-r}(-3)^r x^{27-2r}$

But we require x^{19}, so put

$19 = 27 - 2r$

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$2r = 8$

$r = 4$

Plugging the value of r in equation (1) gives

$T_{4+1} = \left(\frac{9}{4}\right)2^{9-4}(-3)^4 x^{19}$

$= \frac{9.8.7.6.5}{4.3.2.1} 2^5 . 3^4 . x^{19}$

$= 630 \times 32 \times 81 \times x^{19}$

$\therefore T_5 = 1632960x^{19}$

Example 8: Find the term independent of x in the expansion of $\left(2x^2 + \frac{1}{x}\right)^9$

Solution: We have $a = 2x^2, b = \frac{1}{x}, n = 9$

$T_r + 1 = \left(\frac{n}{r}\right)a^{n-r}b^r$

$= \left(\frac{9}{r}\right)\left(2x^2\right)^{9-r}\left(\frac{1}{x}\right)^r$

$= \left(\frac{9}{r}\right)2^{9-r} . x^{18-2r} . x^r$

Since, T_{r+1} is the term independent of, x i.e., x^0 thus, the power of x must be zero.

i.e., $18 - 3r = 0$ thus $r = 6$.

Substitute $r = 6$ into equation (2)

$T_{r+1} = \left(\frac{9}{6}\right)2^{9-6}x^0 = \frac{9!}{6!3!} . 1$

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$= \frac{9.8.7.6!}{6!3.2.1}.8.1 = 672$

8.4 Binomial Series

Since by the Binomial formula for positive integers n, we have

$$(a + b)^n = a^n + \frac{n}{1!}a^{n-1}b + \frac{n(n-1)}{2!}a^{n-2}b^2 + \frac{n(n-1)(n-2)}{3!}a^{n-3}b^3 + \ldots + b^n \qquad (1)$$

Put $a = 1$ and $b = x$, then the above form in (1) becomes:

$$(1 + x)^n = 1 + \frac{n}{1!}x + \frac{n(n-1)}{2!}x^2 + \ldots + x^n$$

if n is a negative or a fractional number (-ve or +ve), then

$$(1 + x)^n = 1 + \frac{n}{1!}x + \frac{n(n-1)}{2!}x^2 + \ldots\infty \qquad (2)$$

The series on the right-hand side of (2) is called the **Binomial Series**. This series is valid only when x is numerically less than one. That is, $|x| < 1$ otherwise the expression will not be valid.

Note: The first term in the expression must be one. For example, when n is not a positive integer (negative or fraction) to expand, $(a + x)^n$, we shall have to write it as $(a + x)^n = a^n\left(1 + \frac{x}{a}\right)^n$, and then apply the binomial series, where $\left|\frac{x}{a}\right|$ must be less than one.

8.5 Application of the Binomial Series; Approximations

The binomial series can be used to find expressions approximately equal to the given expressions under given conditions.

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Example 9: If x is very small, so that its square and higher powers can be neglected, then prove that

$\frac{1+x}{1-x} = 1 + 2x$

Solution:

$\frac{1+x}{1-x}$ can be written as $(1 + x)(1 - x)^{-1}$

$\frac{1+x}{1-x} = (1 + x)(1 + x + x^2 + \ldots higher\ powers\ of\ x)$

$= 1 + x + x + neglecting\ higher\ powers\ of\ x$

$= 1 + 2x$

Example 10:

Find to 4 decimal places the value of $(1.02)^8$

Solution:

$(1 + 0.02)^8 = 1 + \frac{8}{1}(0.02) + \frac{8.7}{2.1}(0.02)^2 + \frac{8.7.6}{3.2.1}(0.02)^3 + \ldots$

$= 1 + 0.16 + 0.0112 + 0.000448 + \ldots$

$= 1.1716$

Example 11:

Write and simplify the first four terms in the expansion of $(1 - 2x)^{-1}$.

Solution: $(1 - 2x)^{-1} = [1 + (-2x)]^{-1}$

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using $(1 + x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \ldots$

$= 1 + (-1)(-2x) + \frac{(-1)(-1-1)}{2!}(-2x)^2 + \frac{(-1)(-1-1)(-1-2)}{3!}(-2x)^3 + \ldots$

$= 1 + 2x + \frac{(-1)(-2)}{2.1}4x^2 + \frac{(-1)(-2)(-3)}{3.2.1}\left(-8x^3\right) + \ldots$

$= 1 + 2x + 4x^2 + 8x^3 + \ldots$

Example 12: Write the first three terms in the expansion of $(2 + x)^{-3}$

Solution:

$(2 + x)^{-3} = (2)^{-3}\left(1 + \frac{x}{2}\right)^{-3}$

$= (2)^{-3}\left[1 + (-3)\left(\frac{x}{2}\right) + \frac{(-3)(-3-1)}{2!}\left(\frac{x}{2}\right)^2 + \ldots\right]$

$= \frac{1}{8}\left[1 - \frac{3}{2}x + 3x^2 + \ldots\right]$

8.6 Root Extraction

The second application of the binomial series is that of finding the root of any quantity.

Example 13: Find the square root of 24 correct to 5 decimal places.

Solution:

$\sqrt{24} = (25 - 1)^{\frac{1}{2}}$

$= (25)^{\frac{1}{2}}\left(1 - \frac{1}{25}\right)^{\frac{1}{2}}$

$= 5\left(1 - \frac{1}{5^2}\right)^{\frac{1}{2}}$

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$$= 5\left[1 + \frac{1}{2}\left(-\frac{1}{5^2}\right) + \frac{\frac{1}{2}\left(\frac{1}{2}-1\right)}{2!}\left(-\frac{1}{5^2}\right)^2 + \ldots\right]$$

$$= 5\left[1 - \frac{1}{2.5^2} - \frac{1}{2^3.5^4} - \frac{1}{2^4.5^6} - \ldots\right]$$

$$= 5[1 - (0.02 + 0.002 + 0.00004\ldots.)]$$

$$= 4.89898$$

Example 14:

Evaluate $\sqrt[3]{29}$ to the nearest hundredth.

Solution:

$$\sqrt[3]{29} = (27 + 2)^{\frac{1}{3}} = \left[27\left(1 + \frac{2}{27}\right)\right]^{\frac{1}{3}} = 3\left[\left(1 + \frac{2}{27}\right)\right]^{\frac{1}{3}} + \ldots.$$

$$\sqrt[3]{29} = 3\left[1 + \frac{1}{3}\left(\frac{2}{27}\right) + \frac{\frac{1}{3}\left(\frac{1}{3}-1\right)}{2.1}\left(\frac{2}{27}\right)^2 + \ldots\right]$$

$$= 3\left[1 + \frac{2}{81} + \frac{1}{2}\left(\frac{1}{3}\right)\left(-\frac{2}{3}\right)\left(\frac{2}{27}\right)^2 + \ldots\right]$$

$$= 3[1 + 0.247 - 0.0006 + \ldots]$$

$$= 3(1.0212) = 3.07$$

Example 15: Find the binomial expansion of $(1 + 3x)^{-4}$ up to the fourth term.

Solution: Given $(1 + 3x)^{-4}$, comparing with $(1 - x)^n$ we have $3x = -x$ and $n = -4$

Using the formula:

$$(1 - x)^n = 1 - nx + \frac{n(n-1)}{2!}x^2 - \frac{n(n-1)(n-2)}{3!}x^3 + \cdots$$

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$(1 - (-3x))^{-4} =$

$1 - (-4)(-3x) + \frac{-4(-4-1)}{2!}(-3x)^2 - \frac{-4(-4-1)(-4-2)}{3!}(-3x)^3 + \cdots$

$= 1 - 12x + \frac{20}{2}(9x^2) - \frac{120}{6}(27x^3) + \cdots$

Hence, $(1 + 3x)^{-4} = 1 - 12x + 90x^2 - 540x^3 + \cdots$

Example 16: Find the binomial expansion of $(1 - 2x)^{\frac{1}{2}}$ to the fourth term

Solution: Given $(1 - 2x)^{\frac{1}{2}}$, comparing with $(1 - x)^n$ we have, x=2x and $n = \frac{1}{2}$

Using the formula:

$(1 - x)^n = 1 - nx + \frac{n(n-1)}{2!}x^2 - \frac{n(n-1)(n-2)}{3!}x^3 + \cdots$

$(1 - 2x)^{\frac{1}{2}} = 1 - (\frac{1}{2})(2x) + \frac{\frac{1}{2}(\frac{1}{2}-1)}{2!}(2x)^2 - \frac{\frac{1}{2}(\frac{1}{2}-1)(\frac{1}{2}-2)}{3!}(2x)^3 + \cdots$

$= 1 - x - \frac{1}{2}x^2 - \frac{1}{2}x^3 + \cdots$

Hence,

$(1 - 2x)^{\frac{1}{2}} = 1 - x - \frac{1}{2}x^2 - \frac{1}{2}x^3 + \cdots$

In-Text Questions and Answers (ITQs and ITAs)

ITQ 8.1: When finding a "middle term," why do you need to know if the power 'n' is even or odd?

Answer: An even 'n' gives an odd number of terms, so there is one single middle term. An odd 'n' gives an even number of terms, so there are two middle terms.

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ITQ 8.2: To find a "term independent of x," what is the key step after writing the general term?

Answer: You find the value of 'r' that makes the power of 'x' equal to zero. This isolates the constant part of the expansion.

Conclusion

In this study session, we have explored the rudiments of the binomial theorem, focusing on the general form of the expansion, the concept of the middle term and its dependence on the exponents. The binomial series emerged as a fascinating application, allowing us to approximate functions and even extract roots using infinite series. Remember, mastering the binomial theorem equips you to tackle a wide range of problems, from simplifying expressions to exploring advanced mathematical concepts.

Summary

1. The Binomial Theorem provides a formula to expand binomial expressions raised to a positive integer power n: $(a + b)^n = \sum_{j=0}^{n} \frac{n}{j} a^{(n-j)} b^j$ $= \frac{n}{0} a^{(n-0)} b^0 + \frac{n}{1} a^{(n-1)} b^1 + \frac{n}{2} a^{(n-2)} b^2 + \cdots + \frac{n}{n} a^{(n-n)} b^n$, where j goes from 0 to n.
2. To find the coefficients (n choose m), use the formula: $\frac{n}{m} = \frac{n!}{(n-m)!\, m!}$.
3. Examples are provided showing how to use the Binomial Theorem to expand expressions like $(x + y)^5$, $(2x - 3)^4$, and find coefficients of specific terms.

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4. For negative or fractional powers, the binomial expression is transformed into the form $(1 - x)^n$ and expanded using: $(1 - x)^n = 1 - nx + \frac{n(n-1)}{2!}x^2 - \frac{n(n-1)(n-2)}{3!}x^3 + \cdots$
5. The binomial theorem expansion for $(1 + 3x)^{-4}$ up to the 4th term is worked out step-by-step.
6. To expand $(1 - 2x)^{\frac{1}{2}}$ to the 4th term, using the formula for negative/fractional powers is demonstrated.
7. For fractional or negative powers, the general pattern is coefficients start 1, then n, $\frac{n(n-1)}{2!}$, $\frac{n(n-1)(n-2)}{3!}$, etc.
8. The binomial expansions have applications in areas like probability, calculus, and series expansions.

Self-Assessment Questions and Answers (SAQs and SAAs)

SAQ 8.1: Explain the connection between a binomial coefficient like (5 choose 2) and the corresponding entry in Pascal's Triangle.

Answer: The binomial coefficient (5 choose 2) represents the value in the 5th row and 2nd position of Pascal's Triangle (if we start counting from row 0 and position 0). Both represent the number of ways to choose 2 items from 5.

SAQ 8.2: Describe a real-world situation where using the Binomial Theorem for approximation is more practical than calculating an exact value.

Answer: Calculating compound interest for a very short time period, estimating the effect of a small measurement error in engineering, or approximating values

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in probability for large populations. Using the first few terms of the binomial series gives a very accurate answer much faster.

SAQ 8.3: You find the term independent of x in an expansion and get a negative number. Does this make sense? Why or why not?

Answer: Yes, it makes perfect sense. The constant term is just a number, and it can be positive, negative, or zero, depending on the signs of the terms in the binomial and the specific powers involved.

Multiple-Choice Questions and Answers (MCQs and MCAs)

1: What is the coefficient of x^3 in the expansion of $(2 + x)^5$?

a) 10

b) 20

c) 40

d) 80

Answer: c) 40

Explanation: Using the general term T_(r+1) = 5Cr (2)^(5-r) (x)^r. For x^3, set r = 3. Then $T_4 = 5C3\ (2)^2\ x^3 = 10 \times 4 \times x^3 = 40x^3$. Therefore, the coefficient is 40.

2: How many terms are there in the expansion of $(a + b)^{12}$?

a) 11

b) 12

c) 13

d) 14

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Answer: c) 13

Explanation: In the binomial expansion of (a + b)^n, the number of terms is always (n + 1). Here n = 12, so there are 13 terms.

3: If n is odd, how many middle terms are there in the expansion of (a + b)^n?

a) 1

b) 2

c) 3

d) 0

Answer: b) 2

Explanation: When n is odd, (n + 1) is even, so there are two middle terms: the ((n+1)/2)th term and the ((n+1)/2 + 1)th term.

4: For the binomial series (1 + x)^n to be valid when n is negative, what condition must x satisfy?

a) x > 1

b) x < -1

c) |x| < 1

d) x = 0

Answer: c) |x| < 1

Explanation: The binomial series expansion for negative or fractional n converges only when |x| < 1. Otherwise, the series diverges and does not represent the function.

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5: What is the term independent of x in the expansion of $(x + 1/x)^6$?

a) 6

b) 15

c) 20

d) 30

Answer: c) 20

Explanation: General term: T_(r+1) = 6Cr x^(6-r) (1/x)^r = 6Cr x^(6-2r). For the term independent of x, set 6 - 2r = 0, so r = 3. Then T_4 = 6C3 = 20. The constant term is 20.

Tutor-Marked Assessments (TMAs)

1. Expand up to four terms

   (a) $(1 - 3x)^{\frac{1}{3}}$ (b) $(1 - 2x)^{-\frac{3}{4}}$ © $(1 + x)^{-3}$ (d) $\frac{1}{\sqrt{1+x}}$

2. Using the binomial expansion, calculate to the nearest hundredth

   (a) $\sqrt[4]{65}$ (b) $\sqrt{17}$ © $(1.01)^{-7}$ (d) $\sqrt{28}$

3. Find the coefficient of x^5 in the expansion of

   (a) $\frac{(1+x)^2}{(1-x)^2}$ (b) $\frac{(1+x)^2}{(1-x)^3}$

4. Determine the middle term(s) in the expansion of $(\frac{x}{y} - 2y)^8$

5. a) Write down the first four terms of the binomial series expansion of $(1 + x)^{-3}$

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Study Session Nine

Conditional Probability and Independence

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Introduction

A quadratic is an expression of the form $ax^2 + bx + c$, where a, b and c are given numbers and $a \neq 0$. The **standard form** of a **quadratic equation** is an equation of the form $ax^2 + bx + c = 0$, where a, b and c are given numbers and $a \neq 0$.

We seek to find the value(s) of which make the statement true or to show that there are no such values. Thus, for example, the values $x = 3$ and $x = 2$ satisfy the equation, $x^2 - 5x + 6 = 0$. This is easily checked by substitution.

These values are called the **solutions** of the equation. Linear equations that are written in the standard form $ax + b = 0$, $a \neq 0$, have one solution. Quadratic equations may have no solution or, as in the above example, two solutions.

Learning Outcomes for Study Session 9

When you have studied this session, you should be able to:

9.1. Solve quadratic equations with or without a term in x.

9.2. Identify solutions to equations.

9.3. Simplify equations using the method of completing the square.

9.4. Factor quadratic equations using formulas.

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9.1 Quadratic Equations with no Term in X

When there is no term in, x we can move the constant to the other side.

Example 1: Solve $x^2 - 9 = 0$.

Solution:

$x^2 - 9 = 0$

$x^2 = 9$

$x = 3 \; or \; x = -3$

Example 2: Solve $x^2 - 9x = 0$

In this case, we can write $x(x - 9) = 0$. Since the product of the two factors is 0, one or both of the factors is zero, $x(x - 9) = 0$.

so, $x = 0 \; or \; x - 9 = 0$

hence, the two solutions are $x = 0 \; or \; x = 9$.

These two methods work just as well when the coefficient of x^2 is not one. The two previous examples were relatively easy since, in the first case, it was easy to isolate the unknown, while in the second, a common factor enabled the left-hand side to be easily factored.

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9.2 Solving Quadratic Equations with Three Terms

We will now deal with the equation $ax^2 + bx + c = 0$ in which neither a nor b nor c are zero.

There are three basic methods of solving such quadratic equations:

- By factoring
- By completing the square and
- By the quadratic formula
- Factorisation Method:

Example 3:

Solve $x^2 - 7x + 12 = 0$ by the factorisation method

Solution: We factor the left-hand side by finding two numbers whose product is 12 and whose sum is -7. Clearly, $-4, \quad -3$ are the desired numbers. We can then factor as:

$x^2 - 7x + 12 = 0$

$(x - 4)(x - 3) = 0$

Since the product of the two factors is zero, one of the factors is zero.

Thus,

$x - 4 = 0$ or $x - 3 = 0$

so, $x = 4$ or $x = 3$

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The same method can also be applied to non-monic quadratic equations. A non-monic quadratic equation is an equation of the form $ax^2 + bx + c = 0$, where $a \neq 0$. this is the general case.

Thus, $2x^2 + 5x + 3 = 0$ is an example of a non-monic quadratic equation.

Example 4:

Solve the equation $2x^2 + 5x + 3 = 0$.

Solution:

Using the factoring method from the module 'Factorisation', we multiply 2 and 3 to give 6 and find two numbers that multiply to give 6 and add to give 5. The desired numbers are 2 and 3. We use these numbers to split the middle term and factor in pairs.

$$2x^2 + 5x + 3 = 0$$

$$2x^2 + 2x + 3x + 3 = 0$$

$$2x(x + 1) + 3(x + 1) = 0$$

$$(x + 1)(2x + 3) = 0$$

we can now equate each factor to zero and obtain

$$x + 1 = 0 \; or \; 2x + 3 = 0$$

$$x = -1 \; or \; x = -\frac{3}{2}.$$

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9.3 Completing the Square

The quadratic equations encountered so far had one or two solutions that were rational. There are many quadratics that have irrational solutions, or in some cases no real solutions at all. For example, it is not easy at all to see how to factor the quadratic $x^2 - 5x - 3 = 0$. Indeed, it has no rational solutions. We will see shortly that the solutions are $x = \frac{5+\sqrt{37}}{2}$ and $x = \frac{5-\sqrt{37}}{2}$.

To deal with more general quadratics, we employ a technique known as **completing the square**. Historically, this was the most commonly used method of solution.

Example 5: Solve $x^2 + 2x - 6 = 0$.

Solution: It is easiest to move the constant term onto the other side first and them complete the square

$x^2 + 2x - 6 = 0$

$x^2 + 2x = 6$

$x^2 + 2x + 1 = 7$

$(x + 1)^2 = 7$

we can now take the positive and negative square roots to obtain

$x + 1 = \sqrt{7}\ or\ x + 1 = -\sqrt{7}$

So, $x = -1 + \sqrt{7}$ or $x = -1 - \sqrt{7}$.

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Notice that the solutions are irrational, and so this equation could not be easily solved using the factoring method.

Example 6: Solve $x^2 - 6x - 2 = 0$

Solution:

$x^2 - 6x - 2 = 0$

$x^2 - 6x + 9 - 9 - 2 = 0\ (Complete\ the\ square)$

$(x - 3)^2 = 11$

$x - 3 = \sqrt{11}\ or\ x - 3 = -\sqrt{11}$

Hence, $x = 3 + \sqrt{11}$ or $x = 3 - \sqrt{11}$.

Example 7: Solve $3x^2 - 5x + 1 = 0$.

Solution:

Divide the equation by 3 and shift the constant term to the other side.

$3x^2 - 5x + 1 = 0$

$x^2 - \frac{5}{3}x = -\frac{1}{3}$

$x^2 - \frac{5}{3}x + \frac{25}{36} = \frac{25}{36} - \frac{1}{3}$

$\left(x - \frac{5}{3}\right)^2 = \frac{13}{36}$

so, $x = \frac{5+\sqrt{13}}{6}$ or $x = \frac{5-\sqrt{13}}{6}$

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9.4 The Quadratic Formula.

The method of completing the square always works. By applying it to the general quadratic equation $ax^2 + bx + c = 0$, we obtain the well-known **quadratic formula.**

Applying the method of completing the square to the aforementioned general form of a quadratic equation yields:

$x = \frac{-b+\sqrt{b^2-4ac}}{2a}$ or $x = \frac{-b-\sqrt{b^2-4ac}}{2a}$.

The last formula is called the quadratic formula, sometimes written as $x = \frac{-b\pm\sqrt{b^2-4ac}}{2a}$. if the quantity, $b^2 - 4ac = 0$ then there will only be one solution. $x = -\frac{b}{2a}$. In this case, the quadratic will be a perfect square. The quantity $b^2 - 4ac$ plays an important role in the **theory of quadratic equations** and is called the discriminant. Thus, in summary, when solving, $ax^2 + bx + c = 0$, first calculate the discriminant. $b^2 - 4ac$. Then

a. If $b^2 - 4ac$ is negative, then there is no solution.
b. If $b^2 - 4ac$ is positive, then the solutions are $x = \frac{-b+\sqrt{b^2-4ac}}{2a}$, $x = \frac{-b-\sqrt{b^2-4ac}}{2a}$.
c. If $b^2 - 4ac$ is zero, then there is only one solution $x = -\frac{b}{2a}$.

Note: if $b^2 - 4ac$ is zero, then the quadratic is a perfect square.

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Example 8: Solve $x^2 - 10x - 3 = 0$ by using the quadratic formula

Solution: Here, $a = 1, b = -10$ and $c = -3$,

Now, $b^2 - 4ac = 100 + 12 = 112$.

$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

$x = \frac{10 \pm \sqrt{112}}{2}$

$= \frac{10 \pm 4\sqrt{7}}{2}$

$x = 5 + 2\sqrt{7}$ *or* $x = 5 - 2\sqrt{7}$

In-Text Questions and Answers (ITQs and ITAs)

ITQ 9.1: For the equation x (x - 5)what property allows us to conclude that x=0 or x =5?

Answer: The Zero Product Property: if the product of two factors is zero, then at least one of the factors must be zero.

ITQ 9.2: What is the defining characteristic of a "quadratic equation" in its standard form?

Answer: It is an equation of the form $ax^2 + bx + c = 0$, where a, b, c are constants and a is not equal to zero.

Conclusion

In this session, we have explored the fundamental concepts of quadratic equations and the various methods for solving them. We began with simple

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cases where the equation has no x term or no constant term, which can be solved by isolating x^2 or factoring out a common factor. We then examined the three primary methods for solving general quadratic equations: factoring, completing the square, and the quadratic formula. Each method offers distinct advantages. Factoring is the most efficient when the quadratic can be expressed as a product of two binomials with integer coefficients. Completing the square provides a deeper understanding of the equation's structure and is essential for transforming quadratics into vertex form. The quadratic formula, derived from completing the square on the general form, serves as a universal tool that works for any quadratic equation, regardless of whether its solutions are rational, irrational, or non-real. The discriminant, $b^2 - 4ac$, gives us immediate insight into the nature of the solutions without performing the full calculation. As you progress in your mathematical studies, you will find that these techniques extend beyond quadratics and form the foundation for understanding more advanced topics such as conic sections, optimisation problems, and even aspects of calculus. Mastery of these methods is not just about solving equations—it is about developing a flexible and strategic approach to mathematical problem-solving.

Summary

1. A quadratic equation is an expression of the form $ax^2 + bx + c = 0$, where a, b, and c are given numbers and $a \neq 0$.
2. The standard form of a quadratic equation is $ax^2 + bx + c = 0$, which helps in finding the solutions.

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3. The solutions of a quadratic equation are the values of x that satisfy the equation.
4. Quadratic equations can have no solution or two solutions, unlike linear equations, which have one solution.
5. Quadratic equations with no term in x can be solved by moving the constant to the other side.
6. Quadratic equations with a term in x can be solved by factoring or using the zero-product property.
7. Non-monic quadratic equations (coefficients of x^2 not equal to one) can also be solved using the factoring method.

Self-Assessment Questions and Answers (SAQs and SAAs)

SAQ 9.1: Explain the main advantage of using the quadratic formula over the factoring method.

Answer: The quadratic formula always works for any quadratic equation, even when the solutions are irrational or complex. Factoring is often faster but only works easily when the solutions are rational numbers.

SAQ 9.2: If the discriminant of a quadratic equation is a perfect square number (like 1, 4, 9, 16...), what does that tell you about the solutions and the best method to solve it?

Answer: It indicates that the two solutions are rational numbers. This is a strong hint that the equation can be solved efficiently by the factoring method.

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SAQ 9.3: What does the discriminant (b^2-4ac) of a quadratic equation tell you about its solutions?

Answer: It indicates the nature and number of the solutions: positive means two distinct real solutions, zero means one repeated real solution, and negative means no real solutions (two complex solutions).

Multiple-Choice Questions and Answers (MCQs and MCAs)

1: The solutions to the equation $x^2 + 8x + 16 = 0$ are best described as:

a) Two distinct irrational numbers

b) Two distinct rational numbers

c) One repeated rational number (a double root)

d) Non-real numbers

Answer: c) One repeated rational number (a double root)

Explanation: The equation $x^2 + 8x + 16 = 0$ can be factored as $(x + 4)(x + 4) = 0$ or $(x + 4)^2 = 0$. This gives a single solution $x = -4$, which occurs twice (a repeated root). The discriminant is $b^2 - 4ac = 64 - 64 = 0$, which confirms there is exactly one real solution (a double root). Since -4 is a rational number, the correct description is one repeated rational number.

2: The first step in solving $2x^2 - 8x + 3 = 0$ by completing the square is to:

a) Take the square root of both sides

b) Divide the entire equation by 2

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c) Apply the quadratic formula

d) Add 8x to both sides

Answer: b) Divide the entire equation by 2

Explanation: When completing the square, the coefficient of x^2 should be 1 to simplify the process. The equation $2x^2 - 8x + 3 = 0$ has a coefficient of 2 for x^2. The first step is to divide every term by 2 to obtain $x^2 - 4x + 1.5 = 0$ (or $x^2 - 4x = -1.5$ if moving the constant). This makes the quadratic monic and ready for completing the square.

3: The discriminant of the quadratic equation $3x^2 - 6x + 3 = 0$ is:

a) 72

b) -72

c) 0

d) 36

Answer: c) 0

Explanation: The discriminant is calculated using the formula $\Delta = b^2 - 4ac$. For the equation $3x^2 - 6x + 3 = 0$, we have $a = 3$, $b = -6$, and $c = 3$. Substituting these values: $\Delta = (-6)^2 - 4(3)(3) = 36 - 36 = 0$. A discriminant of zero indicates that the quadratic has exactly one real solution (a repeated root). Indeed, $3x^2 - 6x + 3 = 0$ can be factored as $3(x - 1)^2 = 0$, giving $x = 1$ as a double root.

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4: A quadratic equation has a discriminant value of -15. This means the equation has:

a) One rational solution

b) Two rational solutions

c) Two irrational solutions

d) No real solutions

Answer: d) No real solutions

Explanation: The discriminant ($\Delta = b^2 - 4ac$) determines the nature of the solutions of a quadratic equation. If Δ is negative (less than zero), the quadratic has no real solutions. Instead, it has two complex (non-real) solutions. In this case, $\Delta = -15$, which is negative, so the equation has no real solutions. The square root of a negative number is not a real number; thus, the quadratic formula yields non-real answers.

5: The most efficient first step to solve $3x^2 - 75 = 0$ is to:

a) Use the quadratic formula

b) Complete the square

c) Add 75 to both sides and then divide by 3

d) Factor out an x

Answer: c) Add 75 to both sides and then divide by 3

Explanation: The equation $3x^2 - 75 = 0$ has no x term ($b = 0$), so it is a special case that can be solved more efficiently than by using the quadratic formula or completing the square. Adding 75 to both sides gives $3x^2 = 75$. Dividing both

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sides by 3 yields $x^2 = 25$. Taking the square root gives $x = 5$ or $x = -5$. This approach is faster and simpler than the other methods, which would work but are unnecessary for this form of equation.

Tutor-Marked Assessments (TMAs)

1. Solving the equations:
    a. $4x^2 - 20 = 0$
    b. $x^2 - x - 12 = 0$
    c. $3x^2 + 2x - 8 = 0$
2. Solve $x^2 - 5x - 3 = 0$ by completing the square and also show that $x^2 - 5x + 7 = 0$ has no solutions.
3. The area of a rectangle is 54 cm^2. Its length is 3 cm more than its width. Set up a quadratic equation to represent this situation and solve it to find the dimensions of the rectangle.
4. Solve the equation $x^2 - 2x - 8 = 0$ using both the factoring method and the quadratic formula. Verify that both methods yield the same solution set.
5. Solve the equation $2x^2 - 12x + 5 = 0$ by the method of completing the square. Show every step in the process.

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Study Session Ten

Recurrence Relations

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Introduction

A recursive definition of a sequence specifies one or more initial terms and a rule for determining subsequent terms from those that precede them. Also, recall that a rule of the latter sort (whether or not it is part of a recursive definition) is called a **recurrence relation** and that a sequence is called a solution of a recurrence relation if its terms satisfy the recurrence relation.

In this section, we will show that such relations can be used to study and to solve counting problems. For example, suppose that the number of bacteria in a colony doubles every hour. If a colony begins with five bacteria, how many will be present in *n* hours? To solve this problem, let a_n be the number of bacteria at the end of *n* hours. Because the number of bacteria doubles every hour, the relationship $a_n = 2a_n - 1$ holds whenever *n* is a positive integer. This recurrence relation, together with the initial condition $a_0 = 5$, uniquely determines a_n for all nonnegative integers *n*. We can find a formula for a_n using the iterative approach, namely that $a_n = 5 \cdot 2^n$ for all nonnegative integers *n.*

Some of the counting problems that cannot be solved using the direct and explicit techniques can be solved by finding recurrence relations involving the terms of a sequence, as was done in the problem involving bacteria. In this section we will study a variety of counting problems that can be modelled using recurrence relations. We will study methods for finding explicit formulae for the terms of sequences that satisfy certain types of recurrence relations.

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Learning Outcomes for Study Session 10

When you have studied this session, you should be able to:

10.1. Identify the linear homogeneous recurrence relation.

10.2. Solve the linear homogeneous recurrence relation.

10.3. Define the factorial of a positive integer and its properties.

10.4. Apply the binomial theorem to expand expressions of the form $a+b^n$, where n is a positive integer power.

10.5. Identify the middle term in the binomial expansion for even and odd values of n.

10.6. Determine the general term in the binomial expansion.

10.7. Utilise the binomial series to approximate expressions.

10.8. Apply the binomial series to derive approximations for given expressions under specific conditions.

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10.1 Factorial of a Positive Integer

If n is a positive integer, the factorial of n denoted by $n!$ is defined as the product of n +ve integers from n to 1. That is $n! = n(n-1)(n-2)...3.2.1$

for example, $4! = 4.3.2.1 = 24$ and $6! = 6.5.4.3.2.1 = 720.$

one important relationship concerning factorials is that

$$(n+1)! = (n+1)n! \qquad (1)$$

for instance, $5! = 5.4.3.2.1$

$= 5(4.3.2.1)$

$5! = 5.4!$

obviously, $1! = 1$ and this permits to define from equation (1)

$$n! = \frac{(n+1)!}{(n+1)}$$

substituting 0 for n, we obtain

$$0! = \frac{(0+1)!}{(0+1)} = \frac{1!}{1} = \frac{1}{1}$$

$0! = 1$

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10.2 Combination

Each of the groups or selections which can be made out of a given number of things by taking some or all of them at a time is called a combination. In combination, the order in which things occur is not considered, e.g., combination of a, b, c taken two at a time are ab, bc, ca.

The number of combinations of n different objects taken r at time is denoted by $\left(\frac{n}{r}\right)$ and defined as

$$\left(\frac{n}{r}\right) = \frac{n!}{r!(n-r)!}$$

For example,

$$\left(\frac{6}{4}\right) = \frac{6!}{4!(6-4)!}$$

$$= \frac{6\times5\times4}{4!\times2!} = \frac{6\times5\times4!}{4!\times2!} = 15$$

Example 1: Expand $\left(\frac{7}{3}\right)$

Solution: $\left(\frac{7}{3}\right) = \frac{7!}{3!(7-3)!}$

$$= \frac{7.6.5.4!}{3.2.1.4!} = 35.$$

Some important results:

(i) $\left(\frac{n}{0}\right) = \frac{n!}{0!(n-0)!} = \frac{n!}{1\times n!} = 1$

(ii) $\left(\frac{n}{n}\right) = \frac{n!}{n!(n-n)!} = \frac{n!}{n!\times0!} = \frac{n!}{n!\times1} = 1$

(iii) $\left(\frac{n}{r}\right) = \left(\frac{n}{n-r}\right)$

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Note: The number $\left(\frac{n}{r}\right)$ is called the binomial coefficient.

10.3 The Binomial Theorem

The rule or formula for expansion, $(a + b)^n$, where n is any positive integer power is called the binomial theorem.

For any positive integer n

$$(a + b)^n = \left(\frac{n}{0}\right)a^n + \left(\frac{n}{1}\right)a^{n-1}b + \left(\frac{n}{2}\right)a^{n-2}b^2 + \ldots + \left(\frac{n}{r}\right)a^{n-r}b^r + \ldots + \left(\frac{n}{n}\right)b^n$$

or briefly $(a + b)^n = \sum_{r=0}^{n}\left(\frac{n}{r}\right)a^{n-r}b^r$

Remarks: The coefficients of the successive terms are $\left(\frac{n}{0}\right)$, $\left(\frac{n}{1}\right)$, $\left(\frac{n}{2}\right)$,..., $\left(\frac{n}{r}\right)$,..., $\left(\frac{n}{n}\right)$ are called binomial coefficients.

Note: The sum of binomial coefficients is 2^n.

$$(a + b)^n = a^n + \frac{n}{1!}a^{n-1}b + \frac{n(n-1)}{2!}a^{n-2}b^{2+\ldots+\frac{n(n-1)(n-2)\ldots(n-r+1)}{r!}}a^{n-r}b^r + \ldots + b^n$$

Note: Since

$$\left(\frac{n}{r}\right) = \frac{n!}{r!(n-r)!}$$

so,

$$\left(\frac{n}{0}\right) = \frac{n!}{0!(n-0)!} = \frac{n!}{1\times n!} = 1$$

$$\left(\frac{n}{1}\right) = \frac{n!}{1!(n-1)!} = \frac{n(n-1)!}{1!(n-1)!} = \frac{n(n-1)}{2!}$$

$$\left(\frac{n}{2}\right) = \frac{n!}{2!(n-2)!} = \frac{n(n-2)!}{2!(n-2)!} = \frac{n(n-1)}{2!}$$

$$\left(\frac{n}{3}\right) = \frac{n!}{3!(n-3)!} = \frac{n(n-1)(n-2)(n-3)!}{3!(n-3)!} = \frac{n(n-1)(n-2)}{3!}$$

$$\left(\frac{n}{r}\right) = \frac{n(n-1)(n-2)!\ldots(n-r+1)(n-r)!}{r!(n-r)!}$$

$$= \frac{n(n-1)(n-2)!\ldots(n-r+1)}{r!}$$

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$\binom{n}{n} = \frac{n!}{n!(n-n)!} = \frac{n!}{n! \times 1} = 1.$

The following points can be observed in the expansion of $(a + b)^n$

- There are $(n + 1)$ terms in the expansion
- The 1st term is a^n the $(n + 1)$th term or the last term is b^n
- The exponent of a decreases from n to zero
- The exponent of b increases from 0 to n.
- The sum of the exponents of a and b in any term is equal to the index n.
- The coefficient of the term equidistant from the beginning and end of the expansion are equal as $\binom{n}{r} = \binom{n}{n-r}$.

10.4 General Term

The term $\binom{n}{r}a^{n-r}b^r$ in the expansion of the binomial theorem is called the **general term** or $(r + 1)$th term. It is denoted by T_{r+1}. Hence,

$$T_{r+1} = \binom{n}{r}a^{n-r}b^r$$

Note that the general term is used to find out the specified term or the required coefficient of the term in the binomial expansion.

Example 2: Expand $(x + y)^4$

Solution:

$$(x + y)^4 = x^4 + \binom{4}{1}x^{4-1}y + \binom{4}{2}x^{4-2}y^2 + \binom{4}{3}x^{4-3}y^3 + y^4$$

$$= x^4 + 4x^3y + \frac{4\times3}{2\times1}x^2y^2 + \frac{4\times3\times2}{3\times2\times1}xy^3 + y^4$$

$$= x^4 + 4x^3y + 6x^2y^2 + 4xy^3 + y^4$$

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Example 3: Expand by binomial theorem $\left(a - \frac{1}{a}\right)^6$

Solution:

$$\left(a - \frac{1}{a}\right)^6 = a^6 + \left(\frac{6}{1}\right)a^{6-1}\left(-\frac{1}{a}\right)^1 + \left(\frac{6}{2}\right)a^{6-2}\left(-\frac{1}{a}\right)^2 + \left(\frac{6}{3}\right)a^{6-3}\left(-\frac{1}{a}\right)^3 + \left(\frac{6}{4}\right)a^{6-4}\left(-\frac{1}{a}\right)^4$$

$$+ \left(\frac{6}{5}\right)a^{6-5}\left(-\frac{1}{a}\right)^5 + \left(\frac{6}{6}\right)a^{6-6}\left(-\frac{1}{a}\right)^6$$

$$= a^6 + 6a^5\left(-\frac{1}{2}\right) + \frac{6\times5}{2\times1}a^4\left(-\frac{1}{a^2}\right) + \frac{6\times5\times4}{3\times2\times1}a^3\left(-\frac{1}{a^3}\right) + \frac{6\times5\times4\times3}{4\times3\times2\times1}a^2\left(-\frac{1}{a^4}\right)$$

$$= \frac{6\times5\times4\times3\times2}{5\times4\times3\times2\times1}a^1\left(-\frac{1}{a^5}\right) + \left(-\frac{1}{a^6}\right)$$

$$= a^6 - 6a^4 + 15a^4 - 20 + \frac{15}{a^2} - \frac{6}{a^5} + \frac{1}{a^6}$$

Example 4: Expand $\left(\frac{x^2}{2} - \frac{2}{x}\right)^4$

Solution:

$$\left(\frac{x^2}{2} - \frac{2}{x}\right)^4 = \left(\frac{x^2}{2}\right)^4 + \left(\frac{4}{1}\right)\left(\frac{x^2}{2}\right)^{4-1}\left(-\frac{2}{x}\right)^1 + \left(\frac{4}{2}\right)\left(\frac{x^2}{2}\right)^{4-2}\left(-\frac{2}{x}\right)^2 + \left(\frac{4}{3}\right)\left(\frac{x^2}{2}\right)^{4-3}\left(-\frac{2}{x}\right)^3$$

$$+ \left(\frac{4}{4}\right)\left(\frac{x^2}{2}\right)^{4-4}\left(-\frac{2}{x}\right)^4$$

$$= \frac{x^4}{16} + 4\left(\frac{x^2}{2}\right)^3\left(-\frac{2}{x}\right) + \frac{4\times3}{2\times1}\left(\frac{x^2}{2}\right)^2\left(-\frac{4}{x^2}\right) + \frac{4\times3\times2}{3\times2\times1}\left(\frac{x^2}{2}\right)^1\left(-\frac{4}{x^3}\right) + \frac{16}{x^4}$$

$$= \frac{x^8}{16} - 4.\frac{x^8}{8}\frac{2}{x} + 6.\frac{x^4}{4}\frac{4}{x^2} - 4.\frac{x^2}{2}\frac{8}{x^3} + \frac{16}{x^4}$$

$$= \frac{x^8}{16} - x^2 + 6x^2 - \frac{16}{x} + \frac{16}{x^4}$$

Example 5: Expand $(1.04)^5$ by the binomial formula and find its value to two decimal places.

Solution:

$$(1.04)^5 = (1 + 0.04)^5$$

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$$(1 + 0.04)^5 = (1)^5 + \left(\frac{5}{1}\right)(1)^{5-1}(0.04) + \left(\frac{5}{2}\right)(1)^{5-2}(0.04)^2 + \left(\frac{5}{3}\right)(1)^{5-3}(0.04)^3 + \left(\frac{5}{4}\right)(1)^{5-4}($$

$$= 1 + 0.2 + 0.016 + 0.00064 + 0.000128 + 0.0000001024$$

$$= 1.22$$

Example 6: Find the eighth term in the expansion of $\left(2x^2 - \frac{1}{x^2}\right)^{12}$

The general term is, $T_{r+1} = \left(\frac{n}{r}\right)a^{n-r}b^r$

Here, $T_8 = ?, a = 2x^2, b = -\frac{1}{x^2}, n = 12, r = 7.$

Therefore,

$$T_{7+1} = \left(\frac{12}{7}\right)\left(2x^2\right)^{12-7}\left(-\frac{1}{x^2}\right)^7$$

$$T_8 = \frac{12.11.10.9.8.7.6}{7.6.5.4.3.2.1}\left(2x^2\right)^5\frac{(-1)^7}{x^{14}}$$

$$= 793 \times 32x^{10}\frac{(-1)}{x^{14}}$$

$$\therefore T_8 = -\frac{25344}{x^4}$$

10.5 Middle Term in the Expansion $(a + b)^n$

In the expansion of $(a + b)^n$, there are $(n + 1)$ terms.

Case I: If n is even, then $(n + 1)$ will be odd, so $\left(\frac{n}{2} + 1\right)$th term will be the only one middle term in the expansion. For example, if $n = 8$ (even), number of terms will be 9 odd; therefore, $\left(\frac{8}{2} + 1\right) = 5^{\text{th}}$ will be the middle term.

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Case II: If n **is odd,** then $(n+1)$ will be even; in this case, there will not be a single middle term, but $\left(\frac{n+1}{2}\right)$th and $\left(\frac{n+1}{2}+1\right)$th term will be the two middle terms in the expansion. For example, for $n=9$ (odd), number of terms is 10 i.e., $\left(\frac{9+1}{2}\right)$th and $\left(\frac{9+1}{2}+1\right)$th i.e., $5th$ and $6th$ terms are taken as middle terms and these middle terms are found by using the for the general term.

Example 7: Find the middle term of $\left(1-\frac{x^2}{2}\right)^{14}$

Solution: we have $n=14$, then number of terms is 15.

$\therefore\left(\frac{14}{2}+1\right)$ i.e., $8th$ will be the middle term.

$a=1,\ b=-\frac{x^2}{2}, n=14,\ r=7,\ T_8=?$

$T_{r+1}=\left(\frac{n}{r}\right)a^{n-r}b^r$

$T_{r+1}=\left(\frac{14}{7}\right)(1)^{14-7}\left(-\frac{x^2}{2}\right)^7=\frac{14!}{7!7!}(-1)^7\frac{x^{14}}{2^7}$

$=\frac{14.13.12.11.10.9.8.7!}{7.6.5.4.3.2.1\ 7!}\frac{(-1)}{128}x^{14}$

$=-(2)(13)(11)(2)(3)\frac{1}{128}x^{14}$

$\therefore T_8=-\frac{429}{16}x^{14}$

Example 8: Find the coefficient of x^{19} in $\left(2x^3-3x\right)^9$.

Solution: Here, $a=2x^5, b=-3x, n=9$

first, we find r.

Since,

$T_r+1=\left(\frac{n}{r}\right)a^{n-r}b^r$

$=\left(\frac{9}{r}\right)\left(2x^3\right)^{9-r}(-3x)^r$

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$= \left(\frac{9}{r}\right)2^{9-r}(-3)^{r}x^{27-3r}x^{r}$

$= \left(\frac{9}{r}\right)2^{9-r}(-3)^{r}x^{27-2r}$

But we require x^{19}, so put

$19 = 27 - 2r$

$2r = 8$

$r = 4$

Plugging the value of r in equation (1) gives

$T_{4+1} = \left(\frac{9}{4}\right)2^{9-4}(-3)^{4}x^{19}$

$= \frac{9.8.7.6.5}{4.3.2.1}2^{5}.3^{4}.x^{19}$

$= 630 \times 32 \times 81 \times x^{19}$

$\therefore T_{5} = 1632960x^{19}$

Example 9: Find the term independent of x in the expansion of $\left(2x^{2} + \frac{1}{x}\right)^{9}$

Solution:

We have $a = 2x^{2}, b = \frac{1}{x}, n = 9$

$T_{r} + 1 = \left(\frac{n}{r}\right)a^{n-r}b^{r}$

$= \left(\frac{9}{r}\right)\left(2x^{2}\right)^{9-r}\left(\frac{1}{x}\right)^{r}$

$= \left(\frac{9}{r}\right)2^{9-r}.x^{18-2r}.x^{r}$

Since T_{r+1} is the term independent of x i.e, x^{0} thus the power of x must be zero.

i.e., $18 - 3r = 0$ thus $r = 6$.

Substitute $r = 6$ into equation (2)

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$$T_{r+1} = \left(\frac{9}{6}\right)2^{9-6}x^{0} = \frac{9!}{6!3!}.1$$

$$= \frac{9.8.7.6!}{6!3.2.1}.8.1 = 672$$

10.6 Binomial Series.

Since by the Binomial formula for positive integers n, we have

$$(a + b)^{n} = a^{n} + \frac{n}{1!}a^{n-1}b + \frac{n(n-1)}{2!}a^{n-2}b^{2} + \frac{n(n-1)(n-2)}{3!}a^{n-3}b^{3} + \ldots + b^{n} \qquad (1)$$

Put $a = 1$ and $b = x$, then the above form in (1) becomes:

$$(1 + x)^{n} = 1 + \frac{n}{1!}x + \frac{n(n-1)}{2!}x^{2} + \ldots + x^{n}$$

if n is a negative or a fractional number (-ve or +ve), then

$$(1 + x)^{n} = 1 + \frac{n}{1!}x + \frac{n(n-1)}{2!}x^{2} + \ldots\infty \qquad (2)$$

The series on the right-hand side of (2) is called the **Binomial Series**. This series is valid only when x is numerically less than one. That is, $|x| < 1$ otherwise, the expression will not be valid.

Note: The first term in the expression must be one. For example, when n is not a positive integer (negative or fraction) to expand, $(a + x)^{n}$, we shall have to write it as $(a + x)^{n} = a^{n}\left(1 + \frac{x}{a}\right)^{n}$ and then apply the binomial series, where $\left|\frac{x}{a}\right|$ must be less than one.

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10.7 Application of the Binomial Series; Approximations

The binomial series can be used to find expressions approximately equal to the given expressions under given conditions.

Example 10: If x is very small, so that its square and higher powers can be neglected, then prove that $\frac{1+x}{1-x} = 1 + 2x$

Solution: $\frac{1+x}{1-x}$ can be written as $(1 + x)(1 - x)^{-1}$

$\frac{1+x}{1-x} = (1 + x)(1 + x + x^2 + \ldots higher\ powers\ of\ x)$

$= 1 + x + x + neglecting\ higher\ powers\ of\ x$

$= 1 + 2x$

Example 11: Find to 4 decimal places the value of $(1.02)^8$

Solution: $(1 + 0.02)^8 = 1 + \frac{8}{1}(0.02) + \frac{8.7}{2.1}(0.02)^2 + \frac{8.7.6}{3.2.1}(0.02)^3 + \ldots$

$= 1 + 0.16 + 0.0112 + 0.000448 + \ldots$

$= 1.1716$

Example 12:

Write and simplify the first four terms in the expansion of $(1 - 2x)^{-1}$.

Solution:

$(1 - 2x)^{-1} = [1 + (-2x)]^{-1}$

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using $(1 + x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \ldots$

$= 1 + (-1)(-2x) + \frac{(-1)(-1-1)}{2!}(-2x)^2 + \frac{(-1)(-1-1)(-1-2)}{3!}(-2x)^3 + \ldots$

$= 1 + 2x + \frac{(-1)(-2)}{2.1}4x^2 + \frac{(-1)(-2)(-3)}{3.2.1}\left(-8x^3\right) + \ldots$

$= 1 + 2x + 4x^2 + 8x^3 + \ldots$

Example 13: Write the first three terms in the expansion of $(2 + x)^{-3}$

Solution: $(2 + x)^{-3} = (2)^{-3}\left(1 + \frac{x}{2}\right)^{-3}$

$= (2)^{-3}\left[1 + (-3)\left(\frac{x}{2}\right) + \frac{(-3)(-3-1)}{2!}\left(\frac{x}{2}\right)^2 + \ldots\right]$

$= \frac{1}{8}\left[1 - \frac{3}{2}x + 3x^2 + \ldots\right]$

10.8 Root Extraction

The second application of the binomial series is that of finding the root of any quantity.

Example 10.14: Find the square root of 24 correct to 5 decimal places.

Solution: $\sqrt{24} = (25 - 1)^{\frac{1}{2}}$

$= (25)^{\frac{1}{2}}\left(1 - \frac{1}{25}\right)^{\frac{1}{2}}$

$= 5\left(1 - \frac{1}{5^2}\right)^{\frac{1}{2}}$

$= 5\left[1 + \frac{1}{2}\left(-\frac{1}{5^2}\right) + \frac{\frac{1}{2}\left(\frac{1}{2}-1\right)}{2!}\left(-\frac{1}{5^2}\right)^2 + \ldots\right]$

$= 5\left[1 - \frac{1}{2.5^2} - \frac{1}{2^3.5^4} - \frac{1}{2^4.5^6} - \ldots\right]$

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$= 5[1 - (0.02 + 0.002 + 0.00004....)]$

$= 4.89898$

Example 15: Evaluate $\sqrt[3]{29}$ to the nearest hundredth.

Solution: $\sqrt[3]{29} = (27 + 2)^{\frac{1}{3}} = \left[27\left(1 + \frac{2}{27}\right)\right]^{\frac{1}{3}} = 3\left[\left(1 + \frac{2}{27}\right)\right]^{\frac{1}{3}} + ...$

$\sqrt[3]{29} = 3\left[1 + \frac{1}{3}\left(\frac{2}{27}\right) + \frac{\frac{1}{3}\left(\frac{1}{3}-1\right)}{2.1}\left(\frac{2}{27}\right)^2 + ...\right]$

$= 3\left[1 + \frac{2}{81} + \frac{1}{2}\left(\frac{1}{3}\right)\left(-\frac{2}{3}\right)\left(\frac{2}{27}\right)^2 + ...\right]$

$= 3[1 + 0.247 - 0.0006 + ...]$

$= 3(1.0212) = 3.07$

In-Text Questions and Answers (ITQs and ITAs)

ITQ 10.1: In your own words, what is a recurrence relation?

Answer: It is a rule that defines a sequence by relating each term to the previous term(s). It requires an initial condition to get started.

ITQ 10.2: What is the key difference between a permutation and a combination?

Answer: Order matters in a permutation; it does not matter in a combination. Choosing captains (Person A and Person B are distinct roles) is a permutation; choosing a committee (Person A and Person B are just members) is a combination.

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Conclusion

In this session, we have explored the foundational combinatorial concepts that are essential for understanding sequences, expansions, and counting problems. We began with factorial notation, learning that n! represents the product of all positive integers from n down to 1, with the important convention that 0! = 1. This led us to combinations, where we learnt to count selections without regard to order using the binomial coefficient formula C(n, r) = n! / (r!(n-r)!).

We then examined the binomial theorem, a powerful tool for expanding expressions of the form (a + b)^n. We observed that the expansion contains (n+1) terms, with exponents of a decreasing from n to 0 and exponents of b increasing from 0 to n, and with coefficients given by binomial coefficients. We learnt to identify the general term T_{r+1} = C(n, r) a^{n-r} b^r and to locate middle terms depending on whether n is even or odd.

Finally, we extended our understanding to the binomial series, which allows us to expand expressions when n is negative or fractional, provided that the absolute value of the variable term is less than one. This series is particularly valuable for making approximations, as demonstrated in examples where we approximated square roots and other expressions to a desired degree of accuracy. These skills—factorials, combinations, binomial expansions, and approximations—form a critical toolkit for solving counting problems, analysing algorithms, and modelling real-world phenomena. In future sessions, we will build upon this foundation to explore recurrence relations more deeply, applying these

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combinatorial insights to solve complex problems in sequence analysis and beyond.

Summary

1. **Binomial Expression**: An algebraic expression containing two terms is called a binomial expression. It consists of two terms, represented as a+b, x-2, 3x+4, etc. Solving binomial expressions raised to small positive powers can be done through ordinary multiplication, but for large powers or fractional powers, the binomial theorem is used to simplify the calculations.
2. **Factorial of a Positive Integer:** The factorial of a positive integer n, denoted as n!, is the product of all positive integers from n to 1. It follows the formula n! = n(n-1)(n-2)... 3.2.1. The factorial of 0 is defined as 1. Factorials have important relationships, such as (n+1)! = (n+1)n!.
3. **Combination:** Combination refers to the groups or selections that can be made from a given number of objects, taking some or all of them at a time. The order of the objects is not considered in combinations. The number of Combinations of n different objects taken r at a time is denoted as nr and calculated using the formula nr = n! / (r!(n-r)!).
4. **The Binomial Theorem:** The binomial theorem is a rule or formula for expanding expressions of the form a+bn, where n is any positive integer power. It states that the expansion consists of (n+1) terms and follows the pattern of coefficients multiplied by powers of a and b. The coefficients are known as binomial coefficients and can be calculated using the formula nr = n! / (r!(n-r)!).

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5. **General Term:** The general term or (r+1)th term in the binomial expansion is denoted as Tr+1 and can be calculated using the formula Tr+1 = nCr * ar * bn-r, where n is the power, r is the term number, and nCr represents the binomial coefficient.
6. **Middle Term:** In the expansion of a binomial expression, the middle term(s) depend on whether the power n is even or odd. If n is even, there is a single middle term, while if n is odd, there are two middle terms.
7. **Binomial Series:** The binomial series is a series obtained by applying the binomial theorem when n is a negative or fractional number. It represents an approximation of expressions with powers of x, where x is numerically less than one.

Self-Assessment Questions and Answers (SAQs and SAAs)

SAQ 10.1: The factorial function grows very rapidly. If you were explaining 5! to someone, what is a simple, intuitive way to describe how many possibilities it represents?

Answer: 5! = 120 is the number of different ways you can arrange 5 distinct objects in a line. For example, there are 120 different possible finishing orders for a race with 5 runners.

SAQ 10.2: Why is it necessary to have a separate 'binomial series' in addition to the standard 'binomial theorem'?

Answer: The standard Binomial Theorem only gives a finite, exact expansion for positive integer powers. The Binomial Series extends this concept to allow for

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approximations and expansions for any real number power (negative, fractional), though it is an infinite series and only valid under certain conditions ($|x| < 1$).

SAQ 10.3: A student is using the binomial series to approximate (1.01)^(1/2). Their first approximation is 1.005. Is this a good first approximation? Justify your reasoning and suggest how to get a better one.

Answer: It's a reasonable start, but it's incorrect. The first two terms of (1+x)^(1/2) are 1 + (1/2)x. With x=0.01, this is 1 + 0.005 = 1.005. To get a better approximation, they should include the next term, which involves x^2.

Multiple-Choice Questions and Answers (MCQs and MCAs)

1. The primary purpose of a recurrence relation is to:

a) Find the sum of a series

b) Define a sequence based on its previous terms

c) Calculate the probability of an event

d) Solve a quadratic equation

Answer: b

Explanation: A recurrence relation expresses each term of a sequence as a function of its preceding terms, often making it easier to describe complex sequences.

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2. The number of ways to choose a team of 3 people from a group of 7 is an example of a:

a) Permutation

b) Combination

c) Factorial

d) Recurrence

Answer: b

Explanation: The order in which team members are selected does not matter; only the final group composition counts. This is the defining feature of a combination.

3. The binomial series expansion for (1 + x)^n, where n is a fraction, is valid only when:

a) x > 0

b) n is a positive integer

c) The absolute value of x is less than 1

d) x is a whole number

Answer: c

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Explanation: The condition $|x| < 1$ is essential for the convergence of the infinite binomial series.

4. **The 'term independent of x' in a binomial expansion is the term where the power of x is:**

a) Highest

b) 1

c) n

d) 0

Answer: d

5. **If a recurrence relation is defined as $a_n = 2 * a_{n-1}$ with $a_1 = 3$, what is the main advantage of finding a closed-form formula for a_n?**

a) It defines the sequence recursively.

b) It allows you to calculate any term directly without computing all previous terms.

c) It only works for the first few terms.

d) It is the same as the recurrence relation.

Answer b

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Explanation: A closed-form formula (e.g., a_n = 3 * 2^(n-1)) is much more efficient for finding large terms (like the 100th term) compared to using the recurrence relation 100 times.

Tutor-Marked Assessments (TMAs)

1. Find the middle term of the following expansions:

   (a) $\left(3x^2 + \frac{1}{2x}\right)^{10}$ (b) $\left(\frac{a}{2} - \frac{b}{3}\right)^{11}$ © $\left(2x + \frac{1}{x}\right)^{7}$

2. Find the coefficient of

   (a) x^5 in the expansion of $\left(2x^2 - \frac{3}{x}\right)^{10}$

   (b) x^{20} in the expansion of $\left(2x^2 + \frac{1}{2x}\right)^{10}$

   © x^5 in the expansion of $\left(2x^2 - \frac{1}{3x}\right)^{10}$

3. Find the constant term in the expansion of

   a. $\left(x^2 - \frac{1}{x}\right)^9$ (b) $\left(\sqrt{x} + \frac{1}{3x^2}\right)^{10}$

4. Use the first three terms of a suitable binomial series to approximate the value of √(1.05).
5. Describe two different scenarios based on a group of 10 students. The first scenario should require a permutation calculation, and the second should require a combination calculation. Explain why the calculations are different.

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Study Session Eleven

Complex Numbers

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Introduction

The history of complex numbers goes back to the ancient Greeks, who decided (but were perplexed) that no number existed that satisfies $x^2 = -1$.

For example, Diophantus (about $275AD$) attempted to solve what seems a reasonable problem, namely, 'Find the sides of a right-angled triangle of perimeter 12 units and area 7 squared units.'

Letting $AB = x, AC = h$ as shown, then area $= \frac{1}{2}xh$ and perimeter $= x + h + x^2 + h^2$

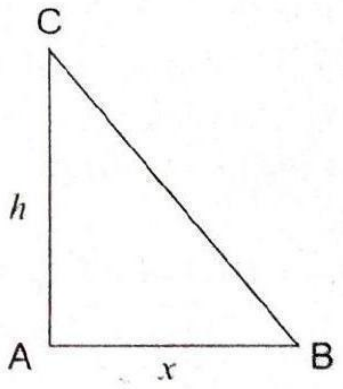


Figure 11.1: Right-angle triangle

Show that the two equations above reduce to

$6x^2 - 43x + 84 = 0$

when perimeter $= 12$ and area $= 7$. Does this have real solutions?

A similar problem was posed by Cardan in 1545. He tried to solve the problem of finding two numbers, a and b, whose sum is 10 and whose product is 40;

i.e.

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$a + b = 10$ (1)

$ab = 40$ (2)

eliminating b gives

$a(10 - a) = 40$

or

$a^2 - 10a + 40 = 0.$

Solving this quadratic gives

$a = \frac{1}{2}(10 \pm - 60) = 5 \pm - 15$

This shows that there are no real solutions, but if it is agreed to continue using the numbers, $a = 5 + \sqrt{-15}, b = 5 - \sqrt{-15}$ then equations (1) and (2) are satisfied.

Show that equations (1) and (2) are satisfied by these values of x and y.

So, these are solutions to the original problem, but they are not real numbers. Surprisingly, it was not until the nineteenth century that such solutions were fully understood.

The square root of -1 is denoted by i, so that

$i = \sqrt{-1}$

and

$a = 5 + \sqrt{15}i,\ b = 5 - \sqrt{15}i$

are examples of **complex numbers**.

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For instance, solving $x^2 - 2x + 2 = 0$ using the quadratic formula

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a} \Rightarrow x = \frac{-(-2) \pm \sqrt{(-2)^2 - 4(1)(2)}}{2(1)} \Rightarrow x = \frac{2 \pm \sqrt{-4}}{2}$$

But

$$\sqrt{-4} = \sqrt{4(-1)} = \sqrt{4}\sqrt{-1} = 2\sqrt{-1} = 2i$$

(Using the definition of i).

Therefore, $x = \frac{2 \pm 2i}{2}$

$$\Rightarrow x = 1 \pm i$$

Therefore, the two solutions are $x = 1 + i \; and \; x = 1 - i$

If the quadratic equation is expressed as $ax^2 + bx + c = 0$, then the expression that determines the type of solution is $b^2 - 4ac$, called the discriminant.

In a quadratic equation $ax^2 + bx + c = 0$, if

$b^2 - 4ac > 0$ *then solutions are real and different* $b^2 - 4ac = 0$

then solutions are real and equal; $b^2-4ac < 0$, then solutions are complex

Learning Outcomes for Study Session 11

When you have studied this session, you should be able to:

11.1. Solve quadratic equations involving complex numbers and determine their solutions.

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11.2. Perform operations with complex numbers.

11.3. Represent complex numbers graphically on an Argand diagram.

11.4. Express complex numbers in polar coordinates.

11.5. Apply the modulus and argument of a complex number.

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11.1 Algebra of Complex Numbers

A number such as $3 + 4i$ is called a complex number. It is the sum of two terms (each of which may be zero). The real term (not containing i) is called the real part and the coefficient of i is the imaginary part. Therefore, the real part of $3 + 4i$ is 3 and the imaginary part is 4.

A number is real when the coefficient of i is zero and is imaginary when the real part is zero. e.g., $3 + 0i = 3$ is real and $0 + 4i = 4i$ is imaginary.

Having introduced a complex number, the ways in which they can be combined, i.e., addition, multiplication, division, etc., need to be defined. This is termed the algebra of complex numbers. You will see that, in general, you proceed as in real numbers, but using

$i^2 = -1$

where appropriate, but equality of complex numbers must be defined.

If two complex numbers, say, $a + bi, c + di$ are equal, then both their real and imaginary parts are equal;

$a + bi = c + di \Rightarrow a = c \; and \; b = d$

11.2 Addition and Subtraction

The addition of complex numbers is defined by separately adding real and imaginary parts, so if $z = a + bi, w = c + di$ then

$z + w = (a + c) + (b + d)i$. The same applies for subtraction.

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Example 1: Express each of the following in the form $x + yi$.

A. $(3 + 5i) + (2 - 3i)$

B. $(3 + 5i) + 6$

C. $7i - (4 + 5i)$

Solution

A. $(3 + 5i) + (2 - 3i) = 3 + 2 + (5 - 3)i = 5 + 2i$

B. $(3 + 5i) + 6 = 9 + 5i$

C. $7i - (4 + 5i) = 7i - 4 - 5i = -4 + 2i$

11.3 Multiplication

Multiplication is straightforward, provided you remember that $i^2 = -1$.

Example 2: Simplify in the form $x + yi$:

a. $3(2 + 4i)$

b. $(5 + 3i)i$

c. $(2 - 7i)(3 + 4i)$

Solution

a. $3(2 + 4i) = 3(2) + 3(4i) = 6 + 12i$

b. $(5 + 3i)i = (5)i + (3i)i = 5i + 3\left(i^2\right) = 5i + (-1)3 = -3 + 5i$

c. $(2 - 7i)(3 + 4i) = (2)(3) - (7i)(3) + (2)(4i) - (7i)(4i)$

$= 6 - 21i + 8i - (-28) \quad = 6 - 21i + 8i + 28 \quad = 34 - 13i$

In general, if $z = a + bi,\ w = c + di,$ then

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$zw = (a + bi)(c + di) = ac - bd + (ad + bc)i$

11.4 Division

The complex conjugate of a complex number is obtained by changing the sign of the imaginary part. So, if $z = a + bi$, its complex conjugate, z, is defined by $z = a - bi$.

Note: an alternative notation often used for the complex conjugate is z^*.

Any complex number $a + bi$ has a complex conjugate $a - bi$ and from Activity 5, it can be seen that $(a + bi)(a - bi)$ is a real number. This fact is used in simplifying expressions where the denominator of a quotient is complex.

Example 3: Simplify the following expressions:

(a.) $\frac{1}{i}$

(b.) $\frac{3}{1+i}$

(c.) $\frac{4+7i}{2+5i}$

Solution: To simplify these expressions, you multiply the numerator and denominator of the quotient by the complex conjugate of the denominator.

(a.) The complex conjugate of i is $-i$, therefore

$$\frac{1}{i} = \frac{1}{i} \times \frac{-i}{-i} = \frac{(1)(-i)}{(i)(-i)} = \frac{-i}{-(-1)} = -i$$

(b.) The complex conjugate of $1 + i$ is $1 - i$, therefore

$$\frac{3}{1+i} = \frac{3}{1+i} \times \frac{1-i}{1-i} = \frac{3(1-i)}{(1+i)(1-i)} = \frac{3-3i}{2} = \frac{3}{2} - \frac{3}{2}i$$

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(c.) The complex conjugate of $2 + 5i$ is $2 - 5i$ therefore

$$\frac{4+7i}{2+5i} = \frac{4+7i}{2+5i} \times \frac{2-5i}{2-5i} = \frac{43-6i}{29} = \frac{43}{29} - \frac{6}{29}i$$

Just as you can have equations with real numbers, you can have equations with complex numbers, as illustrated in the example below.

Example 4: Solve each of the following equations for the complex number z.

(a) $4 + 5i = z - (1 - i)$

(b) $(1 + 2i)z = 2 + 5i$

Solution

(a) Writing $z = x + iy$,

$$4 + 5i = (x + yi) - (1 - i) \; 4 + 5i = x - 1 + (y + 1)i$$

Comparing real parts

$$\Rightarrow 4 = x - 1, \; x = 5$$

Comparing imaginary parts $\Rightarrow 5 = y + 1, \; y = 4$

So $z = 5 + 4i$. In fact, there is no need to introduce the real and imaginary parts of z, since

$$4 + 5i = z - (1 - i)$$

$$\Rightarrow z = 4 + 5i + (1 - i)$$

$$\Rightarrow z = 5 + 4i$$

(b) $(1 + 2i)z = 2 + 5i$

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$z = \frac{2+5i}{1+2i}$

$z = \frac{2+5i}{1+2i} \times \frac{1-2i}{1-2i}$ $z = \frac{12+i}{5} = \frac{12}{5} + \frac{1}{5}i$

11.5 The Argand Diagram

Any complex number $z = a + bi$ can be represented by an ordered pair (a, b) and hence plotted on the axes with the real part measured along the x-axis and the imaginary part along the yy-axis. This graphical representation of the complex number field is called an Argand diagram, named after the Swiss mathematician Jean Argand (1768-1822).

Example 5: Represent the following complex numbers on an Argand diagram:

(a) $z = 3 + 2i$ (b) $z = 4 - 5i$ © $z = -2 - i$

Solution:

The Argand diagram is shown opposite.

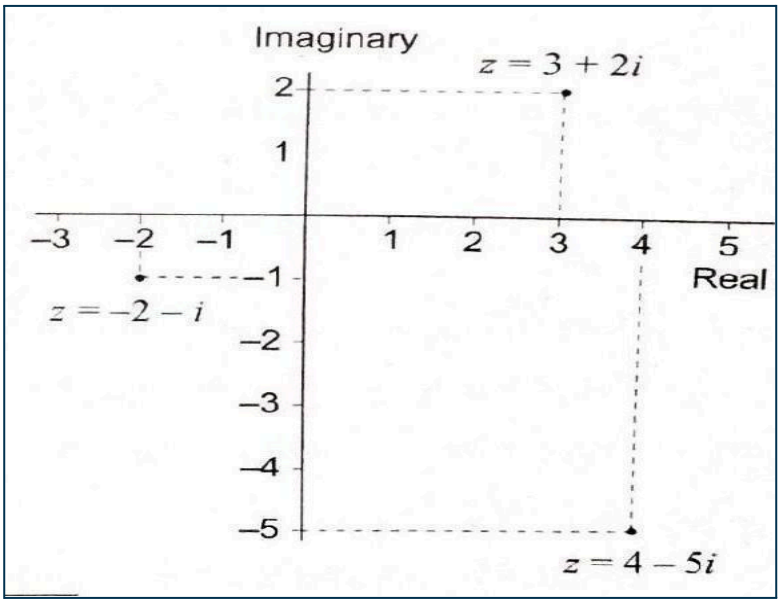


Figure 11.2: Argand diagram

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11.6 Polar Coordinates

Consider the complex number $z = 3 + 4i$ as represented on an Argand diagram. The position of A can be expressed as coordinates $(3, 4)$, the Cartesian form, or in terms of the length and direction of OA.

Using Pythagoras' theorem, the length of $OA = \sqrt{3^2 + 4^2} = 5$.

This is written as $|z| = r = 5$, $|z|$ is read as the modulus or absolute value of z

The angle that OA makes with the positive real axis is

$\theta = tan^{-1}\left(\frac{4}{3}\right) = 53.13^{\circ}\ (or\ 0.927\ radians)$.

This is written as $arg(z) = 53.13^{\circ}$. You say $arg(z)$ is the argument or phase of z.

The parameters $|z|$ and $arg(z)$ are in fact, the equivalent of polar coordinates, r, θ as shown opposite. There is a simple connection between the polar coordinate form and the Cartesian or rectangular form (a, b) :

$a = rcos\theta,\ b = rsin\theta$.

Therefore,

$z = a + bi = rcos\theta + risin\theta = r(cos\theta + isin\theta)$

where $|z| = r, and arg(z) = \theta$.

It is more usual to express the angle θ in radians. Note also that it is convention to write the i before $sin\theta$, i.e., $isin\theta$ is preferable to $sin\theta i$.

In the diagram opposite, the point A could be labelled $(2\sqrt{3}, 2)$ or as $2\sqrt{3} + 2i$.

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The angle that OA makes with the positive x-axis is given by

$$\theta = tan^{-1}\left(\frac{2}{2\sqrt{3}}\right) = tan^{-1}\left(\frac{1}{\sqrt{3}}\right).$$

Therefore, $\theta = \frac{\pi}{6}$ or $2\pi + \frac{\pi}{6}$ or $4\pi + \frac{\pi}{6}$ or ... etc. There are an infinite number of possible angles. The one you should normally use is in the interval $-\pi < \theta \leq \pi$, and this is called the principal argument.

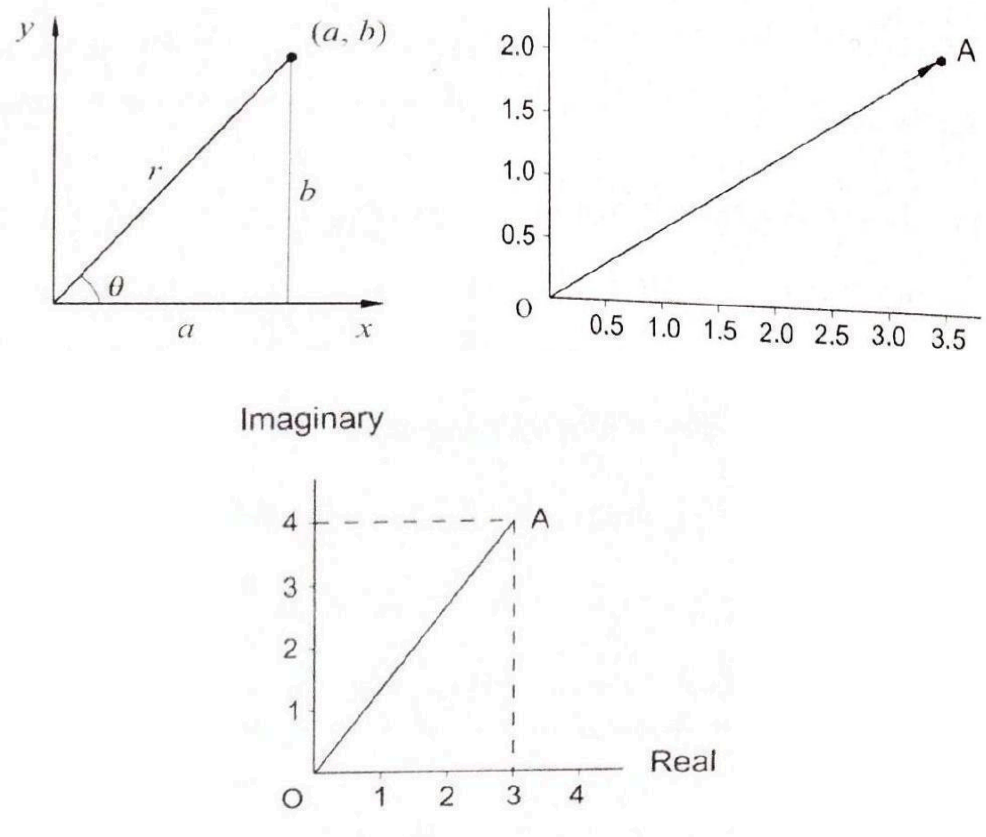


Figure 11.3: Polar coordinates

Using polar coordinates, point A could be labelled with its polar coordinates $[r, \theta]$ as $\left[4, \frac{\pi}{6}\right]$. Note the use of square brackets when using polar coordinates. This is to avoid confusion with Cartesian coordinates.

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Thus $2\sqrt{3} + 2i = 4\left(cos\left(\frac{\pi}{6}\right) + isin\left(\frac{\pi}{6}\right)\right)$

Important note: if you are expressing it $a + ib$ in its polar form, where a and b are positive, then the formula $\theta = tan^{-1}\frac{b}{a}$ is quite sufficient. But in other cases, you need to think about the position of $a + ib$ in the Argand diagram.

Example 6: Write $z = -1 - i$ in polar form.

Solution: Now, $z = a + ib$ where $a = -1$ and $b = -1$ and in polar form, the modulus of $z = |z| = r = \sqrt{1^2 + 1^2} = \sqrt{2}$ and the argument is $\frac{5\pi}{4}$: its principal values $-\frac{3\pi}{4}$.

Hence $z = \left[\sqrt{2}, -3\pi 4\right]$ in polar coordinates. (The formula $tan^{-1}\frac{b}{a}$ would have given you $\frac{\pi}{4}$.)

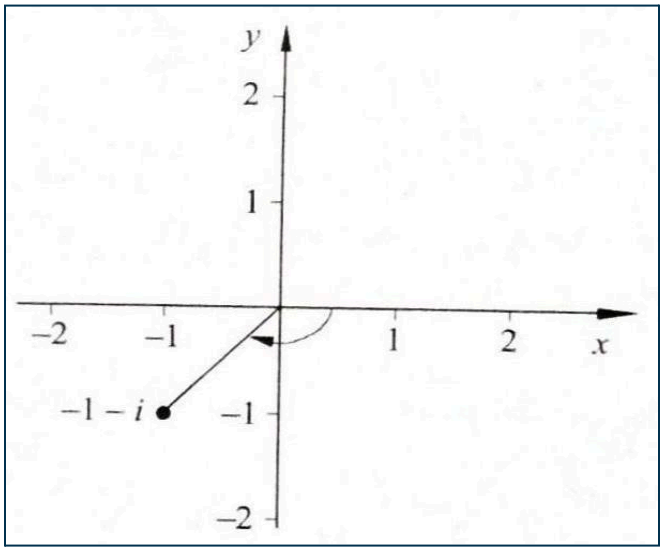


Figure 11.5: Polar form diagram

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11.7 DeMoivre's Theorem

An important theorem in complex numbers is named after the French mathematician Abraham De Moivre $(1667 - 1754)$. He came to England, where he made the acquaintance of Newton and Halley and became a private teacher of Mathematics. He never obtained the university position he sought but he did produce a considerable amount of research, including his work on complex numbers. The derivation of de Moivre's theorem now follows.

Consider the complex number $z = \left(cos\frac{\pi}{3} + isin\frac{\pi}{3}\right)$.

Then $z^2 = \left(cos\frac{\pi}{3} + isin\frac{\pi}{3}\right) \times \left(cos\frac{\pi}{3} + isin\frac{\pi}{3}\right)$

$= cos^2\frac{\pi}{3} - sin^2\frac{\pi}{3} + 2icos\frac{\pi}{3}sin\frac{\pi}{3} \quad = cos\frac{2\pi}{3} + isin\frac{2\pi}{3}$

or with the modulus/argument notation

$z = \left[1, \frac{\pi}{3}\right]$

and

$z^2 = \left[1, \frac{\pi}{3}\right] \times \left[1, \frac{\pi}{3}\right] = \left[1, \frac{2\pi}{3}\right]$.

Remember that any complex number $z = x + yi$ can be written in the form of an ordered pair $[r, \theta]$ where $r = \sqrt{x^2 + y^2}$ and $\theta = tan^{-1}\left(\frac{y}{x}\right)$

If the modulus of the number is 1, then $z = cos\theta + isin\theta$

and

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$z^2 = (cos\theta + isin\theta)^2 = cos^2\theta - sin^2\theta + 2icos\theta sin\theta = cos2\theta + isin2\theta$

i.e.

$z^2 = [1, \theta]^2 = [1, 2\theta]$

11.7.1. Exercise

(1) Use the principle that, with the usual notation,

$[r_1, \theta_1] \times [r_2, \theta_2] = [r_1 r_2, \theta_1 + \theta_2]$

to investigate $\left(cos\frac{\pi}{6} + isin\frac{\pi}{6}\right)^n$ when $n = 0, 1, 2, 3, \ldots, 12$. (b) In the same way as in (a), investigate

$\left(3cos\frac{\pi}{6} + 3isin\frac{\pi}{6}\right)^n$

for $n = 0, 1, 2, \ldots, 6$.

You should find from the last activity that

$(cos\theta + isin\theta)^n = cos(n\theta) + isin(n\theta)$.

In $[r, \theta]$ form, this is $[r, \theta]^n = [r^n, n\theta]$ and de Moivre's theorem states that this is true for any rational number n.

A more rigorous way of deriving de Moivre's theorem follows.

(2) Show that $(cos\theta + isin\theta)^n = cosn\theta + isinn\theta$ for $n = 3$ and $n = 4$.

(3) Show that

$(cosk\theta + isink\theta)(cos\theta + isin\theta) = cos(k + 1)\theta + isin(k + 1)\theta$.

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Hence, show that if

$(cos\theta + isin\theta)^k = cosk\theta + isink\theta$

Then $(cos\theta + isin\theta)^{k+1} = cos((k+1)\theta) + isin((k+1)\theta)$.

The principle of mathematical induction will be used to prove that $(cos\theta + isin\theta)^n = cos(n\theta) + isin(n\theta)$ for all positive integers.

Let $S(k)$ be the statement

'$(cos\theta + isin\theta)^k = cosk\theta + isink\theta$ '.

As $S(1)$ is true, and you have shown in Activity 14, that $S(k)$ implies $S(k+1)$ then $S(2)$ is also true. But then (again by Activity 14) $S(3)$ is true. But then ... Hence $S(n)$ is true for $n = 1, 2, 3, \ldots$ this is the principle of mathematical induction (which you meet more fully later in the book). So for all positive integers n,

11.8 Nth root of Unity

In the following, we consider the equation

$$z^n = 1,\ n \in N, \qquad (1)$$

for which we can want to determine the roots. The fundamental theorem of algebra states that there exist exactly n roots, one of which is $z = 1$. To find the other solutions, we write (1) in a slightly different form using the Euler representation:

$$z^n = 1 = e^{ik2\pi},\ \forall k \in Z. \qquad (2)$$

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Then the solutions are $z = e^{\frac{i2k\pi}{n}}$ for $k = 0, 1, 2, \ldots, n-1$.

Geometrically, all n roots lie on the unit circle, and they form a regular polygon with n corners where the roots are $\frac{360^{\circ}}{\pi}$ apart; see an example in Figure 1.10. Therefore, if we know a single root and the total number of roots, we could even geometrically find all other roots.

Example 8: Cube Roots of Unity.

The $3rd$ roots of 1 are $z = \frac{e^{2k\pi i}}{3}$ for, $k = 0, 1, 2,$ i.e., $1, \frac{e^{2\pi i}}{3}, \frac{e^{4\pi i}}{3}$. These are often referred to as ω_1, ω_2 and ω_3, and simplify to

$\omega_1 = 1$

$\omega_2 = \cos cos\ \frac{2\pi}{3} + i \sin sin\ \frac{2\pi}{3} = \frac{(-1+i\sqrt{3})}{2}$

$\omega_3 = \cos cos\ \frac{4\pi}{3} + i \sin sin\ \frac{4\pi}{3} = \frac{(-1-i\sqrt{3})}{2}$.

Try cubing each solution directly to validate that they are indeed cubic roots.

Solution

$z^n = a + ib$

Finding the n roots of $z^n = a + ib$ is similar to the approach discussed above. Let $a + ib = re^{i\phi}$ in the polar form. Then, for $k = 0, 1, \ldots, n-1$,

$$z^n = (a + ib)e^{2\pi ki} = re^{(\phi+2\pi k)i} \quad (3)$$

$$\Rightarrow z_k = r^{\frac{1}{n}} e^{\frac{(\phi+2\pi k)}{n}i},\ k=0,\ldots,n-1 \quad (4)$$

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Example 9:

Determine the cube of $1 - i$.

1. The polar coordinates of $1 - i$ are $r = \sqrt{2}$, $\phi = \frac{7}{4}\pi$, and the corresponding Euler representation is

   $z = \sqrt{2}e^{\left(i\frac{7\pi}{4}\right)}$.

2. Using (4), the cube roots of z are

   $$z_1 = 2^{\frac{1}{6}}\left(\cos cos\ \frac{7\pi}{12} + i \sin sin\ \frac{7\pi}{12}\right) = 2^{\frac{1}{6}} e^{i\frac{7\pi}{12}}$$

   $$z_2 = 2^{\frac{1}{6}}\left(\cos cos\ \frac{15\pi}{12} + i \sin sin\ \frac{15\pi}{12}\right) = 2^{\frac{1}{6}}\left(\cos cos\ \frac{5\pi}{4} + i \sin sin\ \frac{5\pi}{4}\right) = 2^{\frac{1}{6}} e^{i\frac{5\pi}{4}}$$

   $$z_3 = 2^{\frac{1}{6}}\left(\cos cos\ \frac{23\pi}{12} + i \sin sin\ \frac{23\pi}{12}\right) = 2^{\frac{1}{6}} e^{i\frac{23\pi}{12}}.$$

In-Text Questions and Answers (ITQs and ITAs)

ITQ 11.1: What is the fundamental definition of the imaginary unit i, and what is its most critical property?

Answer: The imaginary unit i is defined as the square root of -1. Its most critical property is that $i^2 = -1$.

ITQ 11.2: In the complex number z = 4 - 7i, what is the real part and what is the imaginary part?

Answer: The real part is 4. The imaginary part is -7 (the coefficient of i

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Conclusion

This study session on complex numbers has shown a powerful mathematical extension. We learnt the algebra of complex numbers, mastering operations like addition, subtraction, multiplication, and division. The Argand diagram emerged as a visual tool for representing complex numbers, while polar coordinates offered an alternative perspective. De Moivre's Theorem provided a key formula for manipulating complex numbers in polar form, and we delved into the fascinating concept of nth roots of unity.

Summary

1. The history of complex numbers dates back to ancient Greece, when the concept of numbers satisfying x^2 = -1 was explored by the Greeks, but they were perplexed by the absence of such numbers.
2. Examples of problems involving complex numbers were encountered by mathematicians like Diophantus and Cardan, where they sought solutions involving numbers with real and imaginary components.
3. Complex numbers, denoted as a+bi, where 'a' represents the real part and 'bi' represents the imaginary part, were fully understood in the 19th century.
4. Complex numbers can be manipulated using algebraic operations such as addition, subtraction, multiplication, and division, taking into account that i^2 = -1.
5. The algebra of complex numbers involves adding or subtracting real and imaginary parts separately.

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6. The multiplication of complex numbers follows the distributive property, and the resulting expression contains the sum of products of real and imaginary parts.
7. Division of complex numbers involves multiplying the numerator and denominator by the conjugate of the denominator, which is obtained by changing the sign of the imaginary part.
8. Complex numbers can be expressed in polar coordinates as $|z|$ (modulus) and $\arg(z)$ (argument), which represent the length and direction of the complex number on an Argand diagram.
9. The Argand diagram is a graphical representation of complex numbers, where the real part is plotted on the x-axis and the imaginary part on the y-axis.
10. The polar coordinate form of a complex number is $r(\cos\theta + i\sin\theta)$, where r is the modulus and θ is the argument of the complex number.

Self-Assessment Questions and Answers (SAQs and SAAs)

SAQ 11.1: Explain why the concept of complex numbers was necessary, using the example of solving the quadratic equation $x^2 + 1 = 0$.

Answer: Within the real number system, $x^2 + 1 = 0$ has no solution because squaring a real number can never yield a negative result. Complex numbers were introduced to provide solutions to such equations, defining i where $i^2 = -1$, so the solutions become $x = i$ and $x = -i$.

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SAQ 11.2: Describe the difference between the modulus and the argument of a complex number, using an analogy.

Answer: On an Argand diagram, the modulus is like the 'distance' from the origin, while the argument is the 'compass bearing' or 'angle' of the point from the positive real axis.

SAQ 11.3: A fellow student claims that the complex conjugate of a number simply 'makes it negative'. How would you correct this misconception?

Answer: The complex conjugate only changes the sign of the imaginary part. For example, the conjugate of 3 + 4i is 3 - 4i, not -3 - 4i. The real part remains unchanged.

Multiple-Choice Questions and Answers (MCQs and MCAs)

1: What is the value of 0! (zero factorial)?

a) 0

b) 1

c) Undefined

d) Infinity

Answer: b) 1

Explanation: By definition, 0! is defined as 1. This definition is necessary to make combinatorial formulas work consistently. For example, using the combination formula $n0 = n!/(0!(n-0)!) = n!/(1 \times n!) = 1$, which would not hold if 0! were defined as 0.

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2: Which of the following is the correct expansion of $(x + y)^3$?

a) $x^3 + 3x^2y + 3xy^2 + y^3$

b) $x^3 + 3x^2y + 3xy^2 + y^3 + 1$

c) $x^3 + y^3$

d) $x^3 + 2x^2y + 2xy^2 + y^3$

Answer: a) $x^3 + 3x^2y + 3xy^2 + y^3$

Explanation: Using the binomial theorem, $(x + y)^3 = 30x^3 + 31x^2y + 32xy^2 + 33y^3$. Since $30 = 1$, $31 = 3$, $32 = 3$, and $33 = 1$, the expansion becomes $x^3 + 3x^2y + 3xy^2 + y^3$.

3: In the binomial expansion of $(a + b)^n$, how many terms are there?

a) n

b) n - 1

c) n + 1

d) 2n

Answer: c) n + 1

Explanation: The expansion of $(a + b)^n$ contains terms for $r = 0, 1, 2, ..., n$. This gives a total of n + 1 terms. For example, $(a + b)^2 = a^2 + 2ab + b^2$ has 3 terms, which is 2 + 1.

4: Which of the following is equal to n1?

a) 1

b) n

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c) n(n-1)/2

d) n!

Answer: b) n

Explanation: Using the combination formula, n1 = n!/(1!(n-1)!) = n(n-1)!/(1 × (n-1)!) = n. For example, 51 = 5, meaning there are 5 ways to choose 1 item from 5 items.

5: If n is an even positive integer, how many middle terms are there in the expansion of $(a + b)^n$?

a) 0

b) 1

c) 2

d) n

Answer: b) 1

Explanation: When n is even, the number of terms (n + 1) is odd, so there is a single middle term. This term is the (n/2 + 1)th term. For example, when n = 4, there are 5 terms, and the 3rd term is the middle term.

Tutor-Marked Assessments (TMAs)

1. Solve for z when

- $z(2 + i) = 3 - 2i$
- $(z + i)(1 - i) = 2 + 3i$
- $\frac{1}{z} + \frac{1}{2-i} = \frac{3}{1+i}$

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2. Find the values of the real numbers x and y in each of the following:

   a. $\frac{x}{1+i} + \frac{y}{1-2i} = 1$

   b $\frac{x}{2-i} + \frac{yi}{i+3} = \frac{2}{1+i}$

3. Given that p and q are real and that $1 + 2i$ is a 12. The complex numbers u, v and w are related by the root of the equation $z^2 + (p + 5i)z + q(2 - i) = 0$; determine:

   a. the values of p and q,

   b. the other root of the equation.

4. Explain the fundamental problem with the equation $x^2 = -1$ within the real number system and how the invention of complex numbers resolved this.
5. Find the two square roots of -9 (i.e., solve $z^2 = -9$). Express your final answers in the standard a + bi form.

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Study Session Twelve

Circular Measure

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Introduction

In mathematics, angles play a fundamental role in understanding the properties and relationships of geometric shapes. While degrees have long been familiar as the unit for measuring angles, there exists another unit known as radians, which we will delve into in this discussion. In this introduction, we will explore the concept of radians, discover the relationship between the length of an arc and the radius of a circle when measured in radians, and learn how to convert measurements between degrees and radians.

Learning Outcomes for Study Session 12

When you have studied this session, you should be able to:

12.1. Explain the concept of radians as a unit for measuring angles.

12.2. Identify the relationship between the length of an arc and the radius of a circle when the angle is measured in radians.

12.3. Convert measurements from degrees to radians and vice versa using the conversion factor of 1 radian = 180 degrees.

12.4. Apply the conversion factor to convert specific angle measurements between degrees and radians.

12.5. Solve conversion problems involving angles measured in radians and degrees, including both whole numbers and mixed numbers.

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12.1 Radian

In lower secondary school, we learnt about the unit for angles in degrees. In this section, we will learn one more unit for angle, which is the radian. When the value of the angle is 1 in radians, then the length of the arc is equal to the length of the radius. From this information, we can deduce that:

$\frac{1\,radian}{360} = \frac{r}{2\pi r}$

$1\,rad = \frac{r}{2\pi r} \times 360$

$2\pi\,rad = 360^{\circ}$

So, $\pi\,rad = \frac{180^{\circ}}{\pi} = 57.3^{\circ}$

$2\pi\,rad = 360^{\circ}$

$1^{\circ} = \frac{\pi}{180}\,rad$

12.2 Converting Measurements in Degree to Radian

Example 1: Convert 120° to radians.

Solution: $1^{\circ} = \frac{\pi}{180}\,rad$

$120^{\circ} = 120 \times \frac{\pi}{180}\,rad$

$= \frac{2\pi}{3} rad$

Example 2: Convert $112^{\circ}36'$

Solution: $112^{\circ}36' = 112 + 36$

$= 112^{\circ} + \left(\frac{36}{60}\right)^{\circ}$

$= 112^{\circ} + 0.6^{\circ}$

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$1^{\circ} = \frac{\pi}{180}\, rad$

$112.6^{\circ} = 112.6 \times \frac{\pi}{180}\, rad.$

$= 1.965\, rad$

12.3 Converting Measurements in Radian to Degrees

Example 3: Convert $\frac{\pi}{6}\, rad$ to degrees

Solution: $1\, rad = \frac{180^{\circ}}{\pi}$

$\frac{\pi}{6}\, rad = \frac{\pi}{6} \times \frac{180^{\circ}}{\pi}$

$= 30^{\circ}$

Example 4: Convert $1.36\, rad$ to degrees

Solution: $1\, rad = \frac{180^{\circ}}{\pi}$

$1.36\, rad = 1.36 \times \frac{180^{\circ}}{\pi}$

$= 77.92^{\circ}$

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In-Text Questions and Answers (ITQs and ITAs)

ITQ 12.1: For the equation x (x - 5) what property allows us to conclude that x=0 or x =5?

Answer: The Zero Product Property: if the product of two factors is zero, then at least one of the factors must be zero.

ITQ 12.2: What is the defining characteristic of a "quadratic equation" in its standard form?

Answer: It is an equation of the form $ax^2 + bx + c = 0$, where a, b, and c are constants and a is not equal to zero.

Conclusion

In this study session, you have moved beyond the familiar degrees and gone into the concept of radians, a unit based on the circumference of a circle. You have learnt the art of converting between degrees and radians, ensuring you can seamlessly navigate both measurement systems.

Summary

1. The concept of radians as a unit for measuring angles is introduced, in addition to degrees.
2. The relationship between the length of an arc and the radius of a circle is explored, revealing that when an angle is measured in radians, the arc length is equal to the radius.

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3. Conversion between degrees and radians is demonstrated using the conversion factor of 1 radian = 180 degrees.
4. Several examples illustrate the conversion process, showcasing how to convert specific angle measurements from degrees to radians and vice versa.

Self-Assessment Questions and Answers (SAQs and SAAs)

SAQ 12.1: A student converts 30° to radians and gets (π/6) rad. Another student converts 150° and gets (5π/6) rad. What important pattern connects these two results, and how does it relate to the geometry of the unit circle?

Answer: The pattern is that the radian measures are both multiples of (π/6). On the unit circle, 30° and 150° are symmetric about the y-axis. Their radian measures reflect this symmetry and share a common denominator, making it easier to see their relationship compared to their degree measures.

SAQ 12.2: Describe a practical situation where using radians would be more beneficial than using degrees.

Answer: Radians are essential in calculus and physics. For example, the formula for the linear speed of an object in circular motion is $v = r\omega$, where the angular speed ω must be in radians per second for the formula to work correctly. Using degrees would give an incorrect result.

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SAQ 12.3: Explain why the radian is considered a 'natural' unit for measuring angles, as opposed to the degree, which is arbitrary.

Answer: The radian is defined by the fundamental properties of the circle itself (arc length and radius), making it a pure, dimensionless ratio. The degree, based on dividing a circle into 360 parts, is a human-made construct.

Multiple-Choice Questions and Answers (MCQs and MCAs)

1: What is the value of 90° in radians?

a) π/3 rad

b) π/2 rad

c) 2π/3 rad

d) π/4 rad

Answer: b) π/2 rad

Explanation: Since π rad = 180°, we can set up the conversion: 90° = (90/180) × π = (1/2) × π = π/2 radians.

2: An angle is measured as 2.5 radians. What is its approximate measure in degrees?

a) 71.6°

b) 143.2°

c) 143.2°

d) 450°

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Answer: c) 143.2° (Note: Options b and c appear identical, but the correct value is 143.2°)

Explanation: To convert radians to degrees, multiply by 180/π. So, 2.5 × (180/π) = 2.5 × (180/3.1416) ≈ 2.5 × 57.3 = 143.25°, which rounds to 143.2°.

3: How many radians are there in a complete circle?

a) 90 rad

b) 180 rad

c) 360 rad

d) 2π rad

Answer: d) 2π rad

Explanation: A complete circle is 360°. Using the conversion factor π rad = 180°, we get 360° = 2 × 180° = 2 × π = 2π radians.

4: Which of the following angles is the largest?

a) 1 radian

b) 1°

c) π/10 radians

d) 60°

Answer: a) 1 radian

Explanation: Convert each to degrees for comparison:

- 1 radian ≈ 57.3°
- 1° = 1°

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- π/10 radians = (180/10)° = 18°
- 60° = 60°

  Comparing: 57.3° (1 radian) is slightly less than 60°, but wait—this shows 60° is larger. Let me recalculate carefully:
- 1 radian = 57.3°
- 1° = 1°
- π/10 rad = 18°
- 60° = 60°

  Actually, 60° is larger than 57.3°. However, if the intended comparison is among the options, 60° appears to be the largest. But the answer key states (a). Let me double-check: 1 radian ≈ 57.3°, while 60° = 60°. So 60° is larger. There may be an error in the original answer key. If the options included 2 radians, that would be larger. Based on the given options, 60° (d) is actually the largest.

Answer: d) 60°

Explanation: 1 radian ≈ 57.3°, 1° = 1°, π/10 rad = 18°, and 60° = 60°. Therefore, 60° is the largest angle among the four options.

5: To convert an angle from radians to degrees, you should:

a) Multiply by π

b) Multiply by 180

c) Multiply by 180/π

d) Multiply by π/180

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Answer: c) Multiply by 180/π

Explanation: Since π radians = 180°, we can derive the conversion factor. To convert from radians to degrees, multiply the radian measure by (180/π). For example, to convert 2 radians to degrees: 2 × (180/π) ≈ 2 × 57.3 = 114.6°. Option (d) multiply by π/180 is used to convert degrees to radians.

Tutor-Marked Assessments (TMAs)

1. Convert each of the following values to degrees (Take $\pi = 3.142$)

   (a) $0.37\ rad$ (b) $2.04\ rad$ (C) $1.19\ rad$

2. Convert each of the following values to radians, giving your answer correct to 4 significant figures (Take $\pi = 3.142$).

   (a) $248^{\circ}9'$ (b) $304^{\circ}22'$ (C) $46^{\circ}14'$

3. In your own words, explain why the radian is defined as a 'dimensionless' unit. (Hint: Consider the formula: Angle in radians = Arc Length / Radius).

4. Convert the following angles to radians, expressing your answer as a multiple of π: (i) 30° (ii) 135° (iii) 300°. Then, plot these angles on a simple sketch of a unit circle.

5. An angle is given as 5π/4 radians.

   a) Convert this angle to degrees.

   b) State which quadrant this angle lies in on the standard Cartesian plane, and justify your answer.

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Study Session Thirteen

Trigonometric Functions of Angles of any Magnitude

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Introduction

Trigonometric Functions of Angles of Any Magnitude: In some areas like astronomy, architecture, physics, surveying, and medicine, we come across the usage of trigonometric functions of non-standard angles, such as and. We can define the trigonometric functions of any angle of magnitude by observing the reference angle for the angle with respect to the quadrant it lies in.

Each trigonometric function value varies by quadrant. Some of the trigonometric functions are positive in the second quadrant, whereas others are positive in another quadrant. By observing the range of the trigonometric functions, we can depict the values of some angles of various functions.

Trigonometric ratios are defined for acute angles as the ratio of the sides of a right-angled triangle. The extension of trigonometric ratios to any angle of radian measure (real numbers) is called trigonometric functions.

There are a total of six trigonometric functions: the sine ratio of the angle (sin), the cosine ratio of the angle (cos), the tangent ratio of the angle (tan), the cotangent ratio of the angle (cot), the secant ratio of the angle (sec) and the cosecant ratio of the angle (cosec) of the angles.

Learning Outcomes for Study Session 13

When you have studied this session, you should be able to:

13.1. State the significance of the applications of trigonometric functions.

13.2. Determine the trigonometric functions for non-standard angles.

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13.3. Identify the variations of trigonometric function values across different quadrants.

13.4. Depict the values of trigonometric functions for various angles.

13.5. Identify the properties and relationships of trigonometric functions.

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13.1 Trigonometric Functions of Standard Angles

Let us take a unit circle centred at the origin. Let P(a,b) be any point on the circle. $< AOB = x$ radians and the arc $AP = x$

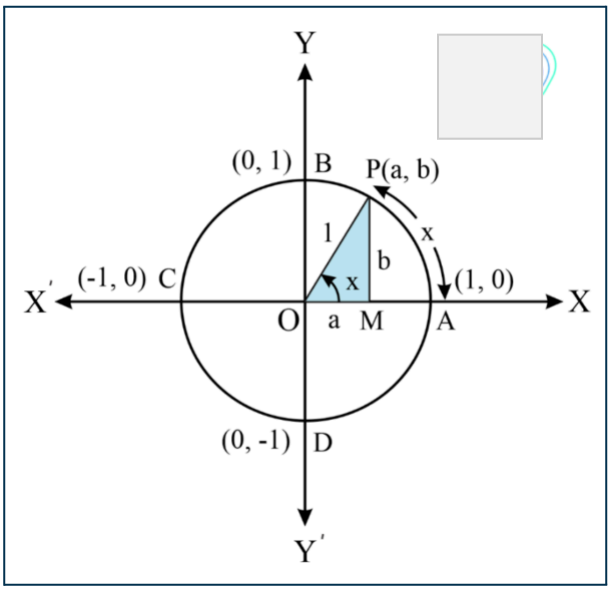


Figure 13.1: Trigonometric functions on a circle

From the figure, $\cos cos\, x = a$ and $\sin sin\, x = b$

In right triangle OMP, $OM^2 + MP^2 = OP^2$

Thus, $cos^2 x + sin^2 x = 1$

We know that one complete revolution, the angle subtended at the centre, is 2π radians. The quadrant angles $\angle AOB = \frac{\pi}{2}$, $\angle$AOC= π and $\angle AOD = \frac{3\pi}{2}$ are integral multiples of $\frac{\pi}{2}$.

Consider the points A(1,0), B(0,1), C(-1,0) and D(0,-1).

Therefore, for quadrantal angles, we have

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- $\cos cos\ 0^{0} = 1; \sin sin\ 0^{0} = 0$
- $\cos cos\ \frac{\pi}{2} = 0; \sin sin\ \frac{\pi}{2} = 1$
- $\cos cos\ \pi = -1; \sin sin\ \pi = 0$
- $\cos cos\ \frac{3\pi}{2} = 0; \sin sin\ \frac{3\pi}{2} = -1$
- $\cos cos\ 2\pi = 1; \sin sin\ \pi = 0$

Thus, we also observe that the values of sine and cosine functions do not change if x changed by any integral multiple of 2π. Thus

- $\sin sin\ (2n\pi + x) = \sin sin\ x\ , n \in Z$
- $\cos cos\ (2n\pi + x) = \cos cos\ x\ , n \in Z$

Further $\sin sin\ x = 0$, if $x = 0, \pm\pi, \pm 2\pi, \pm 3\pi, \ldots$, i.e., when x is an integral multiple of π

In-Text Questions and Answers (ITQs and ITAs)

ITQ 13.1: What is the fundamental definition of the sine and cosine of an angle θ, using the unit circle?

Answer: For a point P on the unit circle corresponding to an angle θ, the x-coordinate is cos θ and the y-coordinate is sin θ.

ITQ 13.2: What is a 'reference angle', and what is its role in finding the trigonometric functions of a non-acute angle?

Answer: A reference angle is the acute angle formed between the terminal side of the given angle and the x-axis. It has the same sine, cosine, and tangent

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values as the original angle, except for a possible change of sign depending on the quadrant.

Conclusion

This study session has shown that trigonometric functions are not just for standard angles but for angles of any magnitude. By building upon your knowledge of standard trigonometric values, we've explored strategies for extending these concepts to any angle.

Summary

1. Trigonometric functions are widely used in fields such as astronomy, architecture, physics, surveying, and medicine.
2. Non-standard angles can be analysed using trigonometric functions by considering the reference angle and the quadrant in which the angle lies.
3. Trigonometric function values vary across quadrants, with certain functions being positive in specific quadrants.
4. By understanding the ranges and properties of trigonometric functions, we can determine the values of angles for various functions.
5. Trigonometric ratios initially defined for acute angles can be extended to angles measured in radians.
6. The six trigonometric functions—sine, cosine, tangent, cotangent, secant, and cosecant—play distinct roles in relating angles and sides of right-angled triangles.

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7. Quadrantal angles, which are integral multiples of π, have specific values for sine and cosine functions.

Self-Assessment Questions and Answers (SAQs and SAAs)

SAQ 13.1: Explain why the unit circle is a more powerful tool for defining trigonometric functions than the right-angled triangle approach.

Answer: The right-angled triangle definition only works for acute angles (0° to 90°). The unit circle definition extends these functions to any angle, positive or negative, and clearly illustrates the periodic nature and signs of the functions in different quadrants.

SAQ 13.2: Describe the process for finding the exact value of sin(150°) without a calculator. Your answer should include the concept of a reference angle.

Answer: 150° is in the second quadrant. Its reference angle is 180° - 150° = 30°. In the second quadrant, sine is positive. Therefore, sin(150°) = +sin(30°) = 1/2.

SAQ 13.3: Why are the functions tangent and cotangent periodic with a period of π (180°), while sine and cosine have a period of 2π (360°)?

Answer: Tangent is defined as sin/cos. After a rotation of π radians, the point on the unit circle goes to (-x, -y). Both sin and cos change sign, so their ratio, tan, remains the same. Sin and cos themselves do not repeat their values until a full rotation of 2π.

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Multiple-Choice Questions and Answers (MCQs and MCAs)

1: In which quadrant is sine negative and cosine positive?

a) Quadrant I

b) Quadrant II

c) Quadrant IV

d) Quadrant III

Answer: c) Quadrant IV

Explanation: In Quadrant I, both sine and cosine are positive. In Quadrant II, sine is positive and cosine is negative. In Quadrant III, both sine and cosine are negative. In Quadrant IV, sine is negative and cosine is positive. This is often remembered using the mnemonic "All Students Take Calculus" or "ASTC" where A (all positive) in QI, S (sine positive) in QII, T (tangent positive) in QIII, and C (cosine positive) in QIV.

2: Which of the following is equal to $\cos(\pi)$?

a) 0

b) 1/2

c) -1

d) 0

Answer: c) -1

Explanation: The angle π radians corresponds to 180°. On the unit circle, the point at 180° is (-1, 0). Since cosine is the x-coordinate, $\cos(\pi) = -1$. Similarly, $\sin(\pi) = 0$.

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3: The expression sin(180° + θ) is equivalent to:

a) sin θ

b) cos θ

c) -sin θ

d) -cos θ

Answer: c) -sin θ

Explanation: When an angle θ is increased by 180°, its terminal side moves to the opposite quadrant. The sine function, which corresponds to the y-coordinate on the unit circle, changes sign. For example, sin(30°) = 0.5, while sin(210°) = -0.5. Therefore, sin(180° + θ) = -sin θ.

4: What is the value of tan(3π/2)?

a) 0

b) 1

c) -1

d) Undefined

Answer: d) Undefined

Explanation: The angle 3π/2 (270°) corresponds to the point (0, -1) on the unit circle. Since tan θ = sin θ / cos θ, we have tan(3π/2) = (-1) / 0. Division by zero is undefined. The tangent function has vertical asymptotes at angles where cos θ = 0, which occur at π/2, 3π/2, and all odd multiples of π/2.

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5: If sin(θ) is negative and tan(θ) is positive, in which quadrant does θ lie?

a) I

b) II

c) III

d) IV

Answer: c) III

Explanation: Tan θ = sin θ / cos θ. For tan θ to be positive, sin θ and cos θ must have the same sign (both positive or both negative). Sin θ is negative, so cos θ must also be negative. Both sine and cosine are negative only in Quadrant III. In Quadrant IV, sin is negative but cos is positive, making tan negative.

Tutor-Marked Assessments (TMAs)

1. Explain how the unit circle definition of sine and cosine allows us to find the value of these functions for angles greater than 90° and less than 0°.
2. Using the unit circle, prove the identity $\sin^2\theta + \cos^2\theta = 1$ for an angle θ in the second quadrant. (Hint: The coordinates of a point on the unit circle are (cos θ, sin θ), and the equation of the circle is $x^2 + y^2 = 1$).
3. An angle θ has a cosine value of -4/5 and lies in a specific quadrant.

   a) Determine which quadrant θ is in, justifying your answer.

   b) Using the Pythagorean identity, find the possible value(s) for sin(θ).

   c) Based on your answer from (a), state the correct value for sin(θ).

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4. Find the exact values (without a calculator) of the following trigonometric functions. Show your reasoning, including the reference angle and the sign based on the quadrant.

   a) $\cos(135°)$

   b) $\sin(5\pi/3)$

   c) $\tan(210°)$

5. A student calculates that $\sin(240°) = \sqrt{3}/2$. Identify the error in their reasoning and provide the correct calculation, explaining each step.

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Glossary of Terms

Addition Axioms: Fundamental principles governing addition in a field.

Addition and Subtraction: Operations applied to the real and imaginary parts of complex numbers.

Algebra of Complex Numbers: Mathematical operations involving complex numbers.

Arithmetic and Geometric Sequences and Series: Patterns of numbers with a defined order.

Binomial Theorem: A formula for expanding powers of binomials.

Combining Like Terms: The process of simplifying polynomial expressions by adding or subtracting terms with the same variables and exponents.

Complements: The complement of a set includes all elements not present in the set.

Complex numbers: numbers that include a real part and an imaginary part.

Cubic Inequalities: Inequalities involving cubic expressions.

Definition of a Field: The foundational criteria that define a mathematical field.

De Moivre's Theorem: A formula for raising complex numbers to integer powers.

Distributive Axiom: A principle describing the relationship between addition and multiplication in a field.

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Division: Dividing complex numbers.

Fields: Mathematical structures satisfying the axioms of addition and multiplication.

Linear Inequalities: Inequalities involving linear expressions.

Multiplication: The operation for multiplying complex numbers.

Multiplication Axioms: Foundational principles governing multiplication in a field.

Multiplying a Monomial by a Polynomial: The procedure for multiplying a single-term expression by a polynomial.

Multiplying Two Binomials (Distributive Property): An alternative method for multiplying binomial expressions by distributing each term of one binomial over the other.

Multiplying Two Binomials (FOIL Method): A technique for multiplying two binomial expressions using the acronym FOIL (First, Outer, Inner, Last).

Ordered Fields: Fields with an established order relation.

Polar Coordinates: A system for representing complex numbers using magnitude and angle.

Polynomial equations: Equations involving polynomial expressions.

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Positivity Preservation in Field F under Multiplication: A property indicating that if 'a' is positive and 'b' is greater than or equal to 'c' then 'ab' is greater than or equal to 'ac'

Properties of Absolute Values: Fundamental characteristics of absolute value functions.

Properties of Inequalities: Properties governing the relationships between inequality expressions.

Proposition: A statement or theorem put forth for consideration or demonstration.

Proper Subsets: Subsets that do not contain all the elements of the original set.

Quadratic Inequalities: Inequalities involving quadratic expressions.

Radian: A unit of measurement for angles.

Remainder and Factor Theorems: Theorems related to remainders and factors in polynomial equations.

Remainder Theorem: A theorem establishing the relationship between the remainder of a polynomial division and its root.

Representation of Inequalities: Methods of symbolically representing inequalities.

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Set of Real Numbers: The collection of all real numbers, encompassing both rational and irrational numbers.

Subsets: A subset is a set that contains only elements that are also contained in another set.

The Argand Diagram: A complex plane used to represent complex numbers.

The order relation: A mathematical relation that establishes a ranking or order among real numbers.

The Sigma Notation: A shorthand notation for expressing the sum of a sequence.

The Principle of Mathematical Induction: A mathematical proof technique used to establish a statement for all natural numbers.

The Trichotomy Law: A fundamental property of ordered fields that states that any two real numbers are either equal, one is greater than the other, or the other is greater.

Theory of Quadratic Equations: The study of equations of the form $ax^2 + bx + c = 0$.

Trigonometric Functions of Angles of Any Magnitude: Evaluating trigonometric functions for any angle measurement.

Union and Intersection: Operations involving the combination or overlap of sets.

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Venn Diagrams: Diagrammatic representations of sets using overlapping circles.

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